

Tarea_03

22 de septiembre de 2025

1. Bibliotecas

```
[1]: # Manipulación de datos
import pandas as pd
import numpy as np
import re
```

```
[2]: # Graficas
import matplotlib.pyplot as plt
from matplotlib import cycler
import seaborn as sns
import plotly.express as px
import plotly.graph_objects as go
from plotly.subplots import make_subplots
```

```
[3]: # Escalamiento
from sklearn.preprocessing import StandardScaler

# Componentes principales
from sklearn.decomposition import PCA

# Modelo de clustering
from sklearn.cluster import KMeans

# Métricas para selección de clusters
from sklearn.metrics import calinski_harabasz_score, davies_bouldin_score, \
    silhouette_score

# Pruebas estadísticas
from scipy.stats import kruskal
```

```
[4]: # !pip install nltk

from scipy import stats
from collections import Counter
import nltk # para trabajar con tokenizacion y stopwords (ejercicio 2)
```

```
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
```

```
[5]: from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.impute import SimpleImputer
```

2. Ejercicio 1

2.1. Cargar datos

```
[6]: songs1 = pd.read_csv(r'https://raw.githubusercontent.com/Wsm-erick/Diplomado/
↪refs/heads/main/Modulo-I/Datasets/features_songs.csv')
```

```
[7]: songs1.shape
```

```
[7]: (175, 40)
```

```
[8]: songs1.head()
```

```
[8]:
```

	archivo	duration	rms	tempo	\
0	02 Maroon.mp3	218.305306	0.269286	[107.66601562]	
1	03 Anti-Hero.mp3	200.724898	0.251603	[97.50884434]	
2	04 Snow On The Beach.mp3	256.156735	0.191276	[109.95678191]	
3	05 You_re On Your Own, Kid.mp3	194.246531	0.261242	[120.18531977]	
4	06 Midnight Rain.mp3	174.837551	0.226225	[69.83741554]	

	zcr	spectral_centroid	spectral_bandwidth	spectral_rolloff	\
0	0.030423	2131.429071	3054.294473	4833.956751	
1	0.042470	2270.523020	2887.932364	4680.853820	
2	0.026679	1691.923217	2516.562246	3338.831279	
3	0.021683	1637.701889	2600.810313	3373.619237	
4	0.021928	1485.590595	2261.139587	3117.497364	

	spectral_contrast	mfcc_1	...	chroma_9	chroma_10	chroma_11	\
0	21.985519	-165.31206	...	0.601635	0.531422	0.506995	
1	20.961313	-116.05038	...	0.481668	0.461698	0.506635	
2	21.885129	-205.33786	...	0.507185	0.615904	0.499524	
3	21.883741	-183.35786	...	0.440590	0.486334	0.450555	
4	21.791955	-223.94078	...	0.479455	0.510774	0.485107	

	chroma_12	tonnetz_1	tonnetz_2	tonnetz_3	tonnetz_4	tonnetz_5	tonnetz_6
0	0.518395	0.275898	0.210128	0.175254	0.011638	0.049738	-0.057944
1	0.596817	0.067301	-0.201125	-0.039244	0.056137	0.023695	-0.028362
2	0.530878	0.140940	-0.188128	-0.098086	-0.042389	0.013691	-0.021691

3	0.462748	0.128289	0.005553	0.065578	-0.140425	0.006879	0.009277
4	0.532159	0.107616	0.223978	-0.027103	0.086051	0.020835	0.057351

[5 rows x 40 columns]

2.2. Descripción de las variables

- **Duration:** duración total de la canción (en segundos).
- **RMS (Root Mean Square Energy):** mide la energía promedio del audio; valores altos indican pasajes fuertes, mientras que valores bajos corresponden a partes suaves.
- **Tempo:** velocidad rítmica estimada de la canción, medida en *beats por minuto (BPM)*.
- **Zero Crossing Rate (ZCR):** frecuencia con la que la señal cruza el eje cero; valores altos suelen asociarse a sonidos ruidosos o percusivos.
- **Spectral Centroid:** centro de masa del espectro; indica si el sonido se percibe como más brillante/agudo (valores altos) o más grave/opaco (valores bajos).
- **Spectral Bandwidth:** anchura espectral; mide la dispersión de las frecuencias alrededor del centroide.
- **Spectral Rolloff:** frecuencia por debajo de la cual se concentra el 85 % de la energía espectral; se relaciona con el timbre del audio.
- **Spectral Contrast:** diferencia entre picos y valles en el espectro; describe cuán plana o marcada es la distribución espectral.
- **MFCCs (Mel-Frequency Cepstral Coefficients):** coeficientes que resumen la envolvente espectral en la escala mel; son representaciones del timbre y muy utilizados en reconocimiento de voz y música.
- **Chroma:** distribución de la energía a lo largo de las 12 notas musicales (do, do#, re, ...); útil para analizar armonía y progresiones de acordes.
- **Tonnetz:** representa relaciones tonales y armónicas (quintas, terceras, consonancias); permite caracterizar la tonalidad de una pieza musical.

2.3. Actividad

Con estas variables se sugiere hacer una segmentación (análisis de clustering) para cada usuario y así encontrar grupos sobre las canciones que escucha y armar playlist personalizadas.

Tu tarea es analizar el conjunto de datos, realizar un análisis estadístico, escalar los datos y preparar un nuevo df reduciendo dimensiones para que posteriormente el equipo realice el análisis de clustering.

¿Qué otras variables agregarías tú y por qué? Justifica

3. Exploración

3.0.1. Tipo de dato por variable

```
[9]: songs1.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 175 entries, 0 to 174
Data columns (total 40 columns):
#   Column                Non-Null Count  Dtype
---  -
0   archivo               175 non-null    object
1   duration              175 non-null    float64
2   rms                   175 non-null    float64
3   tempo                 175 non-null    object
4   zcr                   175 non-null    float64
5   spectral_centroid     175 non-null    float64
6   spectral_bandwidth    175 non-null    float64
7   spectral_rolloff      175 non-null    float64
8   spectral_contrast     175 non-null    float64
9   mfcc_1                175 non-null    float64
10  mfcc_2                175 non-null    float64
11  mfcc_3                175 non-null    float64
12  mfcc_4                175 non-null    float64
13  mfcc_5                175 non-null    float64
14  mfcc_6                175 non-null    float64
15  mfcc_7                175 non-null    float64
16  mfcc_8                175 non-null    float64
17  mfcc_9                175 non-null    float64
18  mfcc_10               175 non-null    float64
19  mfcc_11               175 non-null    float64
20  mfcc_12               175 non-null    float64
21  mfcc_13               175 non-null    float64
22  chroma_1              175 non-null    float64
23  chroma_2              175 non-null    float64
24  chroma_3              175 non-null    float64
25  chroma_4              175 non-null    float64
26  chroma_5              175 non-null    float64
27  chroma_6              175 non-null    float64
28  chroma_7              175 non-null    float64
29  chroma_8              175 non-null    float64
30  chroma_9              175 non-null    float64
31  chroma_10             175 non-null    float64
32  chroma_11             175 non-null    float64
33  chroma_12             175 non-null    float64
34  tonnetz_1             175 non-null    float64
35  tonnetz_2             175 non-null    float64
36  tonnetz_3             175 non-null    float64
```

```

37 tonnetz_4          175 non-null    float64
38 tonnetz_5          175 non-null    float64
39 tonnetz_6          175 non-null    float64
dtypes: float64(38), object(2)
memory usage: 54.8+ KB

```

3.0.2. Valores ausentes

```

[10]: def is_na_df(df):
        #Establecemos el resumen de NAs por columna
        na_summary = (
            df.isna().sum()
            .reset_index()
            .rename(columns={"index": "columna", 0: "n_missing"})
        )

        #Tomamos totales y prporciones
        na_summary["total"] = len(df)
        na_summary["prop_missing"] = na_summary["n_missing"] / na_summary["total"]
        na_summary["prop_original"] = 1 - na_summary["prop_missing"]

        #Filtramos solo columnas con al menos 1 NA
        na_summary = na_summary[na_summary["n_missing"] > 0].reset_index(drop=True)

        #Agregamos fila resumen global
        total_missing = df.isna().sum().sum()
        total_elements = df.size
        prop_missing_global = total_missing / total_elements
        prop_original_global = 1 - prop_missing_global

        resumen = pd.DataFrame({
            "columna": ["TOTAL"],
            "n_missing": [total_missing],
            "total": [total_elements],
            "prop_missing": [prop_missing_global],
            "prop_original": [prop_original_global]
        })

        #Creamos la tabla
        na_summary = pd.concat([na_summary, resumen], ignore_index=True)

        return na_summary

```

```

[11]: is_na_df(songs1)

```

```

[11]:   columna  n_missing  total  prop_missing  prop_original
0    TOTAL           0   7000           0.0           1.0

```

No hay valores ausentes por lo que no hace falta aplicar un método de imputación.

3.0.3. Distribución de variables numéricas

```
[12]: var_cat = songs1.select_dtypes(['object']).columns.tolist()
      var_cat
```

```
[12]: ['archivo', 'tempo']
```

```
[13]: songs1['tempo_num'] = songs1['tempo'].str[1:-1].astype(float)
```

```
[14]: def guess_lang_heuristic(text: str):

    spanish_words = {
        ↪
        ↪ "el", "la", "los", "las", "un", "una", "unos", "unas", "de", "del", "al", "y", "o", "u",
        ↪
        ↪ "que", "como", "con", "por", "para", "sin", "sobre", "entre", "más", "menos", "mi", "tu",
        ↪      "te", "se", "me", "lo", "le", "su", "si", "ya", "no", "sí", "a", "en"
        ↪    }

    tokens = {t for t in re.findall(r"\b\w+\b", text)}
    hits = len(tokens & spanish_words)

    # si tiene acentos/símbolos muy comunes en español, suma
    accent_bonus = 1 if re.search(r"[áéíóúñü]", text.lower()) else 0
    score_es = hits + accent_bonus

    # título muy corto => baja confianza
    conf = min(1.0, 0.2*score_es + 0.2*(len(tokens) >= 3))
    if score_es >= 2:
        return "es", conf

    # si no hay indicios en español y tiene muchas palabras comunes en inglés
    english_words = ↪
    ↪ {"the", "and", "you", "your", "i", "me", "my", "on", "in", "with", "for", "of", "to", "from", "at", "it", "is"}
    hits_en = len(tokens & english_words)
    if hits_en >= 2:
        return "en", max(conf, 0.6)

    # por defecto "en" si todo está en ASCII y sin acentos; si no, desconocido
    return ("en", 0.5) if text.isascii() else ("unk", 0.3)

[15]: def label_from_code(code):
      return {"es": "Español", "en": "Inglés", "unk": "Desconocido"}.
      ↪get(code, "Desconocido")
```

```
[16]: songs1['archivo'] = songs1['archivo'].str.replace('.mp3' , '').str.lower()
```

```
[17]: songs1['archivo'].to_csv('canciones.csv')
```

```
[18]: songs1["lang_code"], songs1["lang_conf"] = zip(*songs1["archivo"].
↳map(guess_lang_heuristic))
songs1["lang"] = songs1["lang_code"].map(label_from_code)
```

```
[19]: songs1.head(3)
```

```
[19]:
```

	archivo	duration	rms	tempo	zcr	\
0	02 maroon	218.305306	0.269286	[107.66601562]	0.030423	
1	03 anti-hero	200.724898	0.251603	[97.50884434]	0.042470	
2	04 snow on the beach	256.156735	0.191276	[109.95678191]	0.026679	

	spectral_centroid	spectral_bandwidth	spectral_rolloff	spectral_contrast	\
0	2131.429071	3054.294473	4833.956751	21.985519	
1	2270.523020	2887.932364	4680.853820	20.961313	
2	1691.923217	2516.562246	3338.831279	21.885129	

	mfcc_1	...	tonnetz_1	tonnetz_2	tonnetz_3	tonnetz_4	tonnetz_5	\
0	-165.31206	...	0.275898	0.210128	0.175254	0.011638	0.049738	
1	-116.05038	...	0.067301	-0.201125	-0.039244	0.056137	0.023695	
2	-205.33786	...	0.140940	-0.188128	-0.098086	-0.042389	0.013691	

	tonnetz_6	tempo_num	lang_code	lang_conf	lang
0	-0.057944	107.666016	en	0.5	Inglés
1	-0.028362	97.508844	en	0.5	Inglés
2	-0.021691	109.956782	en	0.6	Inglés

[3 rows x 44 columns]

```
[20]: class analyze:
    # Inicializamos la clase
    def __init__(self, df, num_vars, cat_vars):
        self.df = df
        self.num_vars = num_vars
        self.cat_vars = cat_vars

    #Método para graficar variables numéricas (cada una en figura independiente)
    def plot_num(self):
        for var in self.num_vars:
            fig = make_subplots(rows=1, cols=2, subplot_titles=(f"{var} - 
↳Boxplot", f"{var} - Histograma"))

            # Boxplot
            fig.add_trace(
```

```

        go.Box(y=self.df[var], name="Boxplot", marker_color="#008080"),
        row=1, col=1
    )

    # Histograma
    fig.add_trace(
        go.Histogram(x=self.df[var], name="Histograma",
                     #marker_color="#00a77d",
                     #marker_color='#722F37',
                     marker_color="rgba(114, 47, 55, 0.5)", # burdeos
→ con 50% transparencia
                     nbinsx=50,
                     histnorm='probability density'),
        row=1, col=2
    )

    fig.update_layout(title_text=f"Variable numérica: {var}",
→ height=400, width=800)
    fig.show()

    #Método para graficar variables categóricas
    def plot_cat(self):
        n = len(self.cat_vars)
        cols = 2
        rows = (n + 1) // cols if n > 1 else 1

        fig = make_subplots(
            rows=rows, cols=cols,
            specs=[[{'type': 'domain'}]*cols for _ in range(rows)],
            subplot_titles=self.cat_vars
        )

        for i, var in enumerate(self.cat_vars):
            row = i // cols + 1
            col = i % cols + 1

            value_counts = self.df[var].value_counts()
            labels = value_counts.index.astype(str)
            values = value_counts.values

            wine_palette = [
                "#722F37", "#B85042", "#E6AF2E", "#F6D55C", "#F4A7B9",
                "#2F4F4F", "#8D6E63", "#A1887F", "#D7CCC8", "#795548"
            ]

            fig.add_trace(
                go.Pie(

```



```

        labels=labels,
        values=values,
        name=var,
        textinfo='percent+label',
        marker=dict(colors=wine_palette)
    ),
    row=row, col=col
)

#fig.add_trace(
#    go.Pie(labels=labels, values=values, name=var,
→textinfo='percent+label'),
#    row=row, col=col
#)

fig.update_layout(title_text="Variables Categóricas", height=400*rows)
fig.show()

#Método para graficar la matriz de correlación de numéricas
def corr_matrix(self, color: str, values: bool = True):
    corr_matrix = self.df[self.num_vars].corr()

    if values:
        fig = go.Figure(data=go.Heatmap(
            z=corr_matrix.values,
            x=corr_matrix.columns,
            y=corr_matrix.columns,
            colorscale=color,
            text=corr_matrix.round(1).values,          # valores de correlación
→con 1 decimal
            texttemplate="%{text}",                    # mostrar texto en cada
→celda
            textfont={"size":10, "color":"black"} # estilo de fuente
        ))
    else:
        fig = go.Figure(data=go.Heatmap(
            z=corr_matrix.values,
            x=corr_matrix.columns,
            y=corr_matrix.columns,
            colorscale=color
        ))

    fig.update_layout(
        title='Matriz de Correlación',
        width=800,
        height=600
    )

```

```
fig.show()
```

```
[21]: var_cat = songs1.select_dtypes('object').columns.tolist()
      var_num = list(set(songs1.columns)-set(var_cat))
      var_cat
```

```
[21]: ['archivo', 'tempo', 'lang_code', 'lang']
```

```
[22]: analisis = analyze(songs1, var_num, ['lang'])
```

```
[23]: songs1.describe()
```

```
[23]:
```

	duration	rms	zcr	spectral_centroid	\
count	175.000000	175.000000	175.000000	175.000000	
mean	242.625580	0.178263	0.058199	2648.488772	
std	67.956199	0.049730	0.018800	635.439223	
min	124.240000	0.055421	0.018877	1036.709252	
25%	191.916952	0.138568	0.044932	2252.657618	
50%	229.111293	0.178693	0.057285	2645.661730	
75%	280.912120	0.209344	0.071291	3040.246637	
max	472.201020	0.307894	0.106914	4261.901169	

	spectral_bandwidth	spectral_rolloff	spectral_contrast	mfcc_1	\
count	175.000000	175.000000	175.000000	175.000000	
mean	2988.567941	5473.065749	21.801889	-158.931074	
std	500.307972	1397.766313	0.858255	38.227926	
min	1320.658488	1975.675843	19.465870	-276.422400	
25%	2654.740590	4612.901963	21.283166	-181.099535	
50%	2989.998973	5520.686528	21.773645	-154.530210	
75%	3308.289221	6251.972851	22.263098	-133.741050	
max	4387.397055	9319.848099	24.342274	-84.056380	

	mfcc_2	mfcc_3	...	chroma_11	chroma_12	tonnetz_1	\
count	175.000000	175.000000	...	175.000000	175.000000	175.000000	
mean	140.614699	-26.212339	...	0.456177	0.466682	0.040485	
std	21.623594	21.760878	...	0.085315	0.070548	0.103817	
min	85.240814	-75.233640	...	0.237338	0.254564	-0.222699	
25%	125.182952	-42.279401	...	0.405111	0.425203	-0.032431	
50%	139.702700	-26.979185	...	0.461059	0.466638	0.036603	
75%	155.288480	-15.626485	...	0.508748	0.517050	0.122870	
max	220.203380	36.474575	...	0.642065	0.682561	0.290125	

	tonnetz_2	tonnetz_3	tonnetz_4	tonnetz_5	tonnetz_6	tempo_num	\
count	175.000000	175.000000	175.000000	175.000000	175.000000	175.000000	
mean	0.032733	0.013377	0.013172	0.002823	-0.002182	111.935760	
std	0.102631	0.069900	0.077798	0.019156	0.021415	22.777153	

min	-0.293091	-0.228473	-0.211357	-0.046127	-0.063731	69.837416
25 %	-0.028935	-0.037906	-0.042667	-0.009250	-0.014530	97.508844
50 %	0.038114	0.008709	0.019528	0.004239	-0.002709	107.666016
75 %	0.099992	0.053448	0.068193	0.012924	0.011443	120.185320
max	0.262957	0.175254	0.220034	0.049738	0.063773	198.768029

	lang_conf
count	175.000000
mean	0.535429
std	0.116962
min	0.300000
25 %	0.500000
50 %	0.500000
75 %	0.600000
max	1.000000

[8 rows x 40 columns]

```
[24]: analisis.plot_num()
```

```
[25]: analisis.plot_cat()
```

```
[26]: songs1.query("lang == 'Desconocido').shape
```

```
[26]: (13, 44)
```

Eliminación de canciones de otro idioma distinto al inglés y al español

```
[27]: songs1.drop(songs1.query("lang == 'Desconocido').index, inplace=True)
```

```
[28]: songs1.shape
```

```
[28]: (162, 44)
```

3.1. Tratamiento de Datos

3.1.1. Escalado

```
[29]: colors = ['aggrnyl', 'agsunset', 'algae', 'amp', 'armyrose', 'balance',
                'blackbody', 'bluered', 'blues', 'blugrn', 'bluyl', 'brbg',
                'brwnyl', 'bugn', 'bupu', 'burg', 'burgyl', 'cividis', 'curl',
                'darkmint', 'deep', 'delta', 'dense', 'earth', 'edge', 'electric',
                'emrld', 'fall', 'geyser', 'gnbu', 'gray', 'greens', 'greys',
                'haline', 'hot', 'hsv', 'ice', 'icefire', 'inferno', 'jet',
                'magenta', 'magma', 'matter', 'mint', 'mrybm', 'mygbm', 'oranges',
                'orrd', 'oryel', 'oxy', 'peach', 'phase', 'picnic', 'pinkyl',
                'piyg', 'plasma', 'plotly3', 'portland', 'prgn', 'pubu', 'pubugn',
                'puor', 'purd', 'purp', 'purples', 'purpor', 'rainbow', 'rdbu',
```

```
'rdgy', 'rdpu', 'rdylbu', 'rdylgn', 'redor', 'reds', 'solar',
'spectral', 'speed', 'sunset', 'sunsetdark', 'teal', 'tealgrn',
'tealrose', 'tempo', 'temps', 'thermal', 'tropic', 'turbid',
'turbo', 'twilight', 'viridis', 'ylgn', 'ylgnbu', 'ylorbr',
'ylorrd']
```

```
[30]: analisis.corr_matrix('cividis', False)
```

```
[31]: X = songs1.drop(columns=var_cat + ['lang_conf'])
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

3.1.2. Componentes Principales

```
[32]: class rPCA:
    #Inicializamos
    def __init__(self, dataframe, cols, n_comp):
        self.dataframe = dataframe
        self.cols = cols
        self.n_comp = n_comp

        if self.cols == None:
            self.dataframe = dataframe
        else:
            self.dataframe = dataframe[cols]

    #Ajustamos el PCA
    def r_pca(self):
        pca = PCA(n_components = self.n_comp)
        pca.fit(self.dataframe)
        self.df_trans = pd.DataFrame(pca.transform(self.dataframe),
        ↪columns=[f'p_{i}' for i in range(len(pca.explained_variance_ratio_))])
        return self.df_trans, pca.explained_variance_ratio_.cumsum()

    #Diagrama del codo
    def plot_elbow(self):
        pca_temp = PCA().fit(self.dataframe)
        var_exp = pca_temp.explained_variance_ratio_
        cum_var_exp = var_exp.cumsum()

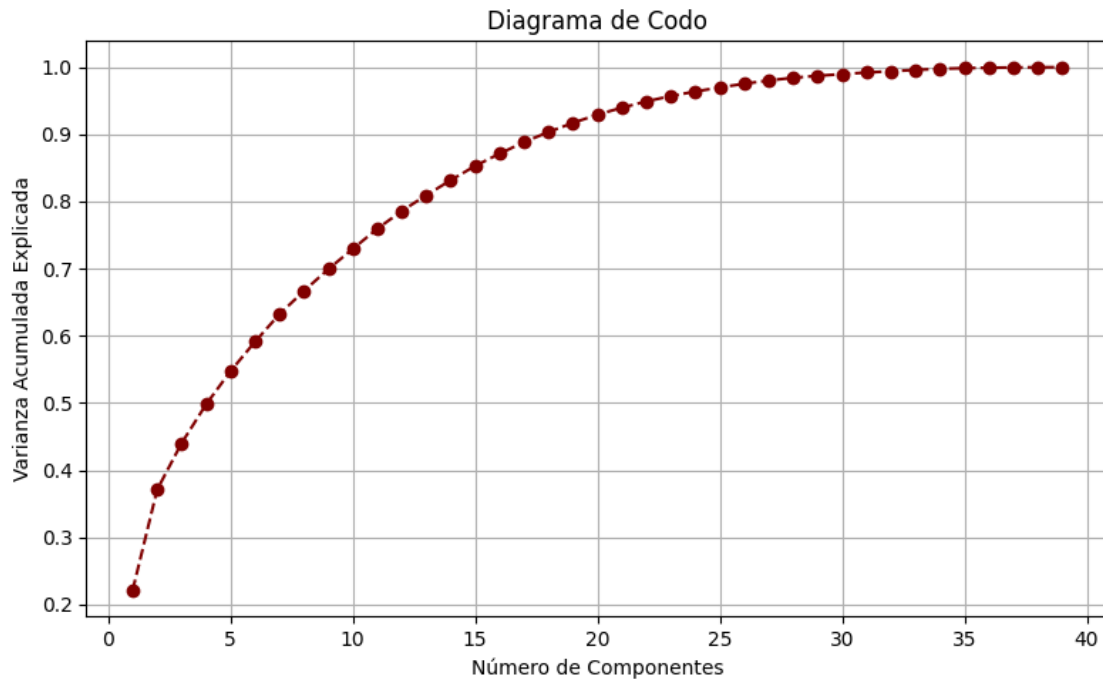
        plt.figure(figsize=(8, 5))
        plt.plot(range(1, len(var_exp) + 1), cum_var_exp, marker='o',
        ↪linestyle='--', color='maroon')
        plt.xlabel('Número de Componentes')
        plt.ylabel('Varianza Acumulada Explicada')
        plt.title('Diagrama de Codo')
```

```
plt.grid(True)
plt.tight_layout()
plt.show()
```

```
[33]: var_num = [x for x in var_num if x != 'lang_conf']
```

```
[34]: pca_01 = rPCA(pd.DataFrame(X_scaled, columns=var_num), var_num, 0.90)
```

```
[35]: pca_01.plot_elbow()
```



```
[36]: print("Componentes:", pca_01.r_pca()[0].shape[1])
```

Componentes: 18

3.1.3. Vecinos Cercanos

```
[37]: class rKMeans:
    #Inicializamos
    def __init__(self, dataframe, cols=None, max_clusters=10, random_state=42):
        self.cols = cols
        self.max_clusters = max_clusters
        self.random_state = random_state
        self.df = dataframe if cols is None else dataframe[cols]
        self.model = None
        self.labels_ = None
```

```

self.centroids_ = None
self.df_plot = None

def optimal_clusters(self, random_state: int | None = None, plot: bool = True):
    if random_state is None:
        random_state = self.random_state

    silhouette_scores = []
    inertias = []
    calinski_scores = []
    davies_scores = []

    K = range(2, self.max_clusters + 1)

    for k in K:
        km = KMeans(n_clusters=k, random_state=random_state)
        labels = km.fit_predict(self.df)

        silhouette_scores.append(silhouette_score(self.df, labels))
        inertias.append(km.inertia_)
        calinski_scores.append(calinski_harabasz_score(self.df, labels))
        davies_scores.append(davies_bouldin_score(self.df, labels))

    df_eval = pd.DataFrame({
        'Clusters': list(K),
        'Silueta': silhouette_scores,
        'Inercia': inertias,
        'Calinski-Harabasz': calinski_scores,
        'Davies-Bouldin': davies_scores
    })

    if plot:
        fig, ax = plt.subplots(2, 2, figsize=(14, 10))

        ax[0, 0].plot(K, silhouette_scores, marker='o', color='blue')
        ax[0, 0].set_title('Coeficiente de Silueta')
        ax[0, 0].set_xlabel('Número de Clusters')
        ax[0, 0].set_ylabel('Silueta')

        ax[0, 1].plot(K, inertias, marker='o', color='green')
        ax[0, 1].set_title('Inercia')
        ax[0, 1].set_xlabel('Número de Clusters')
        ax[0, 1].set_ylabel('Inercia')

        ax[1, 0].plot(K, calinski_scores, marker='o', color='navy')

```

```

        ax[1, 0].set_title('Índice de Calinski-Harabasz')
        ax[1, 0].set_xlabel('Número de Clusters')
        ax[1, 0].set_ylabel('CH')

        ax[1, 1].plot(K, davies_scores, marker='o', color='maroon')
        ax[1, 1].set_title('Índice de Davies-Bouldin')
        ax[1, 1].set_xlabel('Número de Clusters')
        ax[1, 1].set_ylabel('DB')

        plt.tight_layout()
        plt.show()

    return df_eval

def fit(self, n_clusters, random_state: int | None = None):
    if random_state is None:
        random_state = self.random_state

    self.model = KMeans(n_clusters=n_clusters, random_state=random_state)
    self.labels_ = self.model.fit_predict(self.df)
    self.centroids_ = pd.DataFrame(self.model.cluster_centers_, columns=self.
→df.columns)
    #return self.labels_

    def plot(self, df, x='p_0', y='p_1'):
        self.df_plot = df.copy()
        self.df_plot['cluster'] = self.labels_
        sns.lmplot(data=self.df_plot, x=x, y=y, hue='cluster', fit_reg=False,
→palette='viridis')

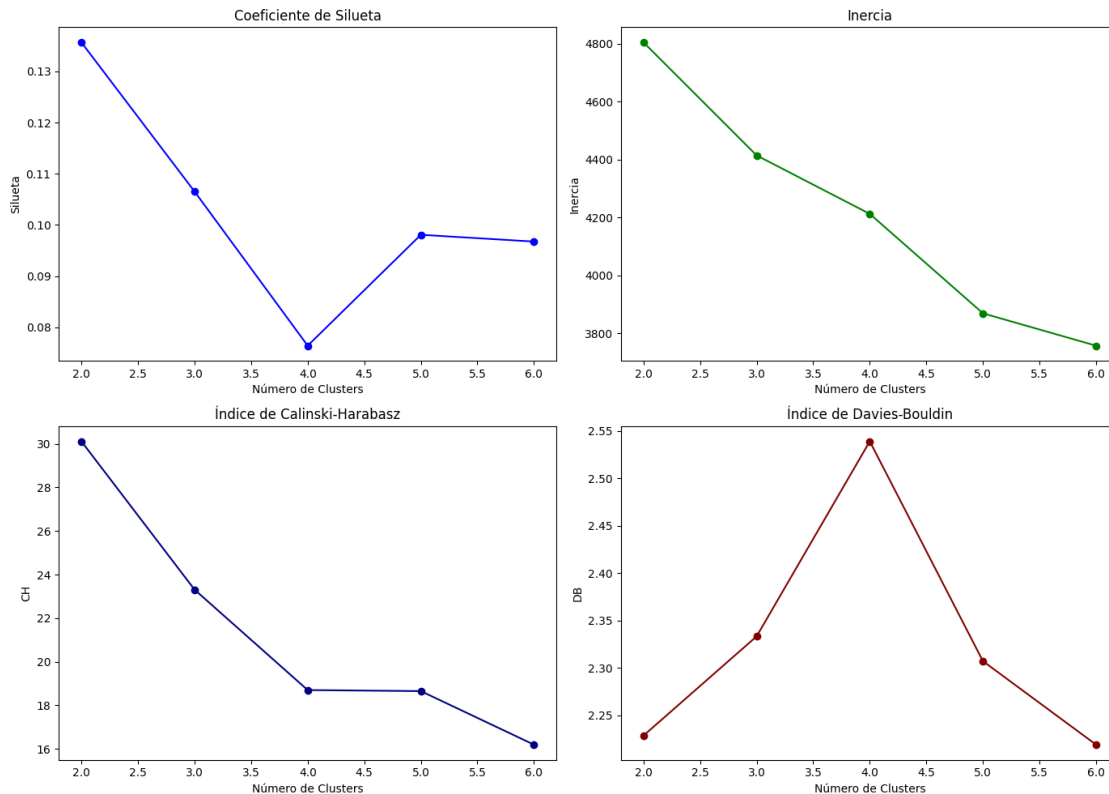
    def get_df(self):
        return self.df_plot

```

```
[38]: data = pca_01.r_pca()[0]
```

```
[39]: kmeans = rKMeans(dataframe=data, cols=None, max_clusters=6, random_state=42)
```

```
[40]: kmeans.optimal_clusters()
```



```
[40]: Clusters  Silueta      Inercia  Calinski-Harabasz  Davies-Bouldin
0       2  0.135649  4804.410852      30.116167        2.228400
1       3  0.106535  4414.329244      23.311480        2.333131
2       4  0.076432  4212.772515      18.701909        2.538843
3       5  0.098087  3869.642521      18.653927        2.306946
4       6  0.096759  3757.972941      16.195832        2.218842
```

3.1.4. Interpretación de Gráficos

- **Coeficiente de Silueta.** Un valor más alto habla de una mejor separación de los clusters, en este caso aunque mejora un poco el de 5 clusters no supera al de 2.
- **Inercia.** Mide la separación de los puntos con respecto sus centroides, siempre disminuye al aumentar los clusters aunque en este caso el nodo está entre 3 y 4.
- **Índice de Calinski–Harabasz (CH).** Un valor más alto habla de clusters mejor definidos, nuevamente dos clusters vuelve a ser el mejor grupo de clusters a elegir.
- **Índice de Davies–Bouldin (DB).** Entre más bajo hay una mejor separación entre clusters, en este caso 2 y 6 son los mejores valores, siendo el peor el de 4 clusters.

```
[41]: pca = PCA(n_components=0.9) # conserva 90% de varianza
X_pca = pca.fit_transform(X_scaled)
```



```

X_data = X_pca # si ya usaste PCA

best_score = -1
best_state = None
scores = []

# Probar múltiples semillas
for rs in range(100):
    kmeans_ = KMeans(n_clusters=2, random_state=rs, n_init=10)
    labels = kmeans_.fit_predict(X_data)
    score = silhouette_score(X_data, labels)
    scores.append((rs, score))

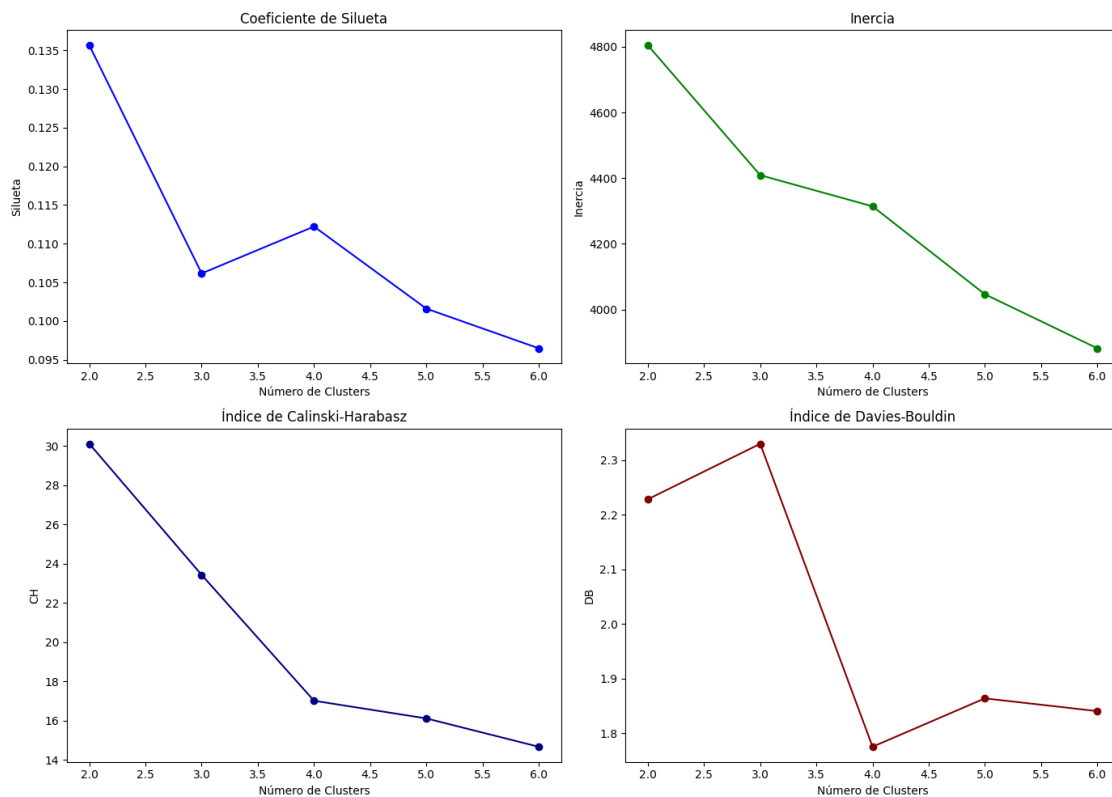
    if score > best_score:
        best_score = score
        best_state = rs

print(f"- Mejor silueta: {best_score:.4f} con random_state = {best_state}")

```

- Mejor silueta: 0.1372 con random_state = 11

```
[42]: kmeans.optimal_clusters(11)
```

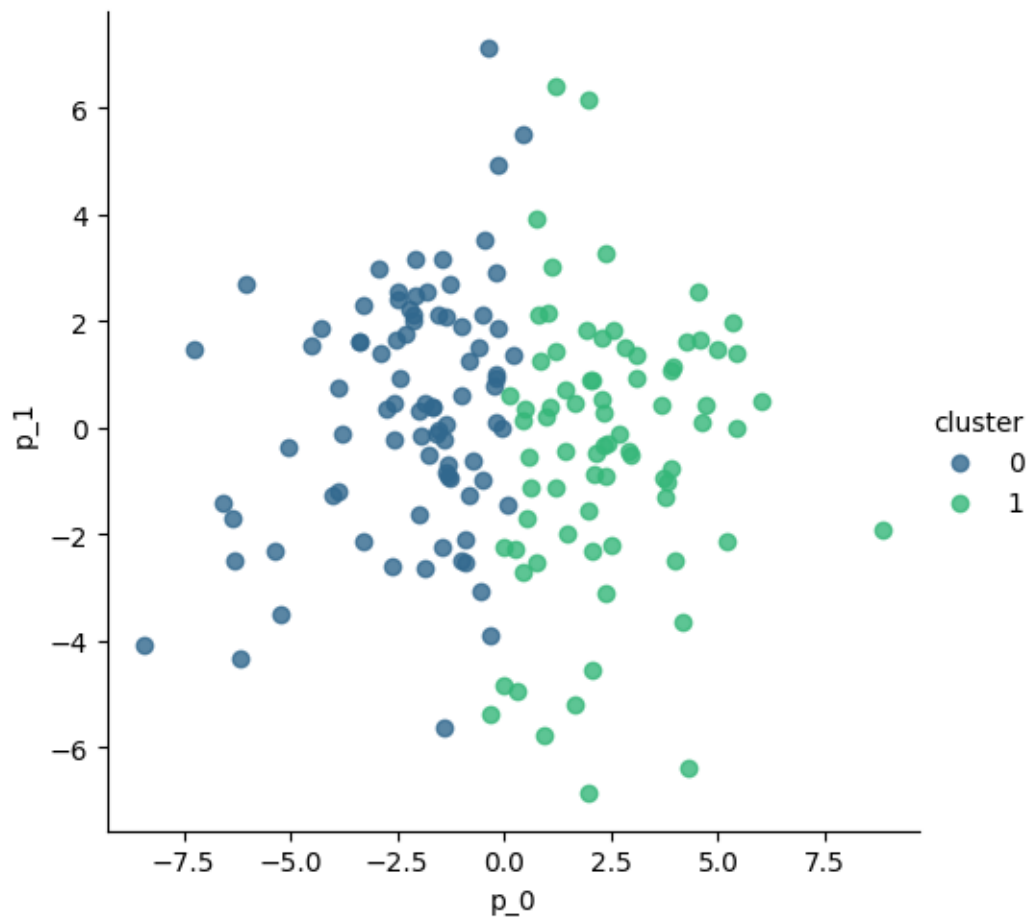


```
[42]:
```

	Clusters	Silueta	Inercia	Calinski-Harabasz	Davies-Bouldin
0	2	0.135649	4804.410852	30.116167	2.228400
1	3	0.106163	4409.072854	23.434050	2.329979
2	4	0.112222	4314.311821	17.022217	1.775183
3	5	0.101610	4046.592685	16.121894	1.863789
4	6	0.096486	3882.283845	14.678215	1.840408

```
[43]: kmeans.fit(n_clusters=2, random_state=11)
```

```
[44]: kmeans.plot(data)
```



```
[45]: songs1.reset_index(drop=True, inplace=True)
```

```
[46]: songs1['cluster'] = kmeans.get_df()['cluster']
```

```
[47]: pd.concat([songs1['cluster'].value_counts(), songs1['cluster'].
    ↳value_counts(normalize=True).apply(lambda x: '{:.1%}'.format(x))], axis=1)
```

```
[47]:          count proportion
cluster
0          85         52.5%
1          77         47.5%
```

3.2. Playlist personalizada

3.2.1. Variables significativas para una mejor separación de clusters

```
[48]: #Definimos un objeto para perfilamiento
class Perfil:
    def __init__(self, df, vars_cat, vars_cont, group_var='cluster'):
        self.df = df
        self.vars_cat = vars_cat
        self.vars_cont = vars_cont
        self.group_var = group_var

    def plot_dist(self, vars_cont: list | None = None):
        if vars_cont is None:
            vars_cont = self.vars_cont

        for var in vars_cont:
            plt.figure(figsize=(6, 4))
            sns.kdeplot(data=self.df, x=var, hue=self.group_var,
                ↪common_norm=False, fill=True, alpha=0.4, palette='viridis')
            plt.title(f'Densidad de {var} por grupos')
            plt.xlabel(var)
            plt.ylabel(None)
            plt.tight_layout()
            plt.show()

    def kw_test(self):
        significativas = []
        for var in self.vars_cont:
            grupos = [grupo[var].dropna() for _, grupo in self.df.groupby(self.
                ↪group_var)]
            stat, p = kruskal(*grupos)
            #if p < 0.05:
                ↪significativas.append((var, round(p, 5)))

            significativas.append((var, round(p, 5)))

        return pd.DataFrame(significativas, columns=['Variable', 'p-value'])

    def group_stats(self, vars):
        df_stats = self.df.groupby('cluster')[vars].agg(['mean', 'median',
            ↪'min', 'max'])
```

```
df_stats.columns = ['_'.join(col) for col in df_stats.columns]
df_stats.reset_index(inplace=True)
return df_stats
```

```
[49]: perfil = Perfil(songs1, var_cat, var_num)
```

```
[50]: aux = perfil.kw_test().sort_values(by='p-value', ascending=False).
      ↪reset_index(drop=True)

var_sig = aux.loc[aux['p-value'] < 0.05, "Variable"].tolist()
var_nosig = aux.loc[aux['p-value'] >= 0.05, "Variable"].tolist()

print('\nVariables no significativas\n')
display(aux.query(f"Variable == {var_nosig}"))

print('\nVariables significativas\n')
display(aux.query(f"Variable == {var_sig}"))
```

Variables no significativas

	Variable	p-value
0	tonnetz_6	0.96923
1	duration	0.70844
2	tempo_num	0.56595
3	tonnetz_4	0.55388
4	mfcc_5	0.49280
5	tonnetz_5	0.28999
6	mfcc_9	0.27942
7	tonnetz_3	0.20191
8	mfcc_3	0.08383
9	tonnetz_2	0.06198
10	mfcc_11	0.05922
11	tonnetz_1	0.05877

Variables significativas

	Variable	p-value
12	zcr	0.03014
13	mfcc_7	0.02369
14	mfcc_1	0.00234
15	rms	0.00058
16	mfcc_13	0.00047
17	mfcc_4	0.00036
18	spectral_centroid	0.00000
19	chroma_10	0.00000

```

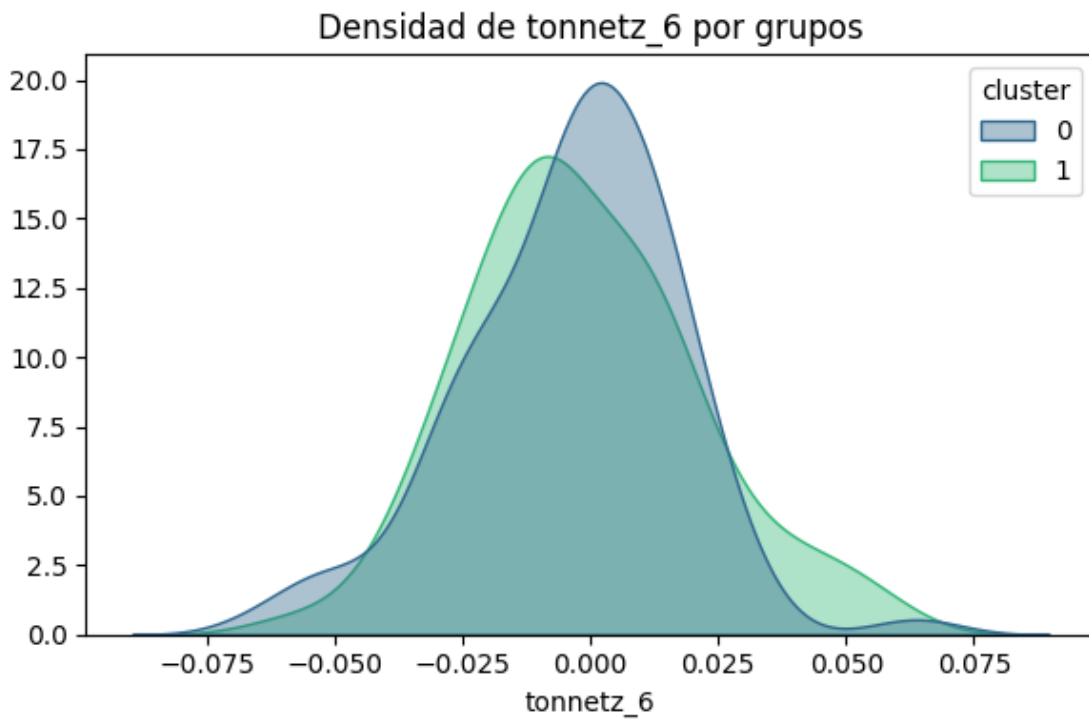
20         mfcc_12  0.00000
21         chroma_12 0.00000
22         chroma_1  0.00000
23         chroma_11 0.00000
24 spectral_bandwidth 0.00000
25         mfcc_10   0.00000
26         chroma_5   0.00000
27         chroma_6   0.00000
28         chroma_2   0.00000
29         mfcc_6     0.00000
30         chroma_3   0.00000
31 spectral_contrast  0.00000
32 spectral_rolloff  0.00000
33         mfcc_8     0.00000
34         chroma_9   0.00000
35         chroma_8   0.00000
36         chroma_4   0.00000
37         mfcc_2     0.00000
38         chroma_7   0.00000

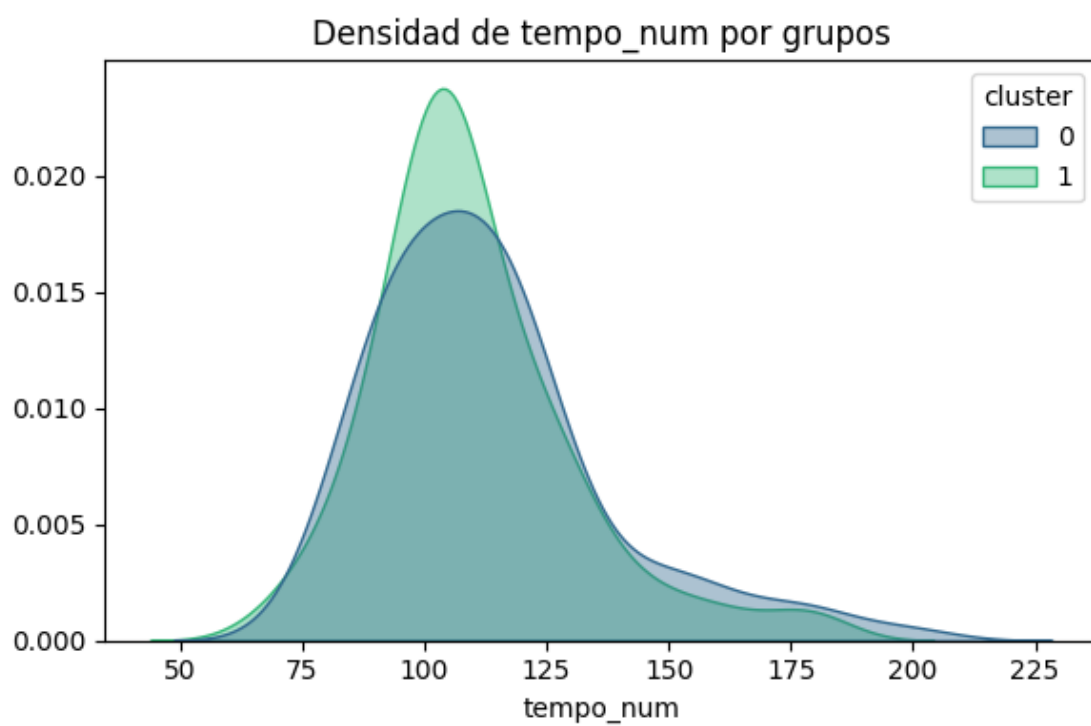
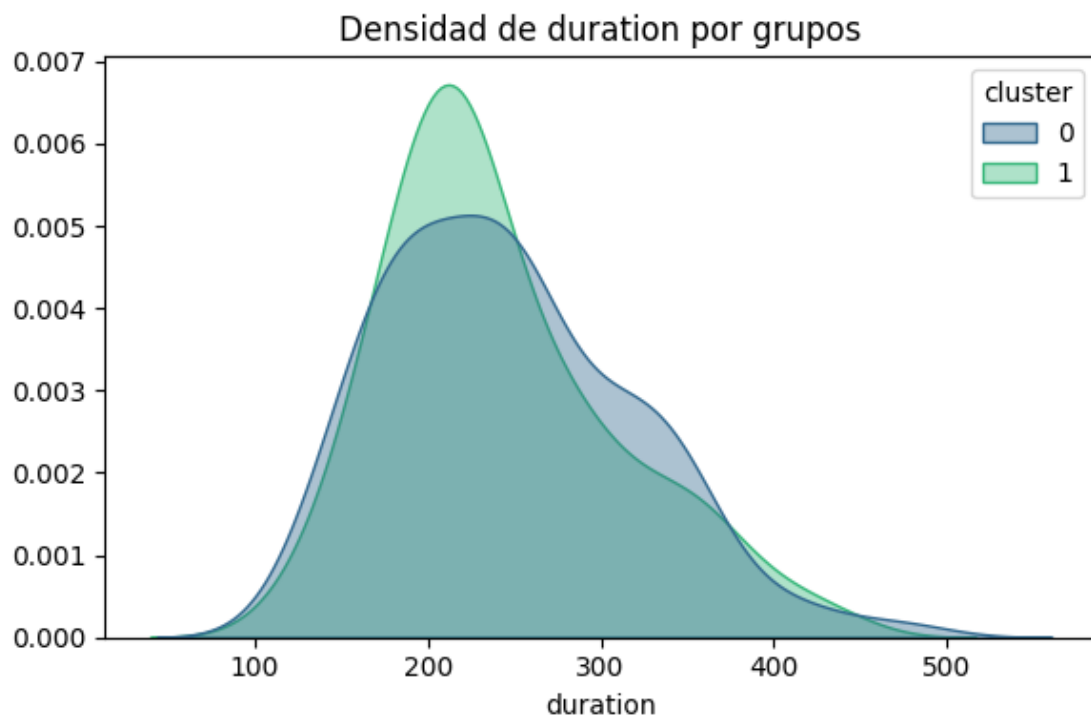
```

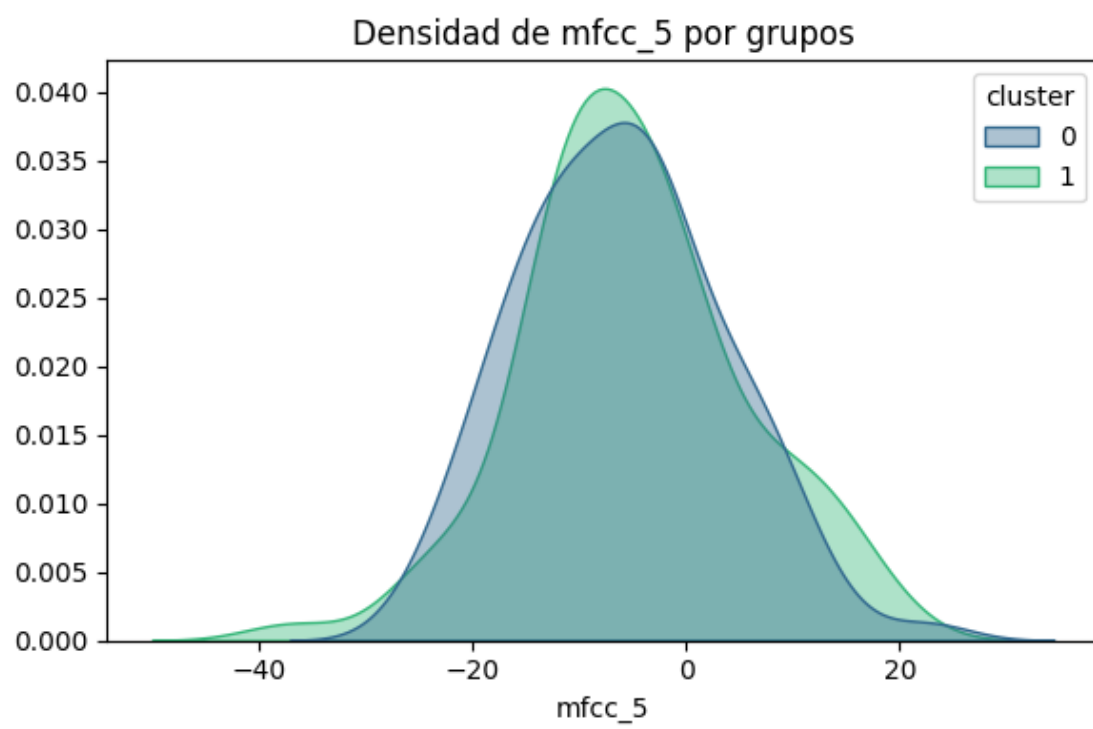
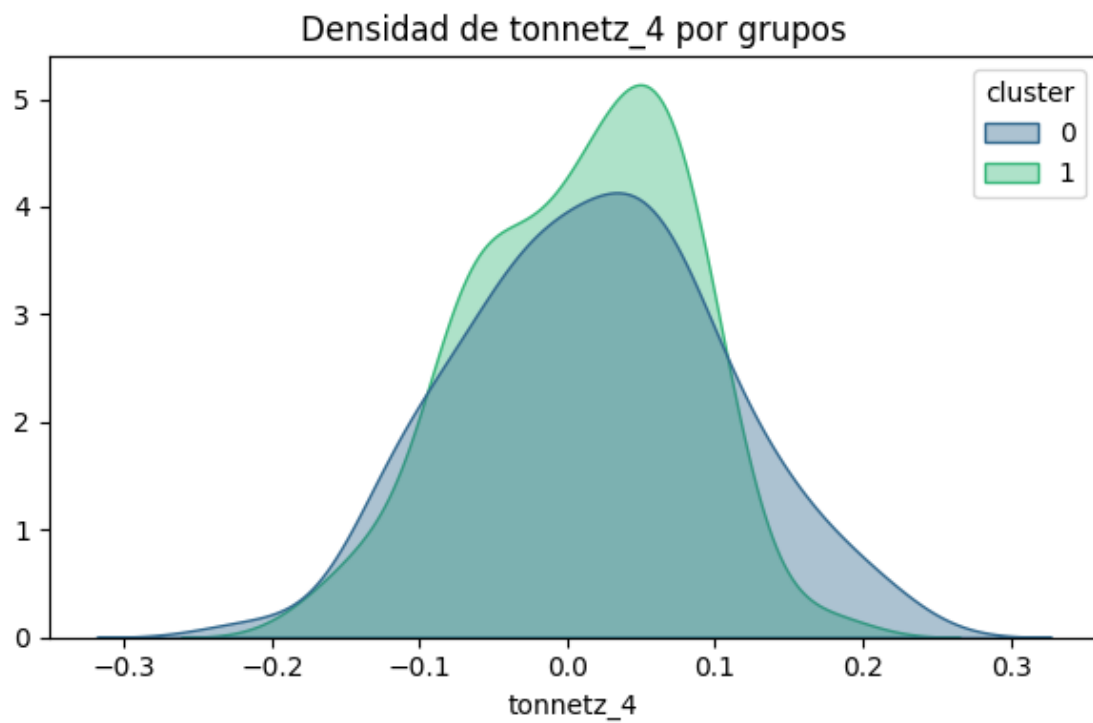
```

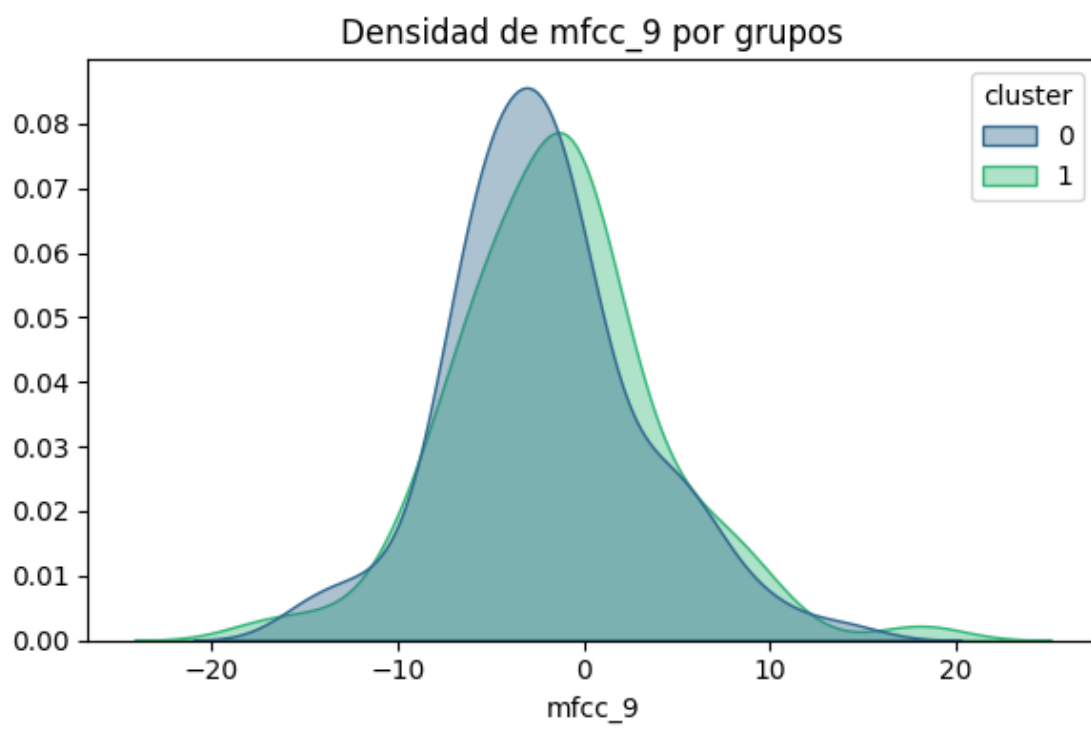
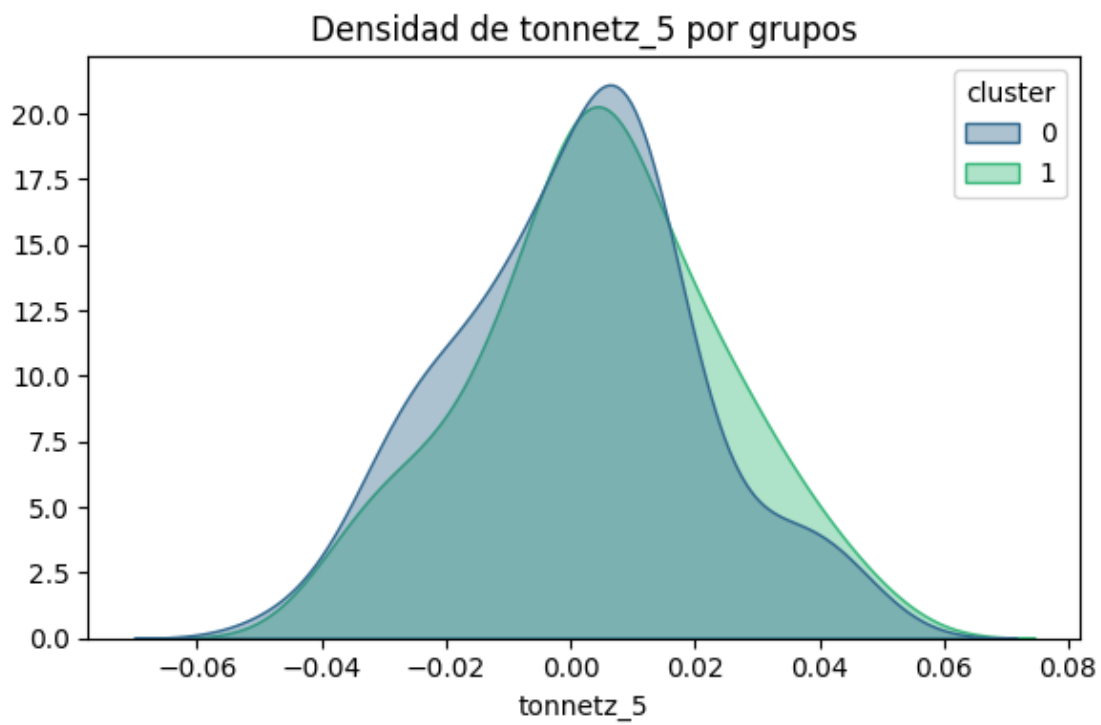
[51]: # Distribución de variables no significativas
      perfil.plot_dist(var_nosig)

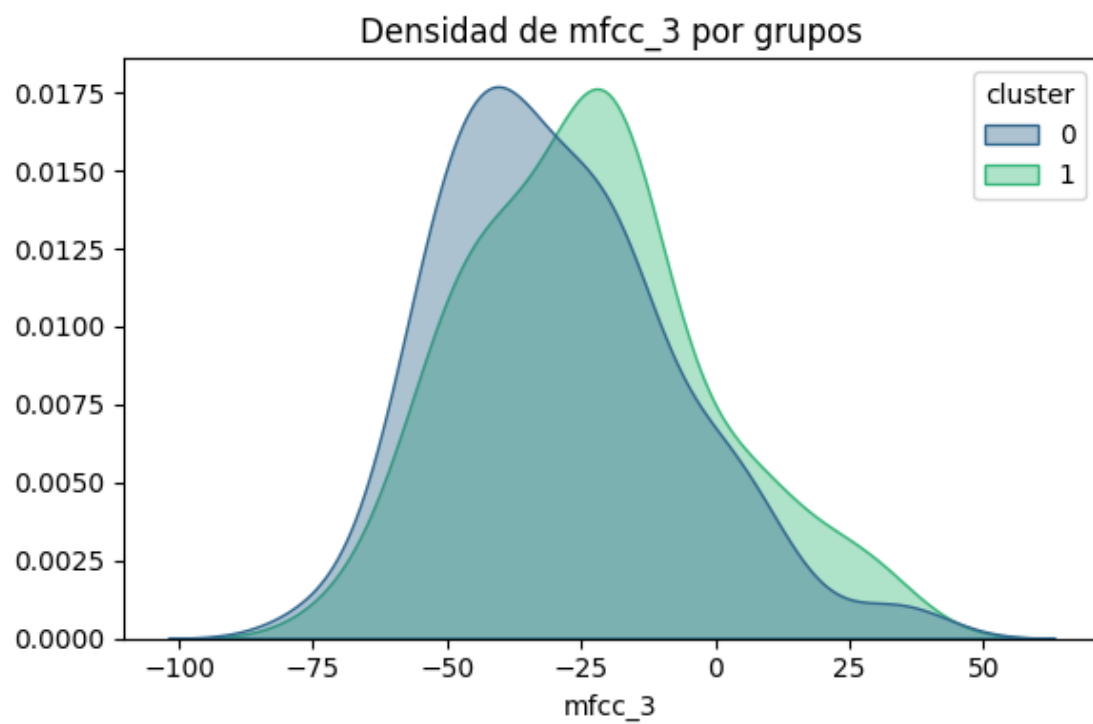
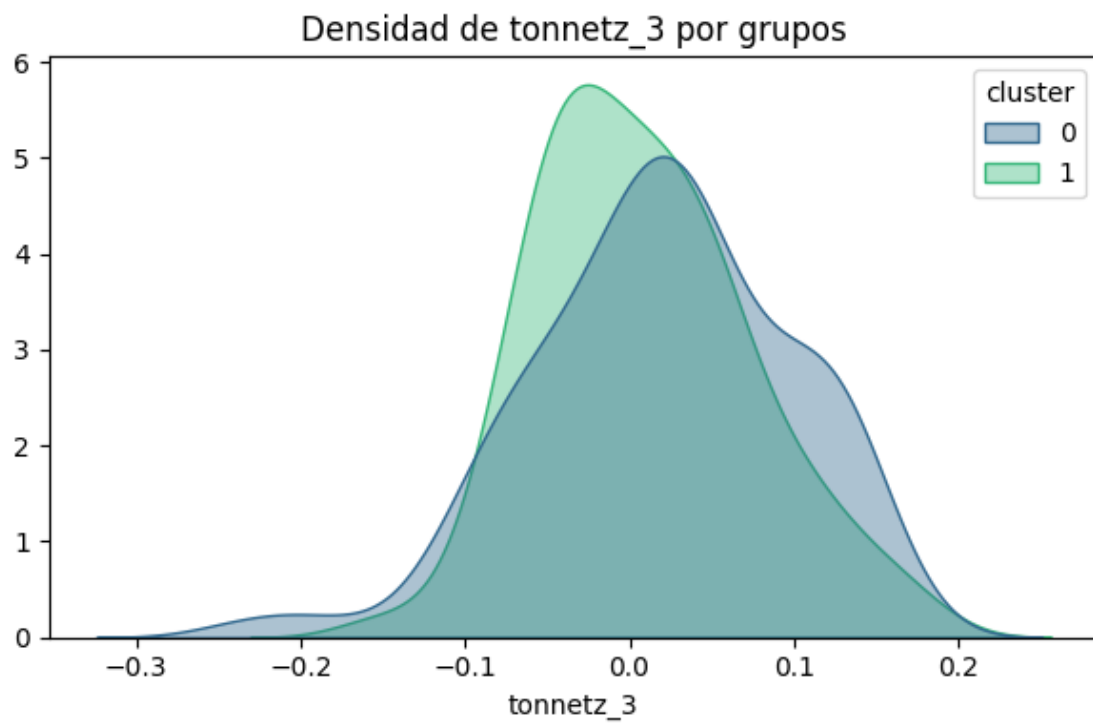
```

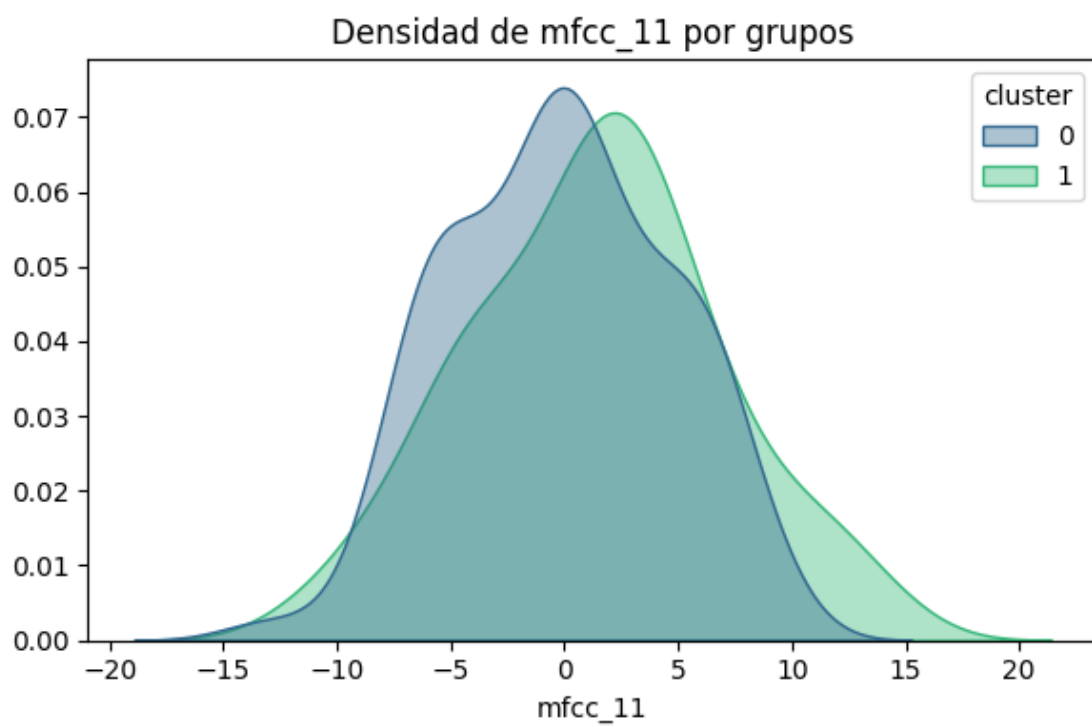
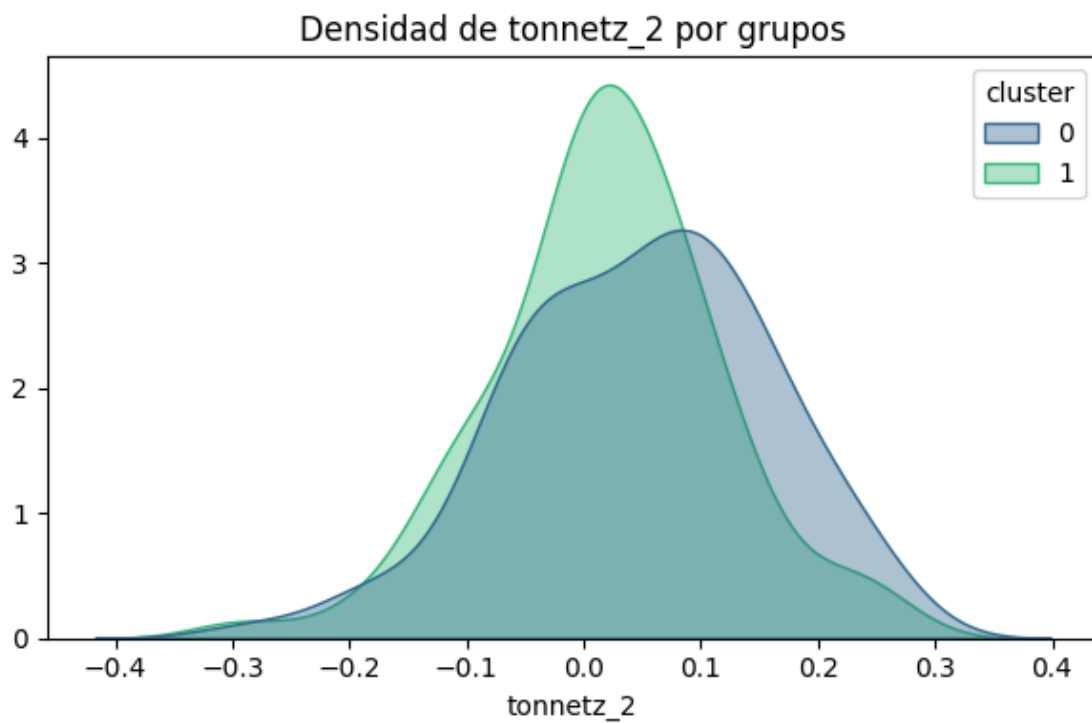


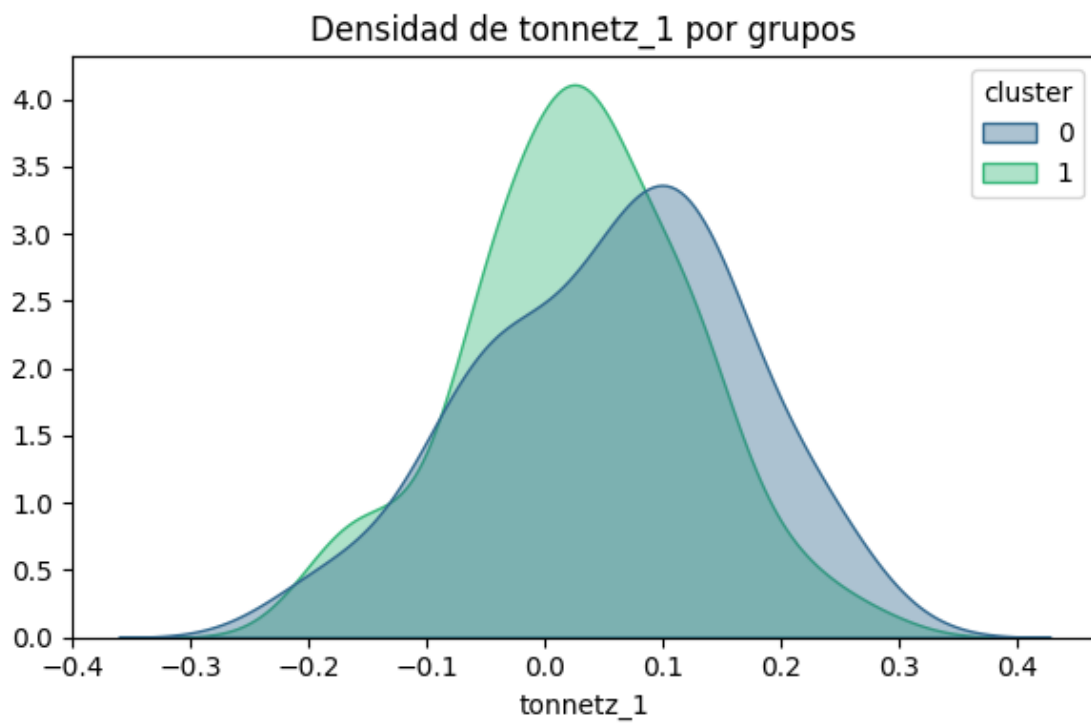




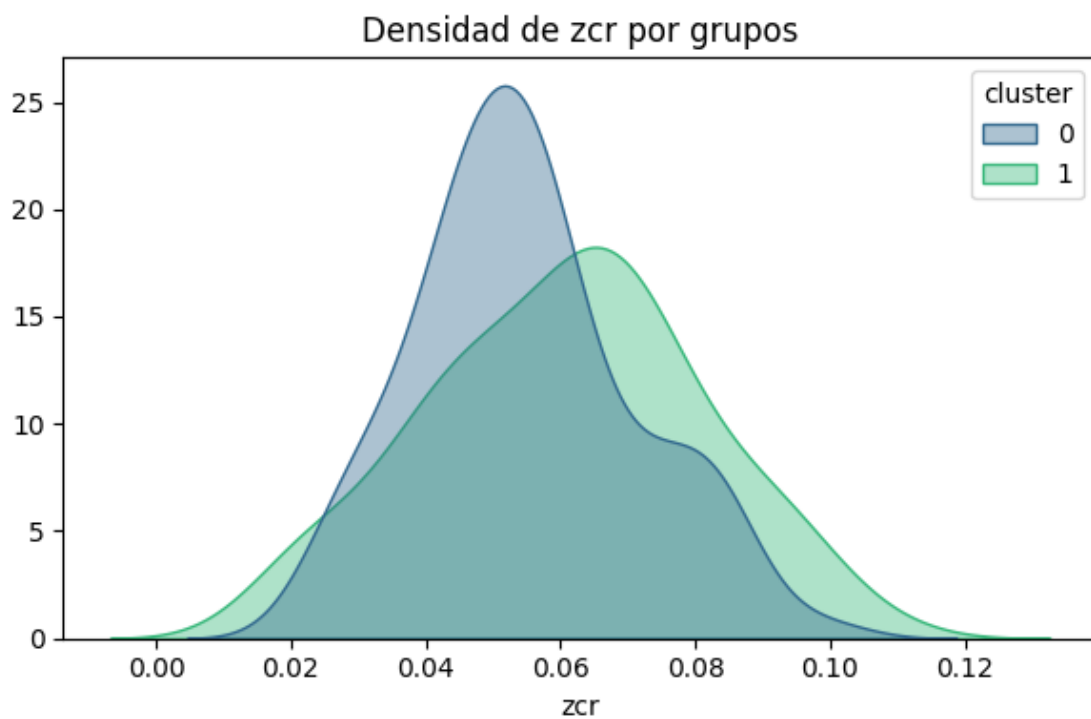


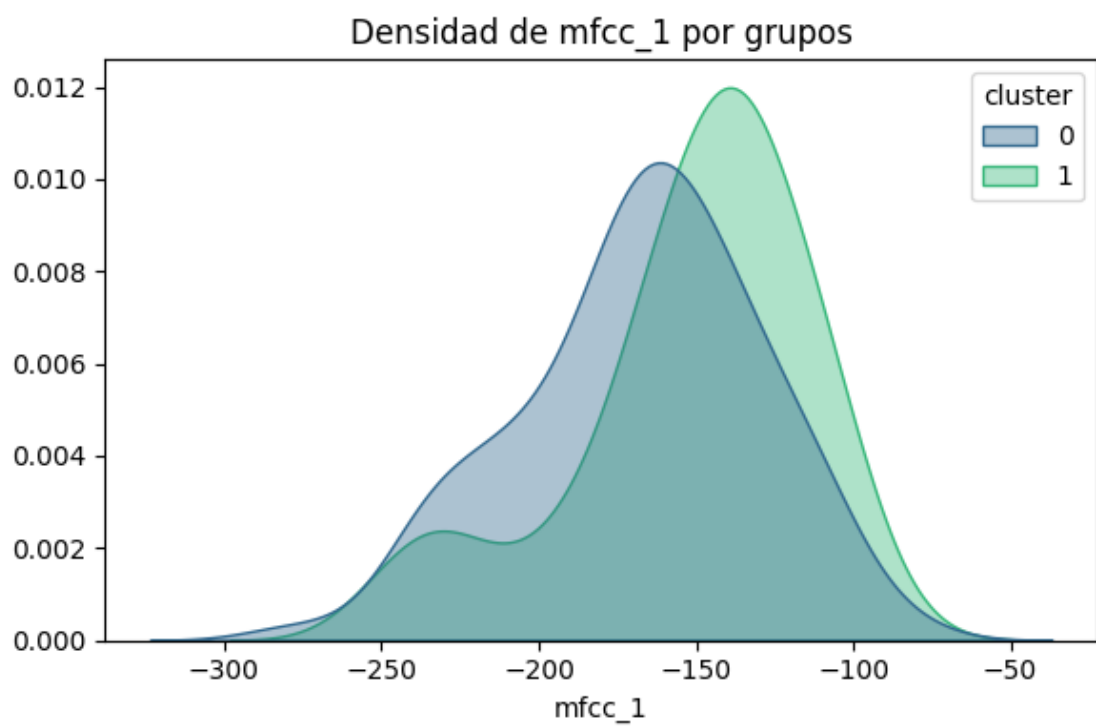
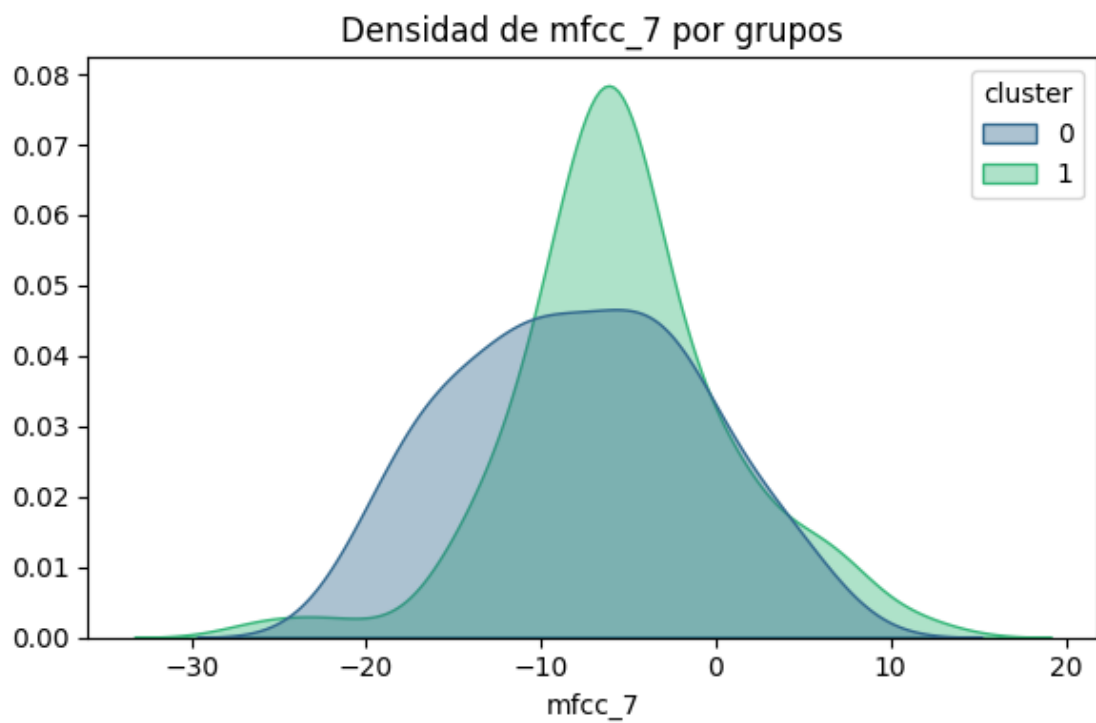


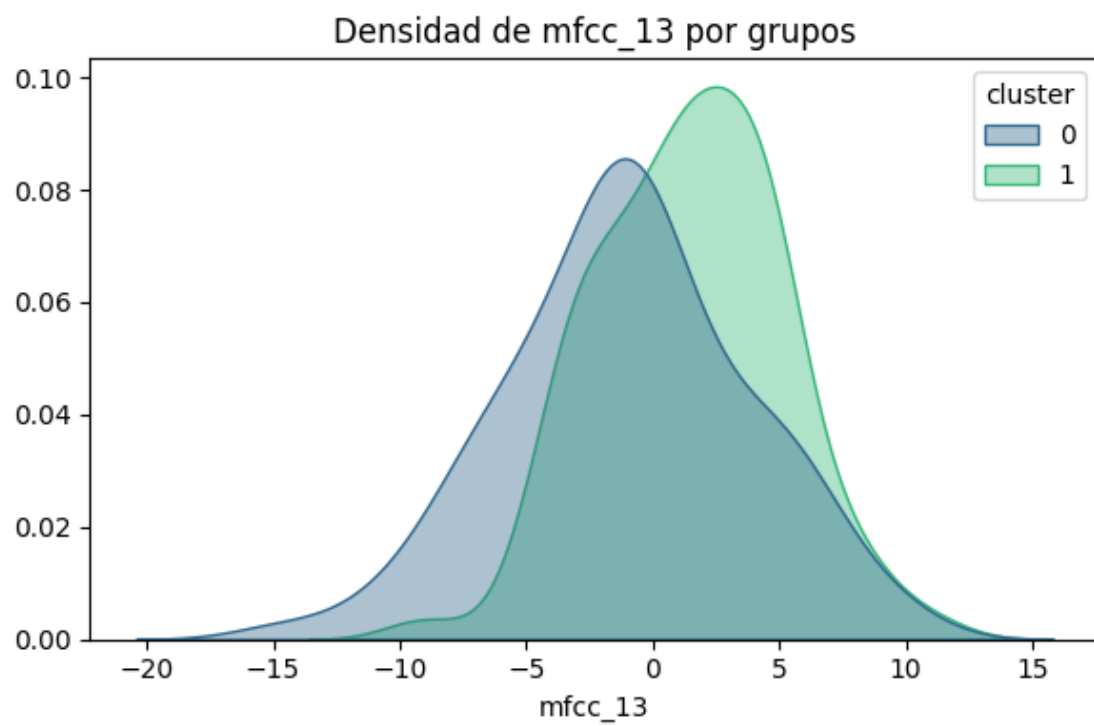
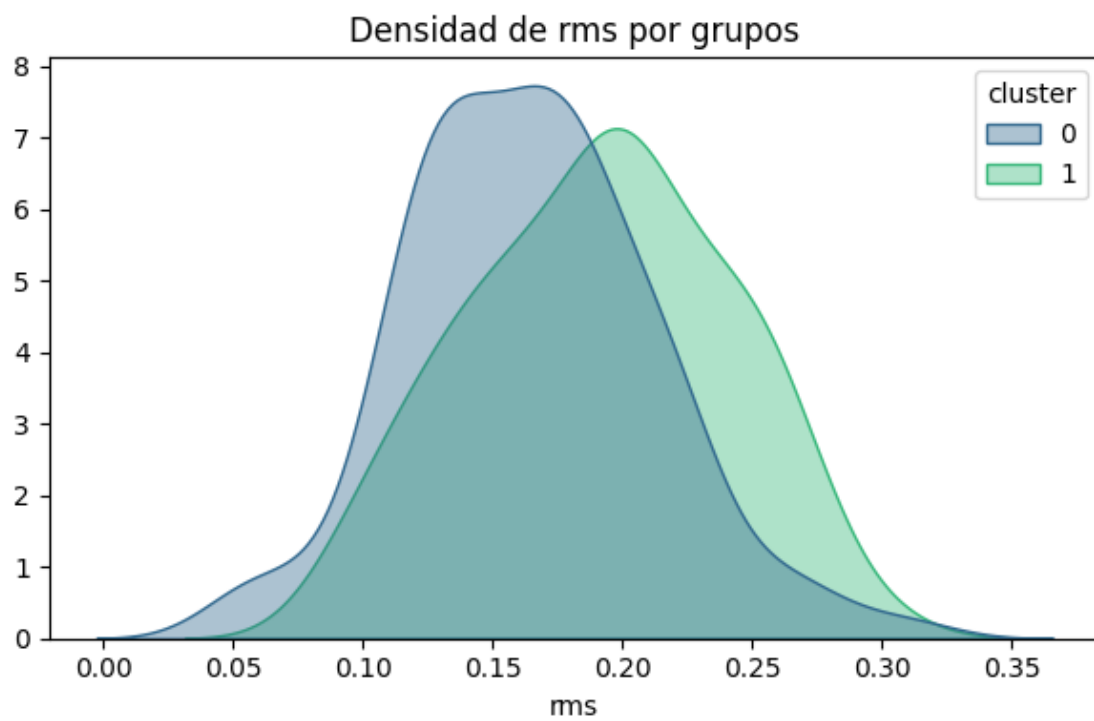


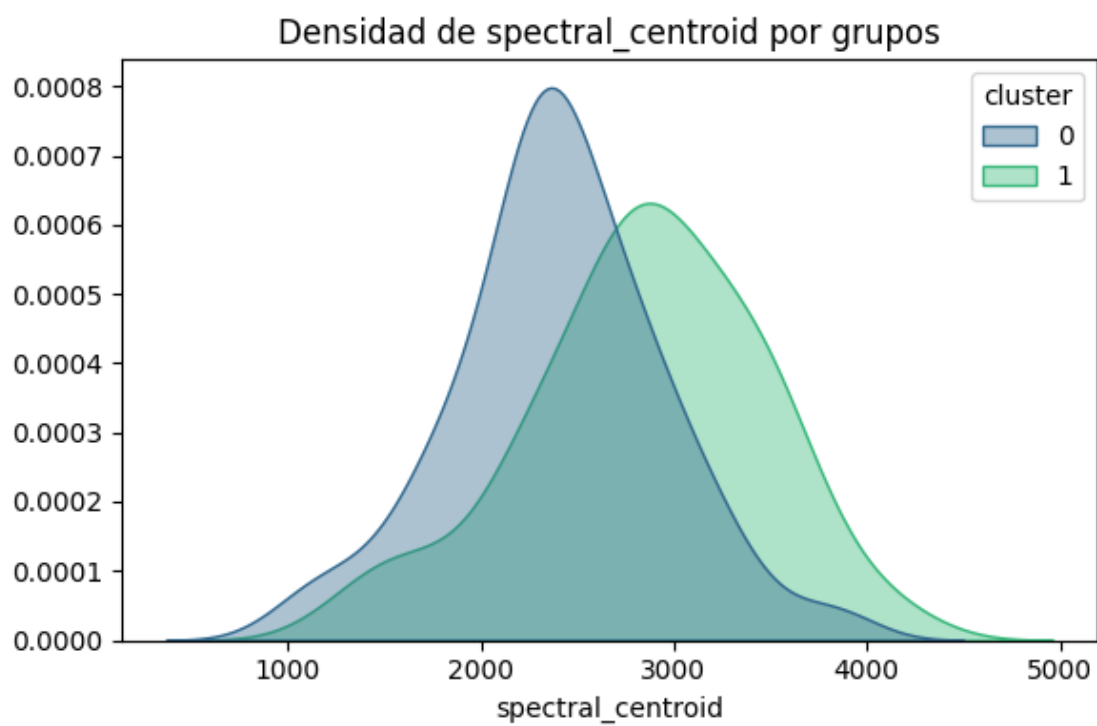
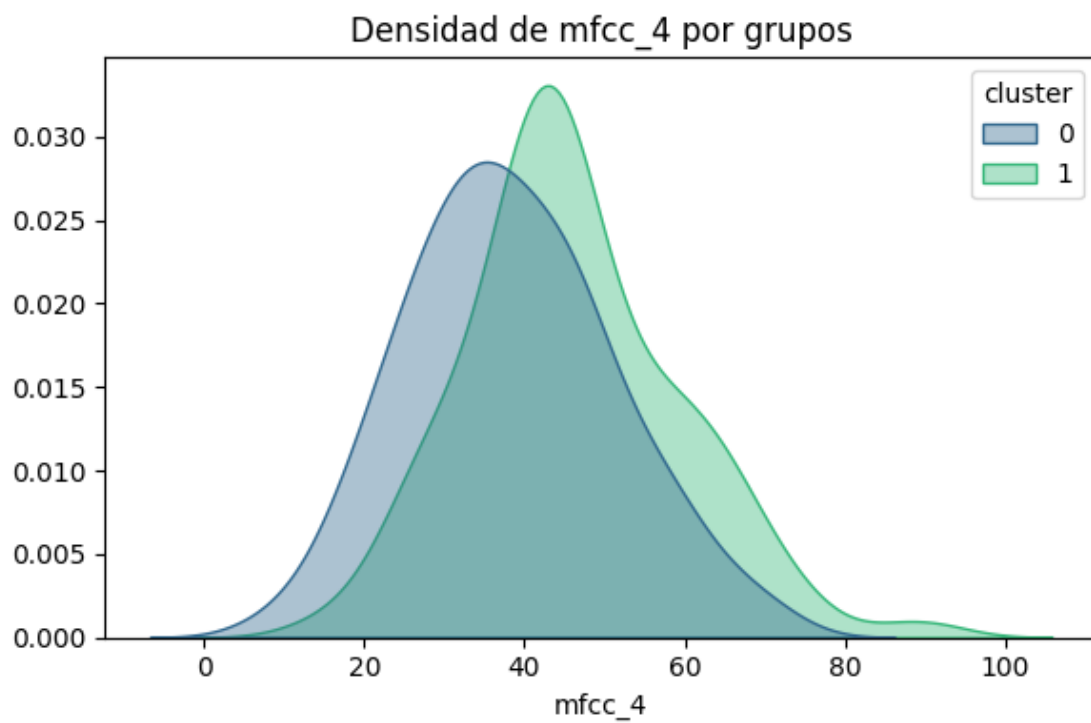


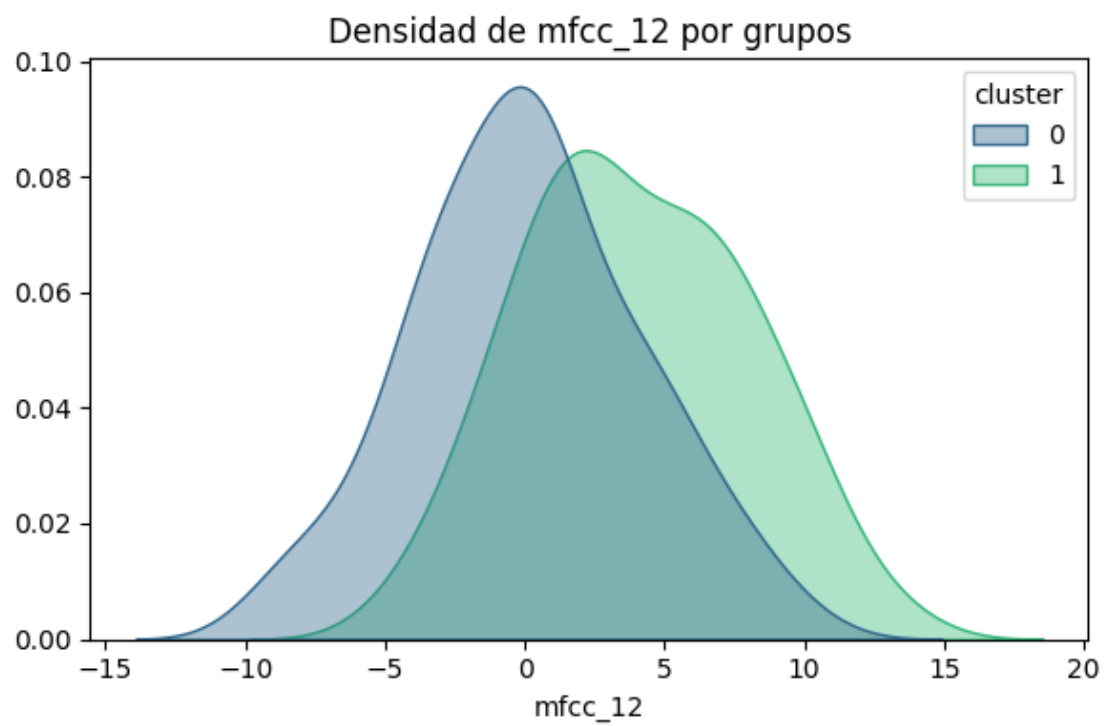
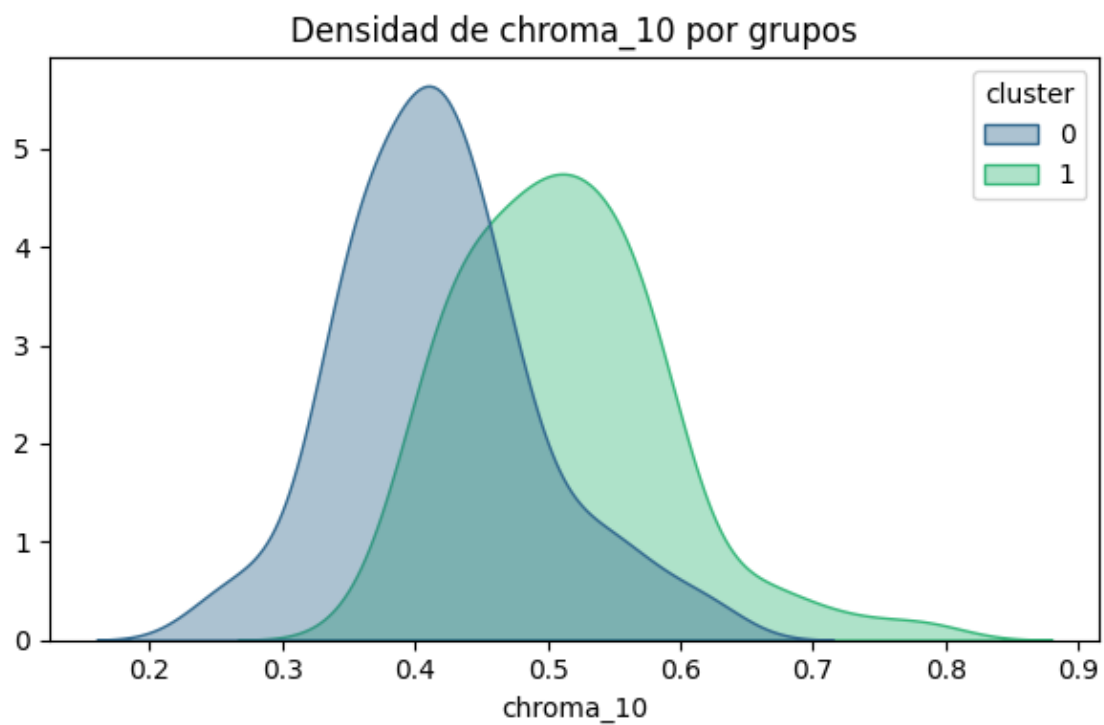
```
[52]: # Distribución de variables significativas  
perfil.plot_dist(var_sig)
```

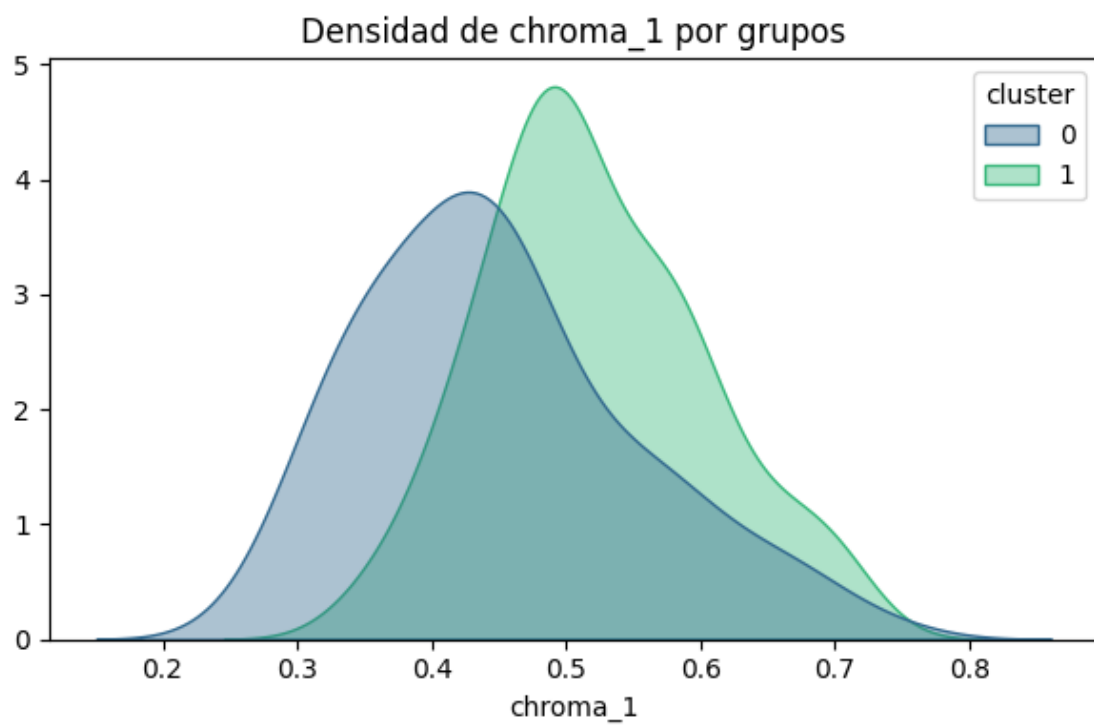
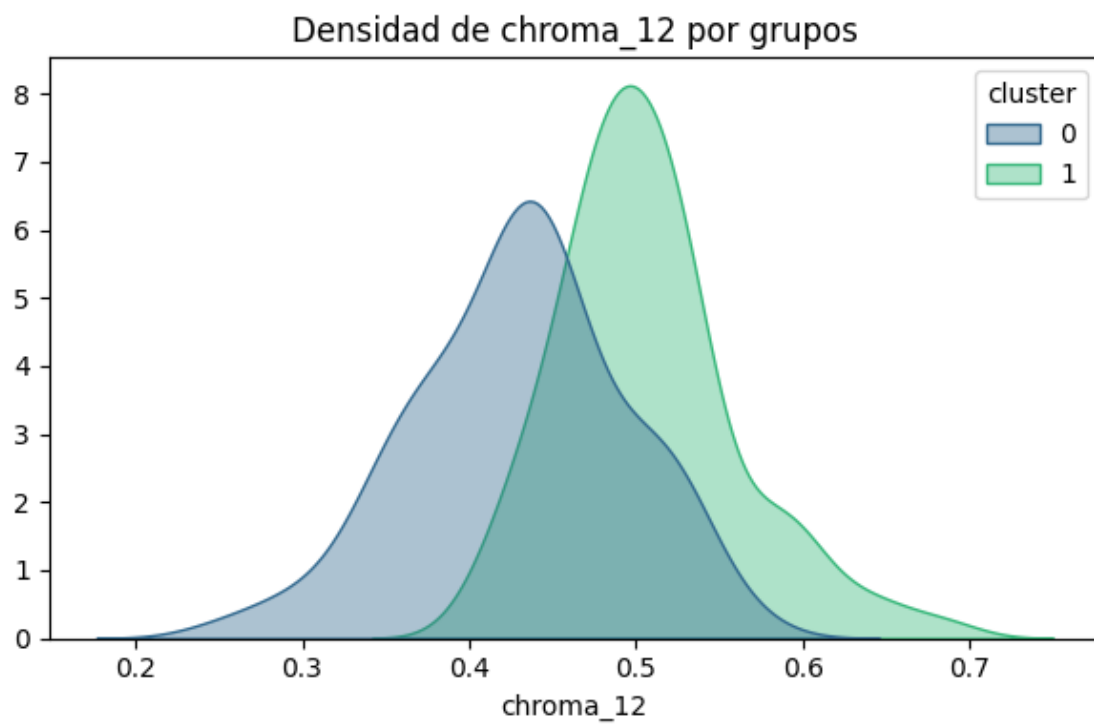


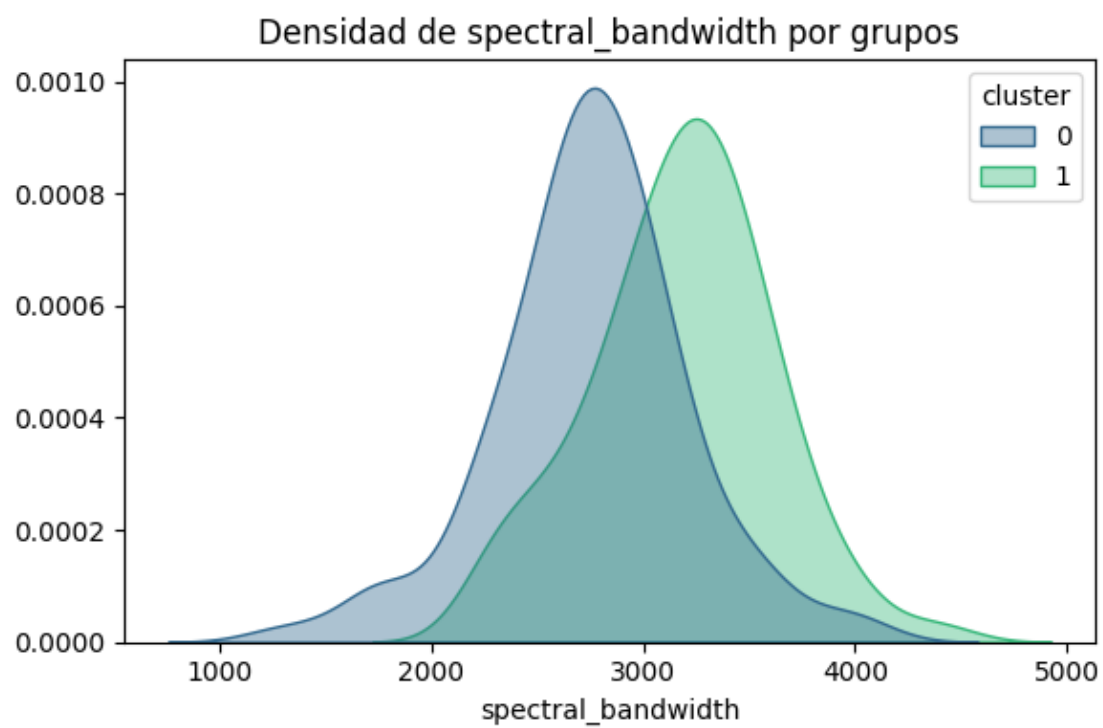
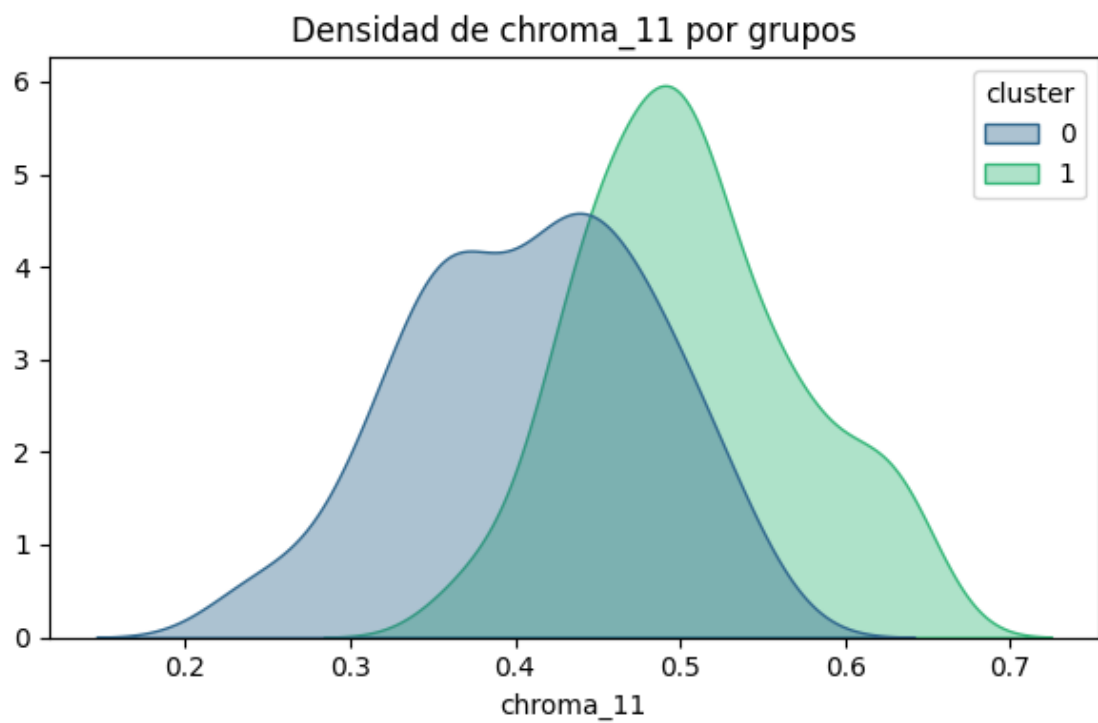


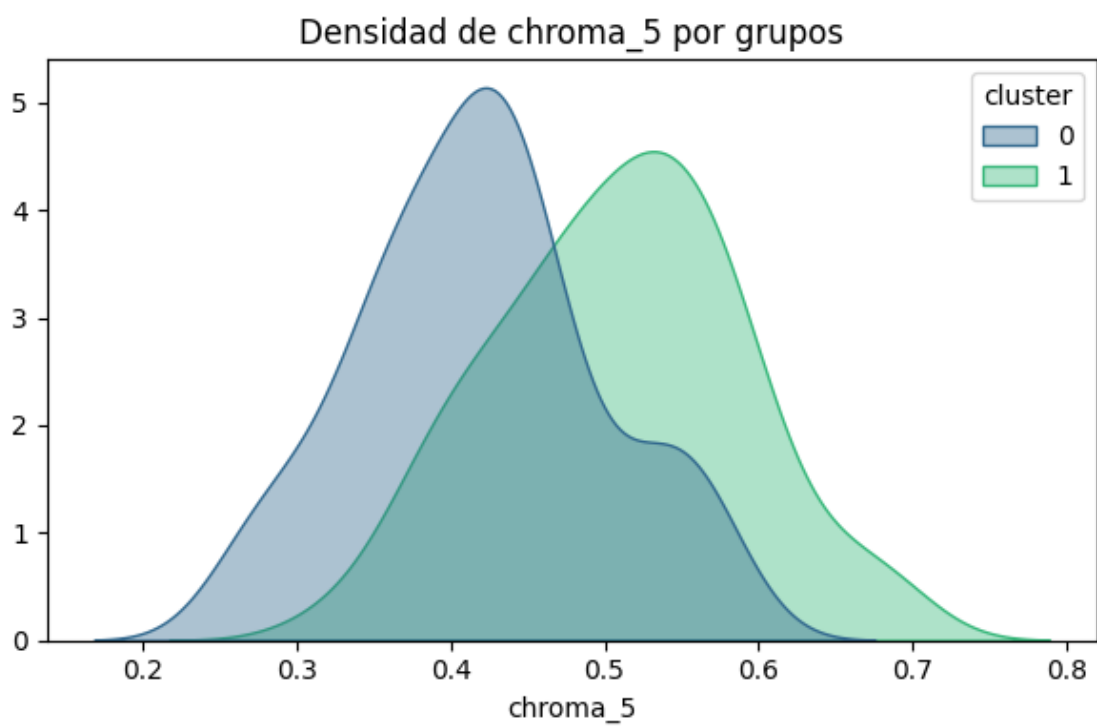
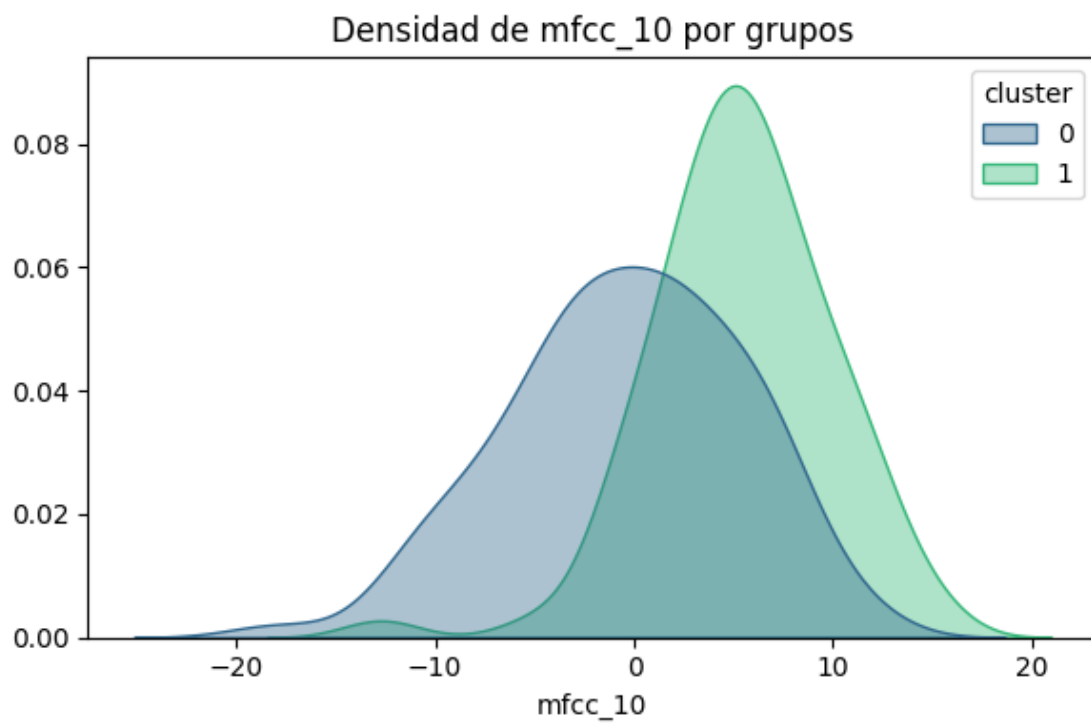


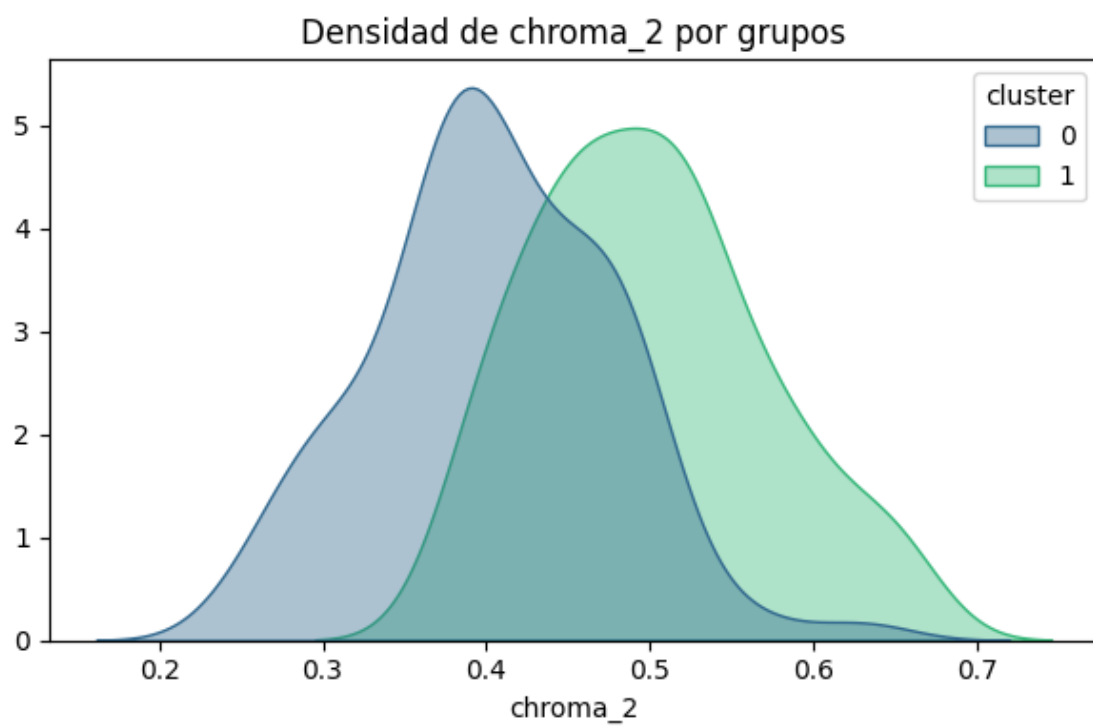
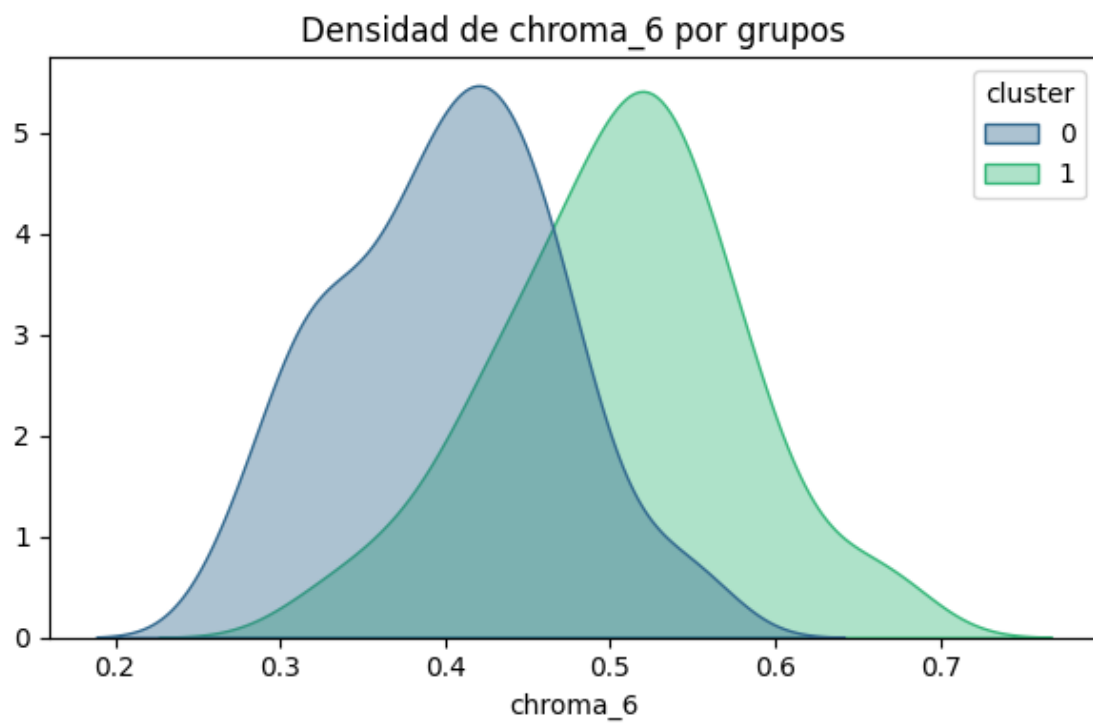


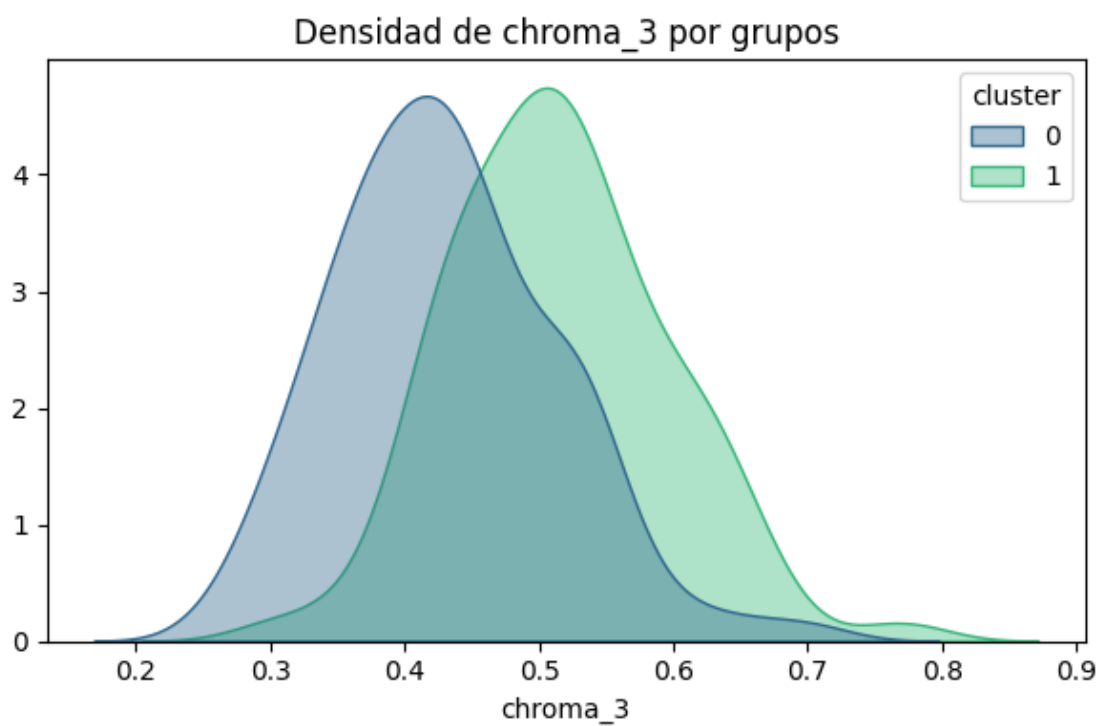
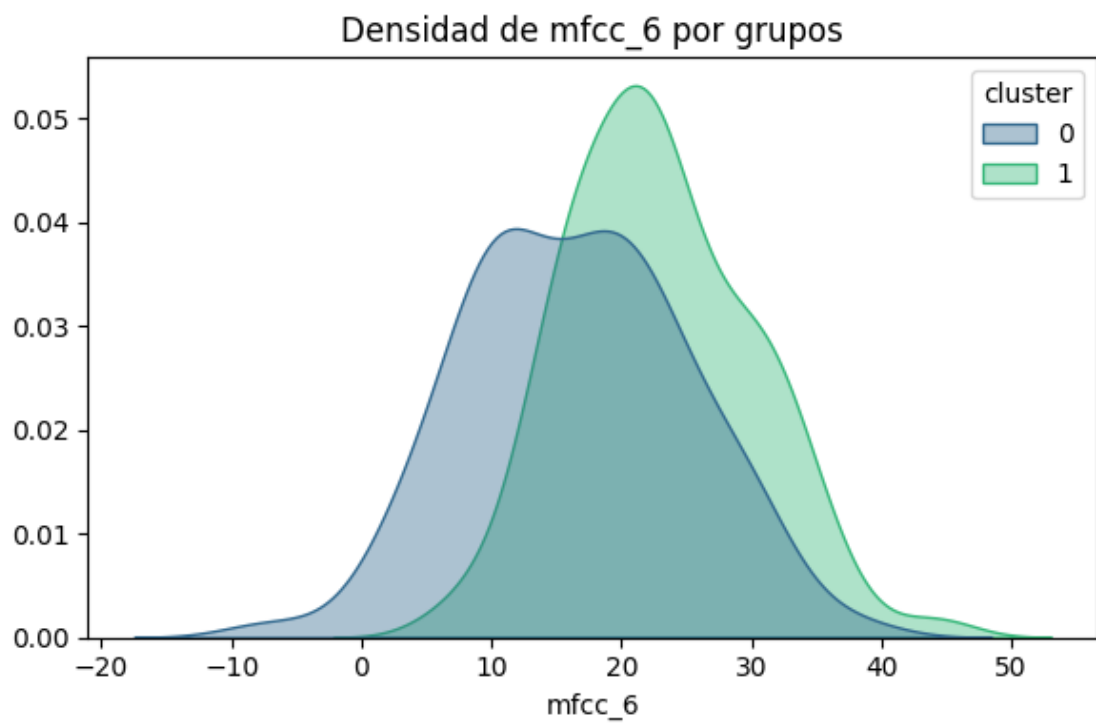


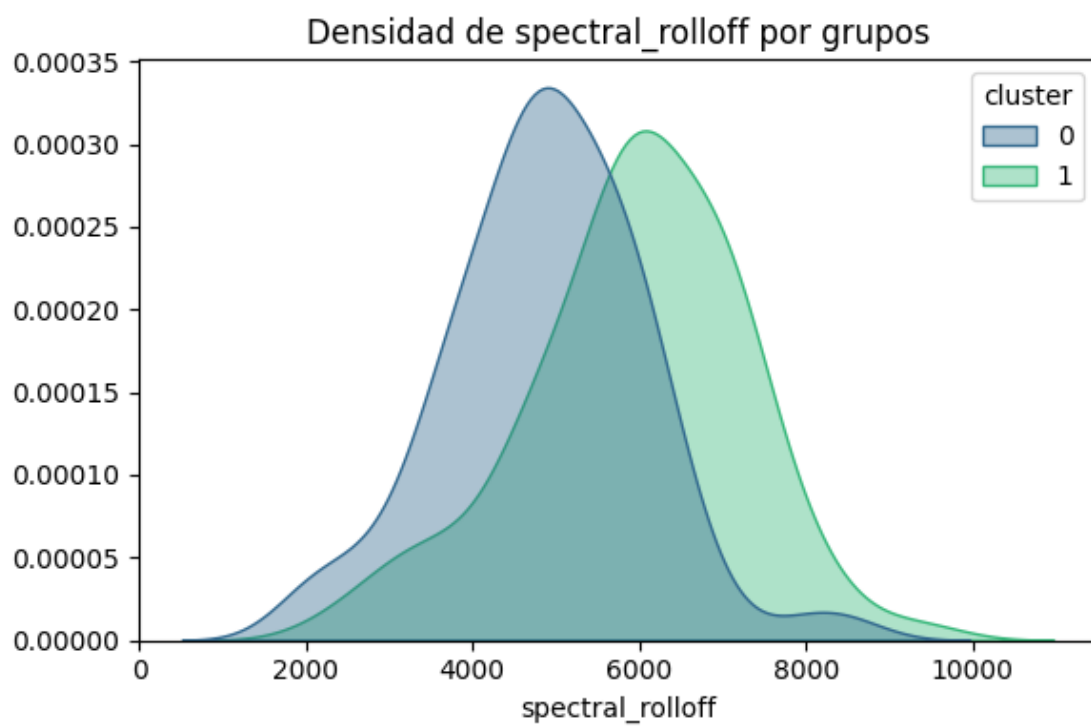
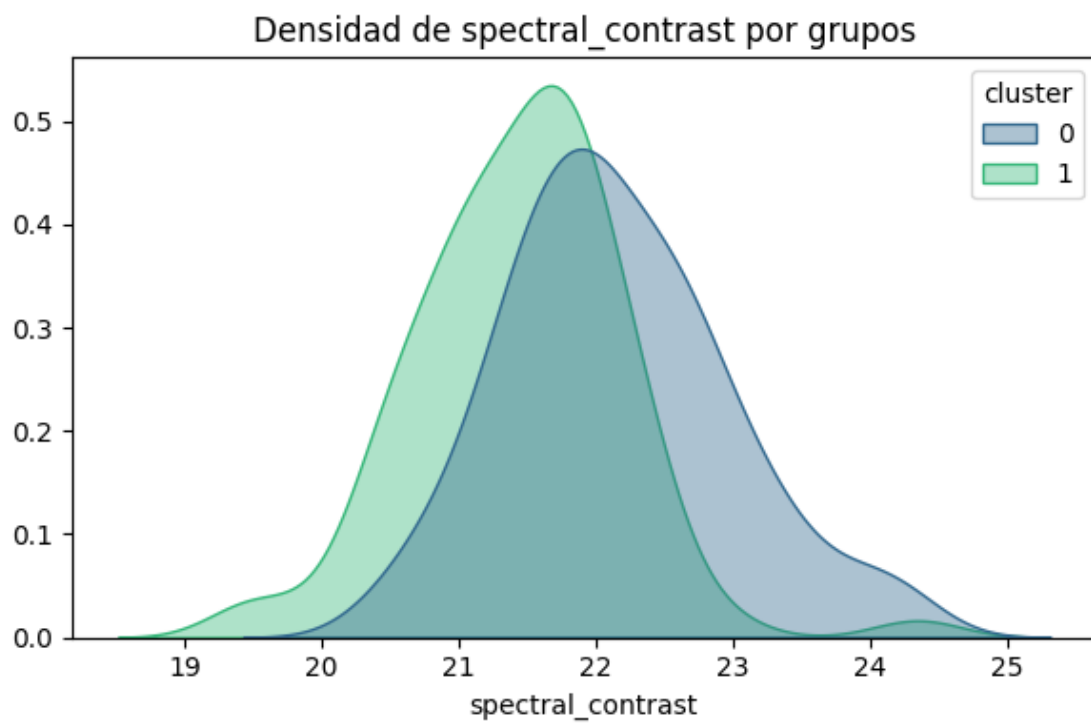


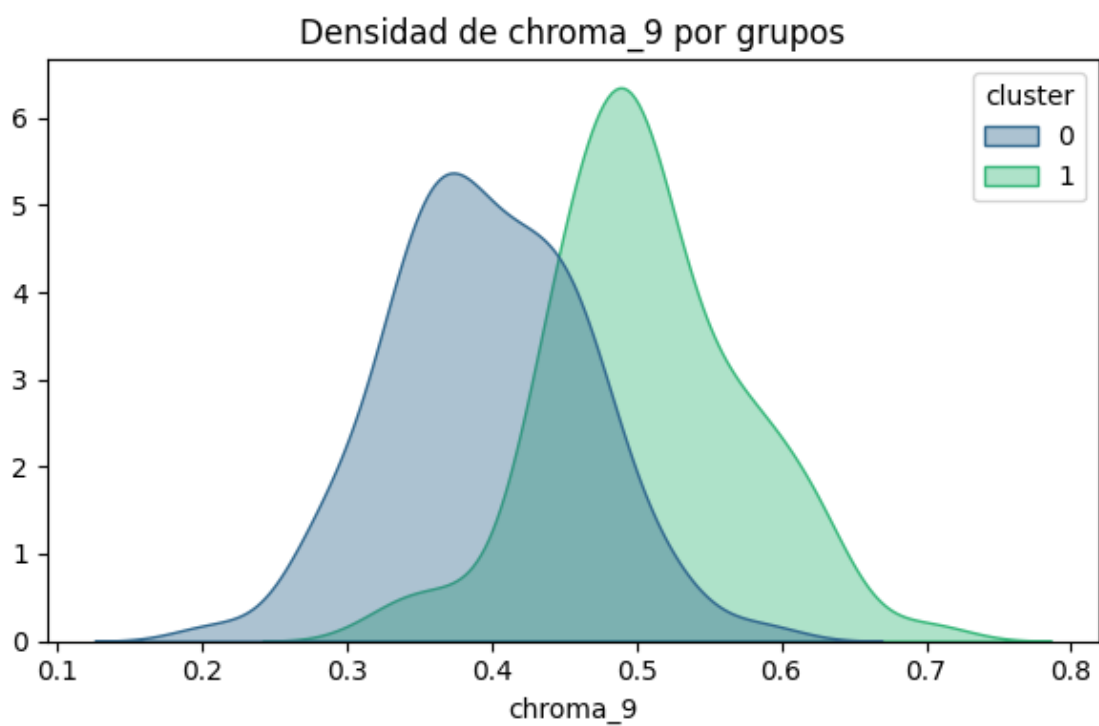
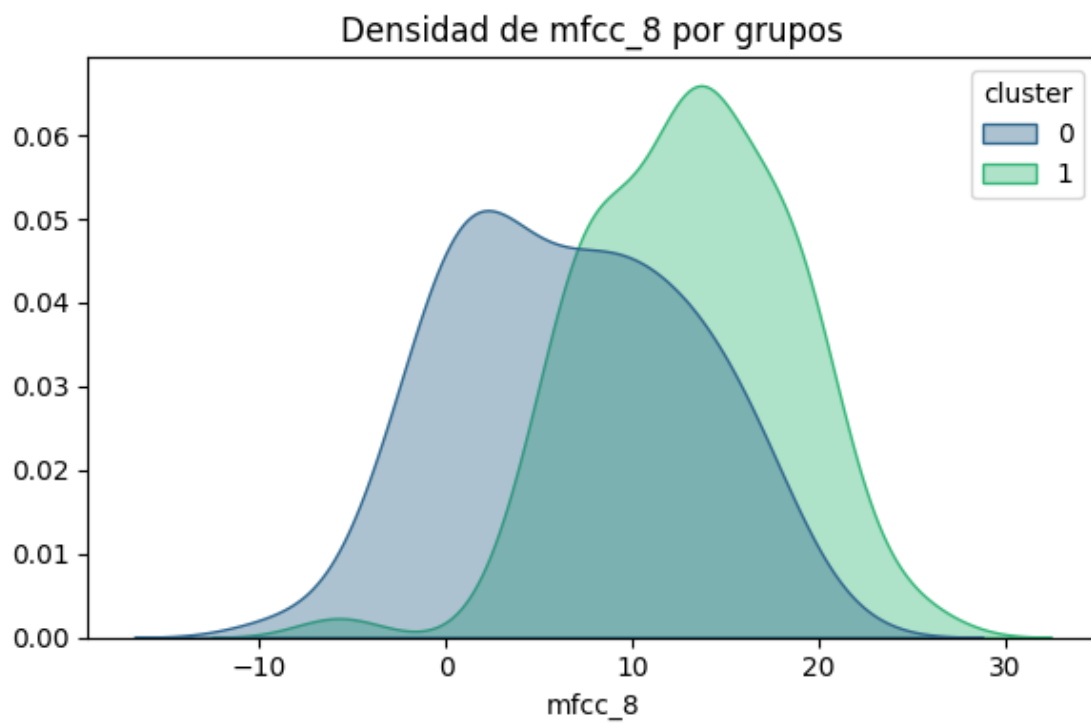


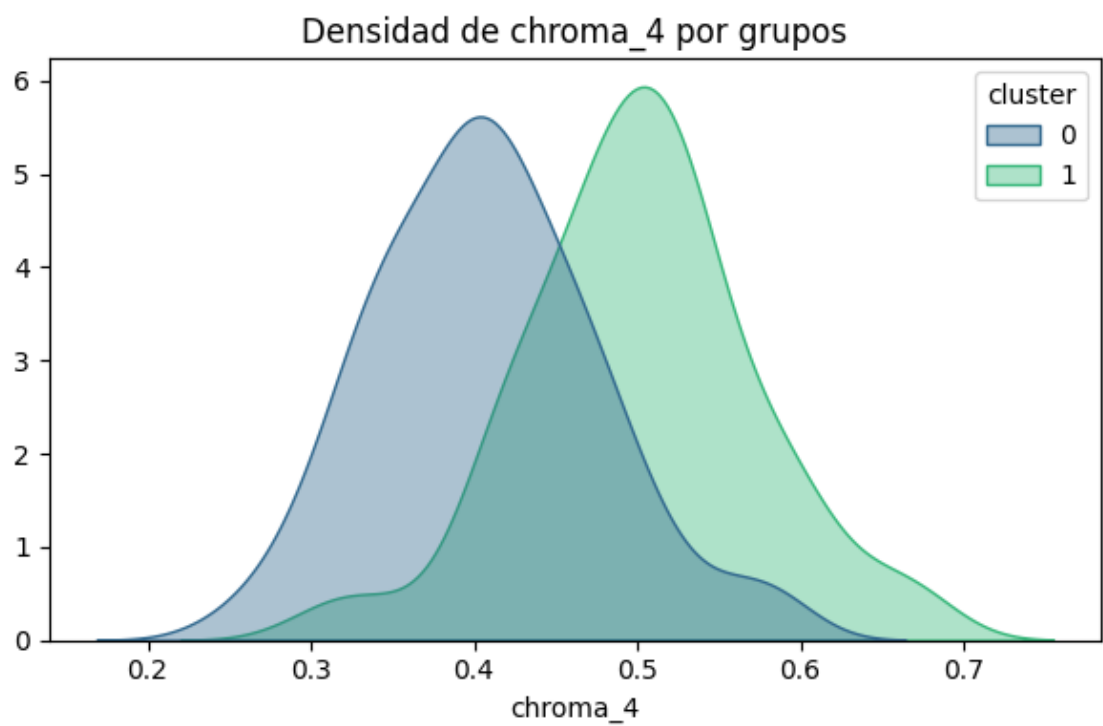
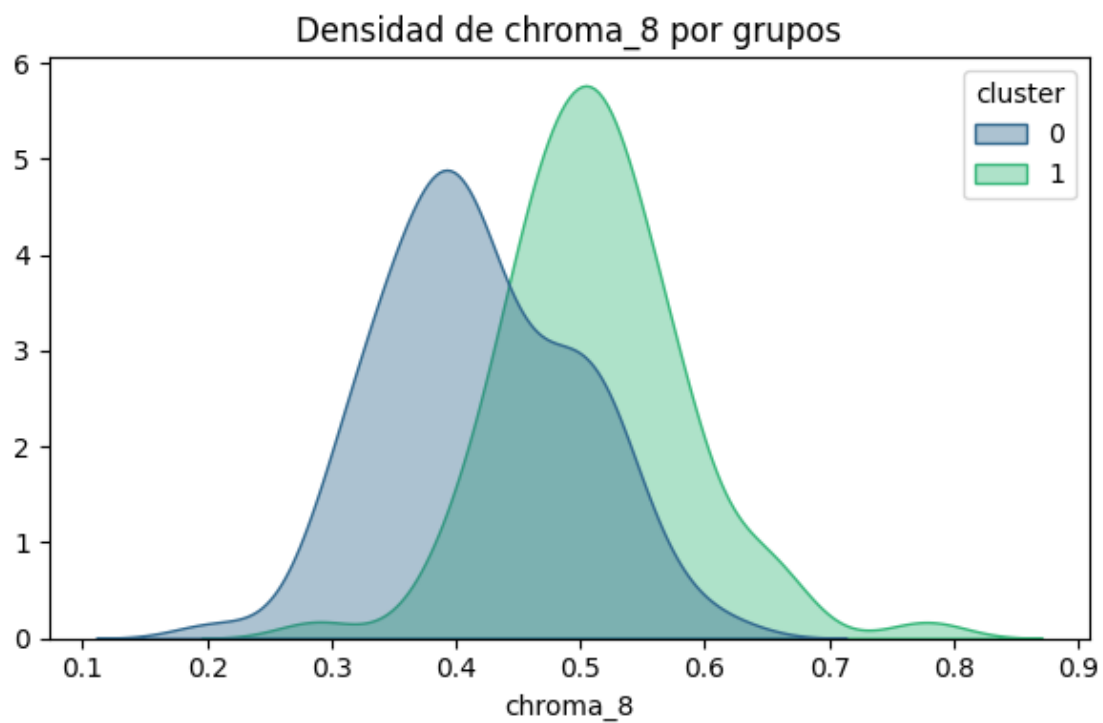


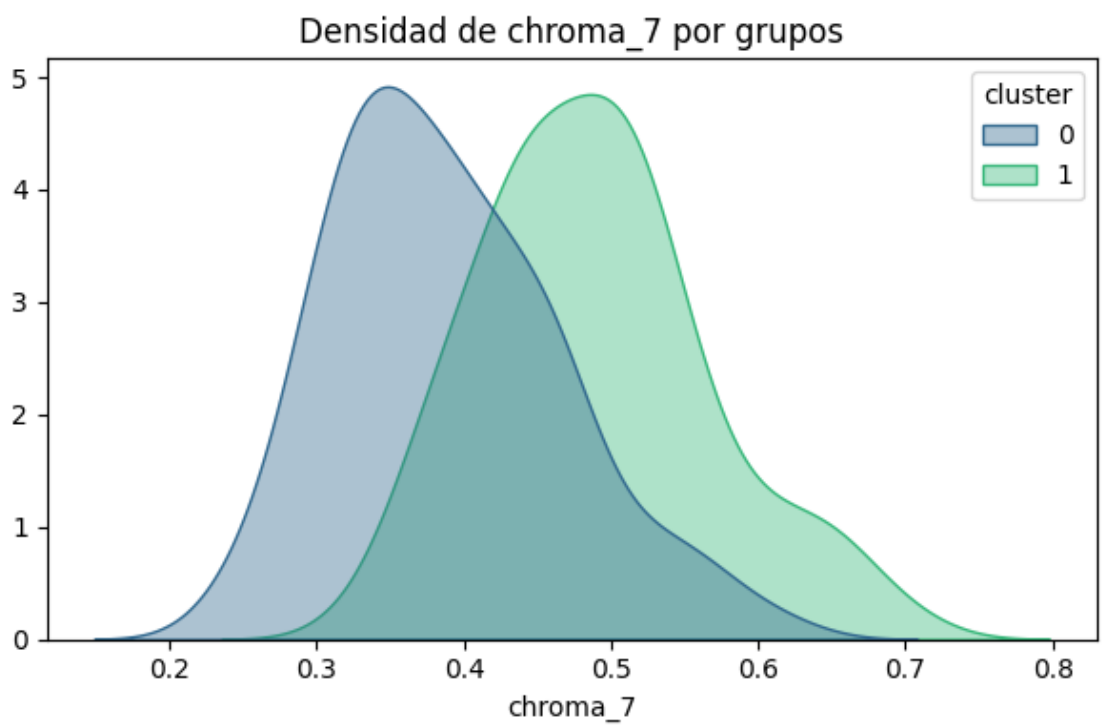
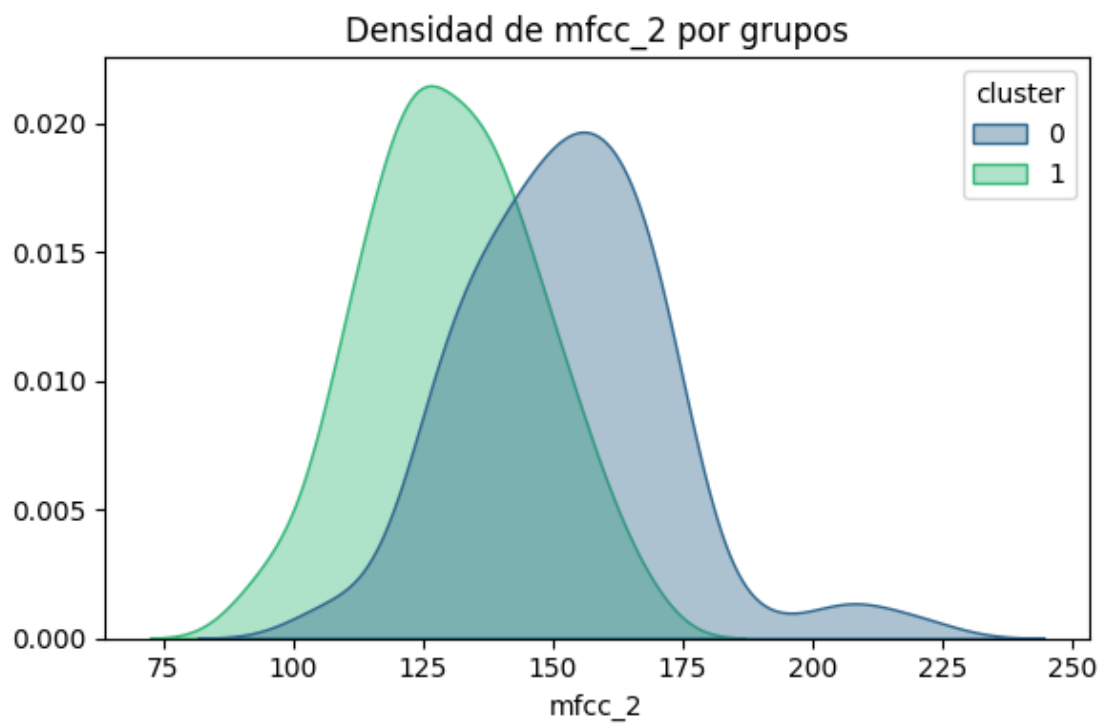












3.2.2. Creación de variables categóricas para interpretación de resultados

3.2.3. Tempo (BPM)

- Música popular: 60–180 BPM.
- Baladas: 60–90 BPM
- Pop/Rock: 100–140 BPM
- Electrónica/Dance: 120–150 BPM
- Trap/Drill/Reguetón: 70–100 BPM (a veces medido en doble tempo $\rightarrow$ 140–200).

3.2.4. Zero Crossing Rate (ZCR)

- Normalizado entre 0 y 1.
- Voces limpias o instrumentos suaves: 0.02 – 0.1.
- Sonidos ruidosos/percusión fuerte: 0.1 – 0.3.
- Señales muy ruidosas pueden superar 0.4.

3.2.5. Spectral Centroid

- Medido en Hz.
- Voces suaves: 1–2 kHz
- Música pop/rock promedio: 2–4 kHz
- Sonidos brillantes (platillos, electrónica): >5 kHz.
- Suele correlacionar con la “brillantez” del sonido.

```
[53]: def inferir_genero(row):  
    t = row["tempo_num"]  
    zcr = row["zcr"]  
    sc = row["spectral_centroid"]  
  
    # Balada  
    if t <= 90 and zcr <= 0.1 and sc <= 2000:  
        return "Balada / Música lenta"  
  
    # Trap / Drill / Reguetón  
    if (70 <= t <= 100 or 140 <= t <= 200) and 0.1 <= zcr <= 0.25 and sc <= 3000:  
        return "Trap / Drill / Reguetón"  
  
    # Pop / Rock  
    if 100 < t <= 140 and 0.1 < zcr <= 0.3 and 2000 < sc <= 4000:  
        return "Pop / Rock"  
  
    # Electrónica / Dance  
    if 120 <= t <= 150 and zcr >= 0.2 and sc > 4000:  
        return "Electrónica / Dance"
```

```
# Genérico
return "Popular / Variada"
```

```
[54]: # Aplicar al DataFrame
songs1["genero_inferido"] = songs1.apply(inferir_genero, axis=1)
```

```
[55]: # Categorías para la variable tempo

bins_tempo = [0, 90, 120, 150, 1000] # cortes
labels_tempo = ["Lento (<90)", "Medio (90-120)", "Rápido (120-150)", "Muy rápido,
↳(>150)"]

songs1["tempo_cat"] = pd.cut(songs1["tempo_num"], bins=bins_tempo,
↳labels=labels_tempo, ordered=True)
```

```
[56]: # Categorías para variable Zero crossing rate

bins_zcr = [0, 0.1, 0.3, 1.0]
labels_zcr = ["Bajo (0.1)", "Medio (0.1-0.3)", "Alto (>0.3)"]

songs1["zcr_cat"] = pd.cut(songs1["zcr"], bins=bins_zcr, labels=labels_zcr,
↳ordered=True)
```

```
[57]: # Categorías para variable centroid

bins_centroid = [0, 2000, 4000, 10000]
labels_centroid = ["Oscuro (2kHz)", "Equilibrado (2-4kHz)", "Brillante (>4kHz)"]

songs1["centroid_cat"] = pd.cut(songs1["spectral_centroid"], bins=bins_centroid,
↳labels=labels_centroid, ordered=True)
```

```
[58]: from pandas.api.types import CategoricalDtype

class playlist:
    def __init__(self, df: pd.DataFrame, selection: list, sig_var: list,
↳other_var: list, group_var: list | None = None):
        """
        Genera un resumen estadístico agrupado por una o varias columnas.

        Parámetros:
        -----
        df : pd.DataFrame
            Dataset con las variables.
        other_var : list
            Columna(s) para agregar en la salida agrupada.
        group_var : str o list
            Columna(s) para agrupar (ej. 'cluster' o ['cluster', 'lang']).
```

```

"""

self.df = df
self.selection = selection
self.sig_var = sig_var
self.other_var = other_var
self.group_var = ['cluster'] if group_var is None else group_var

def agrupado(self, vars_num: list | None = None, agg: str = "mean", observed:
→ bool = True):
    """
    Genera un resumen estadístico agrupado por una o varias columnas.

    Parámetros:
    -----
    vars_num : list
        Lista de variables numéricas a resumir (si None, usa todas las
→ numéricas).
    agg : str o list
        Agregación a aplicar ('mean', 'median', 'std', 'mode' o lista de
→ funciones).

    Retorna:
    -----
    pd.DataFrame con estadísticas agrupadas.
    """

    if vars_num is None:
        vars_num = list(set(self.df.select_dtypes(include="number").columns.
→ tolist())-set(self.group_var))

    # Si es moda, tratamos distinto porque mode() devuelve varias filas
    if agg == "mode":
        result = self.df.groupby(self.group_var,
→ observed=observed)[vars_num].apply(lambda x: x.mode().iloc[0])
    else:
        result = self.df.groupby(self.group_var,
→ observed=observed)[vars_num].agg(agg)

    #return result.reset_index()
    i = 0
    x = self.df.columns[i]

    while x in self.group_var:
        i += 1
        x = self.df.columns[i]

```

```

        return pd.concat([self.df.groupby(self.group_var, observed=observed).
→count()[x], result], axis=1) \
                        .rename(columns={x: 'count'}).reset_index()

        #return pd.concat([self.df.groupby(self.group_var, observed=observed).
→count()[], result], axis=1) \
                        #.reset_index()

    def resumen(self, agg="mean", observed=True):
        pcolumns = self.other_var + list(set(self.sig_var)-set(self.
→other_var)-set(self.group_var))

        return self.agrupado(agg=agg, observed=observed)

    def select(self, i, return_count: bool = True):
        features = self.resumen().iloc[i, :].iloc[: len(self.group_var)].
→to_dict()

        # Crear la máscara booleana con "&"
        if self.group_var != ['cluster']:
            mask = (
                (self.df["cluster"] == features["cluster"]) &
                (self.df["lang"] == features["lang"]) &
                (self.df["genero_inferido"] == features["genero_inferido"]) &
                (self.df["tempo_cat"] == features["tempo_cat"])
            )
        else:
            mask = (
                self.df["cluster"] == features["cluster"]
            )

        df = self.df[mask].copy()

    def str_duration(unidad: str, x: float):
        match unidad:
            case 's':
                return f'{int(x/60)} min and {int(x%60)} s'
            case 'ms':
                return f'{int(x/1000/60)} min and {int((x/1000)%60)} s'

        if 'duration' in df.columns:
            df['duration_min'] = df['duration'].apply(lambda x:
→str_duration('s', x))

        if 'spectral_centroid' in df.columns:
            df['spectral_centroid_kHz'] = df['spectral_centroid']/1000

```

```

        if 'spectral_rolloff' in df.columns:
            df['spectral_rolloff_kHz'] = df['spectral_rolloff']/1000

        if 'duration_ms' in df.columns:
            df['duration_min'] = df['duration_ms'].apply(lambda x:↳
↳str_duration('ms', x))

        return df.loc[:, self.selection]

```

```

[59]: clase = playlist(
        df=songs1,
        selection=['archivo', 'lang', 'genero_inferido', 'tempo_cat',↳
↳'duration_min', 'zcr', 'spectral_centroid', 'mfcc_7', 'mfcc_1', 'rms',↳
↳'mfcc_13', 'spectral_centroid_kHz', 'spectral_rolloff_kHz'],
        sig_var=var_sig,
        other_var=['tempo_num', 'spectral_centroid'],
        group_var=['cluster', 'lang', 'genero_inferido', 'tempo_cat']
    )

```

```

[60]: clase.agrupado(agg='mean')

```

```

[60]:
   cluster  lang      genero_inferido      tempo_cat  count  \
0         0 Español  Popular / Variada      Lento (90)      2
1         0 Español  Popular / Variada      Medio (90-120)    21
2         0 Español  Popular / Variada      Rápido (120-150)   3
3         0 Inglés  Balada / Música lenta      Lento (90)      1
4         0 Inglés  Popular / Variada      Lento (90)      8
5         0 Inglés  Popular / Variada      Medio (90-120)    29
6         0 Inglés  Popular / Variada      Rápido (120-150)   14
7         0 Inglés  Popular / Variada  Muy rápido (>150)      7
8         1 Español  Popular / Variada      Lento (90)      3
9         1 Español  Popular / Variada      Medio (90-120)    7
10        1 Español  Popular / Variada  Muy rápido (>150)      1
11        1 Inglés  Balada / Música lenta      Lento (90)      1
12        1 Inglés          Pop / Rock      Medio (90-120)    1
13        1 Inglés  Popular / Variada      Lento (90)      4
14        1 Inglés  Popular / Variada      Medio (90-120)    41
15        1 Inglés  Popular / Variada      Rápido (120-150)   16
16        1 Inglés  Popular / Variada  Muy rápido (>150)      3

   duration      rms      mfcc_4  tempo_num  chroma_12  ...  chroma_10  \
0  191.884002  0.135923  48.199354   88.347800   0.436333  ...   0.424563
1  252.504737  0.173147  43.689703  102.984730   0.418787  ...   0.421501
2  269.740423  0.161002  37.388587  125.248071   0.459106  ...   0.400266
3  184.041361  0.224655  26.937143   86.132812   0.524760  ...   0.469983

```

4	243.139663	0.148802	39.981222	83.643979	0.433606	...	0.423544
5	242.418344	0.158530	35.904603	106.537676	0.433243	...	0.426969
6	247.453567	0.168425	36.517796	131.501717	0.434878	...	0.403819
7	263.444536	0.172428	31.092083	172.149318	0.403283	...	0.382129
8	218.567203	0.213299	52.145625	86.181503	0.489627	...	0.524737
9	219.629397	0.161042	46.714186	99.900158	0.464074	...	0.459733
10	346.325624	0.195833	44.112064	178.205819	0.464466	...	0.427802
11	174.837551	0.226225	46.260887	69.837416	0.532159	...	0.510774
12	213.043084	0.202872	62.741670	103.359375	0.486284	...	0.487347
13	278.633832	0.193677	44.852663	80.108810	0.583022	...	0.579056
14	245.423473	0.193847	44.259870	104.192930	0.506157	...	0.516771
15	229.026023	0.185304	47.072489	129.278927	0.515611	...	0.501334
16	285.727627	0.214229	52.413543	165.436652	0.487423	...	0.519151

	zcr	spectral_centroid	tonnetz_4	chroma_5	chroma_11	\
0	0.069367	2744.209058	0.001885	0.294393	0.504910	
1	0.056751	2466.219131	0.042088	0.425515	0.403926	
2	0.055584	2175.617833	0.068696	0.445772	0.432470	
3	0.034992	1862.020089	0.053310	0.472580	0.520098	
4	0.057418	2520.394853	0.001604	0.376035	0.381875	
5	0.054139	2442.431392	-0.003242	0.422260	0.405923	
6	0.052285	2242.195715	0.031007	0.445410	0.408565	
7	0.049383	2267.383886	-0.001810	0.393233	0.416404	
8	0.060058	2836.643502	-0.009711	0.490979	0.517017	
9	0.065021	2976.125451	-0.020939	0.433873	0.496584	
10	0.039399	2657.051709	0.102739	0.440252	0.447284	
11	0.021928	1485.590595	0.086051	0.522615	0.485107	
12	0.106914	3720.442344	0.073463	0.579250	0.491300	
13	0.063025	3083.519071	0.031590	0.570401	0.585032	
14	0.061857	2840.310998	0.008704	0.504904	0.505890	
15	0.056827	2794.573232	-0.002225	0.533651	0.498352	
16	0.053674	2680.157737	0.022435	0.536965	0.478382	

	spectral_bandwidth	mfcc_5	mfcc_10	tonnetz_3
0	2927.815384	-5.262213	-2.987007	0.035136
1	2735.968027	-5.124884	1.365021	0.013578
2	2383.529186	-3.033284	1.769222	0.059218
3	2612.166207	-6.117527	6.262030	0.040727
4	2787.981519	-9.507221	-3.103913	0.050455
5	2878.859949	-7.944314	-0.057413	-0.002409
6	2582.045832	-1.494251	-0.552511	0.018393
7	2692.007068	-7.617173	-5.611111	0.069671
8	3175.858852	-4.132424	2.448277	0.039002
9	3261.073596	-6.705578	5.482506	0.042829
10	3281.270538	-14.242673	8.185412	0.108686
11	2261.139587	14.555250	-1.344500	-0.027103
12	3031.988547	-25.031971	-4.886665	-0.061647

13	3325.422671	3.910268	8.829753	-0.008696
14	3167.287907	-4.416376	5.253988	-0.004323
15	3206.888216	-8.783607	5.567748	0.015261
16	3142.565632	-0.208631	7.851516	0.049965

[17 rows x 45 columns]

3.2.6. Selección de grupo de canciones

```
[61]: print(clase.select(5).shape[0], "canciones del grupo de la fila 5 de la tabla_
      ↪ anterior, a continuación se muestra el top 5 ordenado por la métrica ZCR.\n")
      clase.select(5).sort_values(by=['zcr']).reset_index(drop=True).head() #_
      ↪ selección de una fila de la tabla anterior
```

29 canciones del grupo de la fila 5 de la tabla anterior, a continuación se muestra el top 5 ordenado por la métrica ZCR.

```
[61]:
```

	archivo	lang	\
0	8 my boy only breaks his favorite toys (first ...	Inglés	
1	04 snow on the beach	Inglés	
2	10 labyrinth	Inglés	
3	12 sweet nothing	Inglés	
4	20 snow on the beach (feat. more lana del rey)	Inglés	

	genero_inferido	tempo_cat	duration_min	zcr	\
0	Popular / Variada	Medio (90-120)	3 min and 11 s	0.026073	
1	Popular / Variada	Medio (90-120)	4 min and 16 s	0.026679	
2	Popular / Variada	Medio (90-120)	4 min and 8 s	0.028861	
3	Popular / Variada	Medio (90-120)	3 min and 8 s	0.028935	
4	Popular / Variada	Medio (90-120)	3 min and 50 s	0.032701	

	spectral_centroid	mfcc_7	mfcc_1	rms	mfcc_13	\
0	1206.772573	-2.028812	-207.29584	0.212049	-8.893036	
1	1691.923217	-5.396293	-205.33786	0.191276	0.654356	
2	1512.256993	-5.439651	-239.80269	0.116779	-0.800612	
3	1549.530534	-10.776519	-276.42240	0.134137	-2.024282	
4	1847.973731	-6.170126	-175.40880	0.194155	-1.098595	

	spectral_centroid_kHz	spectral_rolloff_kHz
0	1.206773	2.200230
1	1.691923	3.338831
2	1.512257	2.888892
3	1.549531	3.173259
4	1.847974	3.743683

```
[62]: def str_duration(x: float):  
       return f'{int(x/60)} min and {int(x%60)} s'
```

```
[63]: songs1['duration'].describe().loc[['mean', 'min', '25%', '50%', '75%', 'max']].  
      ↪map(str_duration)
```

```
[63]: mean      4 min and 4 s  
      min      2 min and 4 s  
      25%     3 min and 11 s  
      50%     3 min and 50 s  
      75%     4 min and 46 s  
      max      7 min and 52 s  
      Name: duration, dtype: object
```

3.3. Conclusión

El estudio muestra que la mayoría de las canciones analizadas están en idioma inglés y presentan una duración promedio de 3 a 4 minutos, en concordancia con los estándares de la música popular contemporánea. Al examinar los resultados de la segmentación, se observa que, dentro de ciertos clusters que agrupan entre 20 y 40 canciones en inglés, al menos 10 corresponden a artistas reconocidos del pop moderno, como Taylor Swift o Sabrina Carpenter. Este hallazgo no solo refuerza la pertinencia de la categoría inferida “popular/variada”, sino que además aporta evidencia empírica que vincula los clusters obtenidos con tendencias musicales actuales y artistas de referencia en el género.

Algunas variables que podrían enriquecer la interpretación son:

1. Artista
2. Género musical
3. Año de lanzamiento
4. Idioma de la canción (Español, Inglés, etc.)
5. Reproducciones mensuales

Y otras variables que podrían enriquecer el clustering son:

1. **Loudness (LUFS o dBFS)**. Nivel de volumen percibido promedio, ayuda a diferenciar baladas suaves vs. música muy producida y comprimida.
2. **Danceability (0–1)**. Mide cuán adecuada es una pista para bailar según ritmo, estabilidad y beat.
3. **Valence (0–1)**. Mide si la canción transmite emociones positivas (alegría) o negativas (tristeza).
4. **Energy (0–1)**. Similar a RMS pero ajustado a percepción humana; distingue baladas tranquilas de temas electrónicos potentes.
5. **Instrumentalness (0–1)**. Probabilidad de que la pista sea instrumental (sin voz).
6. **Speechiness (0–1)**. Proporción de voz hablada/rap frente a canto melódico.
7. **Acousticness (0–1)**. Probabilidad de que la canción sea acústica vs. electrónica/producida.
8. **Pitch-related features (si quisieras derivar más de Chroma/Tonnetz)**. Porcentaje de notas mayores vs menores, que puede asociarse a emociones.

4. Ejercicio 2

4.1. Cargar datos

```
[64]: # !pip install openpyxl
songs2 = pd.read_excel(r'https://raw.githubusercontent.com/Wsm-erick/Diplomado/
↳refs/heads/main/Modulo-I/Datasets/audio_features_lyrics.xlsx')
```

```
[65]: songs2.shape
```

```
[65]: (18172, 32)
```

```
[66]: songs2.head()
```

```
[66]:      album_name  ep album_release  track_number  track_name  artist \
0  Taylor Swift  0.0    2006-10-24           1.0  Tim McGraw  Taylor Swift
1  Taylor Swift  0.0    2006-10-24           1.0  Tim McGraw  Taylor Swift
2  Taylor Swift  0.0    2006-10-24           1.0  Tim McGraw  Taylor Swift
3  Taylor Swift  0.0    2006-10-24           1.0  Tim McGraw  Taylor Swift
4  Taylor Swift  0.0    2006-10-24           1.0  Tim McGraw  Taylor Swift

      featuring  bonus_track  promotional_release  single_release  ... \
0         NaN           0.0                NaT    2006-06-19  ...
1         NaN           0.0                NaT    2006-06-19  ...
2         NaN           0.0                NaT    2006-06-19  ...
3         NaN           0.0                NaT    2006-06-19  ...
4         NaN           0.0                NaT    2006-06-19  ...

      time_signature  duration_ms  explicit  key_name  mode_name  key_mode  line \
0                4.0    232107.0        0.0         C    major    C major    1
1                4.0    232107.0        0.0         C    major    C major    2
2                4.0    232107.0        0.0         C    major    C major    3
3                4.0    232107.0        0.0         C    major    C major    4
4                4.0    232107.0        0.0         C    major    C major    5

      lyric  element  element_artist
0    He said the way my blue eyes shined  Verse 1    Taylor Swift
1  Put those Georgia stars to shame that night  Verse 1    Taylor Swift
2                I said, "That's a lie"  Verse 1    Taylor Swift
3                Just a boy in a Chevy truck  Verse 1    Taylor Swift
4    That had a tendency of gettin' stuck  Verse 1    Taylor Swift
```

```
[5 rows x 32 columns]
```

4.2. Descripción de las variables

El conjunto de datos contiene las siguientes variables:

- `album_name`: El nombre del álbum. NA si la canción fue lanzada por separado de uno de los álbumes de estudio o EPs de Taylor.
- `ep`: Logical. True si el álbum es un EP y False si es un álbum de estudio.
- `album_release`: La fecha en que el álbum fue lanzado (AAAA-MM-DD).
- `track_number`: El orden de la canción en el álbum o EP.
- `track_name`: El nombre de la canción.
- `artist`: El nombre del artista de la canción. Por lo general, Taylor Swift, pero mostrará otros artistas para canciones en las que Taylor solo aparece.
- `featuring`: Cualquier artista que aparezca en la pista.
- `bonus_track`: Logical. True si la canción es un bonus track, False en otro caso.
- `promotional_release`: La fecha en que la canción fue lanzada como sencillo promocional. NA si la canción nunca fue lanzada como sencillo promocional.
- `single_release`: La fecha en que la canción se lanzó como sencillo oficial. No disponible si la canción nunca se lanzó como sencillo oficial.

- `track_release`: La fecha en que la canción se lanzó públicamente por primera vez. Esta es la fecha más temprana entre `album_release`, `promotional_release` y `single_release`.
- `danceability`: cuán adecuada es una pista para bailar. 0.0 es lo menos bailable, mientras que 1.0 es lo más bailable.
- `energy`: medida perceptual de intensidad y actividad. 0.0 es igual a menor energía y 1.0 es igual a mayor energía.
- `key`: la tonalidad en la que está la pista. Los mapas enteros se corresponden con la notación estándar de Pitch Class.
- `loudness`: volumen de la pista en decibelios (dB), promediado en toda la pista.
- `mode`: modalidad de una pista (mayor/menor). 0 es menor, 1 es mayor.
- `speechiness`: presencia de palabras habladas en una pista. Los valores superiores a 0.66 indican que la pista probablemente esté compuesta completamente de palabras habladas. Los valores entre 0.33 y 0.66 indican tanto música como habla. Los valores inferiores a 0.33 indican que la pista probablemente sea música u otras pistas sin habla.
- `acousticness`: confianza en que la pista es acústica. 0.0 representa baja confianza, 1.0 representa alta confianza.
- `instrumentalness`: Confianza en que la pista es instrumental (es decir, sin voces). 0.0 es baja confianza y 1.0 es alta confianza.
- `liveness`: Confianza en que la pista es una grabación en vivo (es decir, hay público presente). 0.0 es baja confianza, 1.0 es alta confianza.
- `valence`: Positividad musical que transmite la pista. 0.0 equivale a baja valencia (p. ej., triste, deprimido, enojado), 1.0 equivale alta valencia (p. ej., feliz, alegre, eufórico).
- `tempo`: Tempo estimado de la pista en pulsaciones por minuto (PPM).

- `time_signature`: Compás general estimado.
- `duration_ms`: Duración de la pista en milisegundos.
- `explicit`: Logical. True si la canción contiene letras explícitas, False en otro caso.
- `key_name`: corresponde directamente a la clave, pero el entero se convierte al nombre de la clave usando la notación de clase de tono (p. ej., 0 se convierte en C).
- `mode_name`: corresponde directamente al modo, pero el entero se convierte al nombre del modo (p. ej., 0 se convierte en menor).
- `key_mode`: una combinación de las variables `key_name` y `mode_name` (p. ej., C menor).
- `line`. El número de la línea en una canción.
- `lyric`. Verso en esa línea (string).
- `element`. Hace referencia a si esa línea es parte del coro, puente, intro, outro, verso 1, verso 2, etc.

4.3. Actividad

Dado lo anterior, realiza lo siguiente:

1. Encuentra si hay valores ausentes. ¿Es necesario imputarlos?
2. Elige las variables que den información y elimina variables que no sean informativas o transfórmalas para tener información.
3. Utiliza la variable `lyric` para obtener información lírica, por ejemplo el número de líneas por canción, número de palabras, proporción de sílabas por canción, densidad léxica, longitud media de palabras, etc. Es importante considerar el tokenizado y remoción de stopwords.
4. Realiza un análisis estadístico. ¿Qué nos dicen las variables? ¿Qué podemos inferir de las variables?
5. Normaliza las variables categóricas que consideres necesarias.

6. Finalmente escala y codifica las variables.

Al final escribe tus conclusiones con respecto a tu análisis. Si el equipo te pide puntos clave sobre las canciones, ¿qué podrías decir? Por ejemplo, hablar sobre la duración de las canciones, propiedades de las letras, el tono, la bailabilidad, etc.

¿Qué variables agregarías tú a los datos para considerar en un análisis más complejo? (popularidad en redes, viralidad, análisis de sentimientos, momentos sociales, etc).

¿Qué variable definirías como una variable objetivo? Justifica (En el dataset no existe una variable objetivo, por lo que podrías definirla tú desde tu punto de vista)

```
[67]: songs2.tail(2)
```

```
[67]:      album_name  ep album_release  track_number  \
18170      NaN NaN      NaT      NaN
18171      NaN NaN      NaT      NaN

      track_name      artist featuring  \
18170  You'll Always Find Your Way Back Home  Hannah Montana      NaN
18171  You'll Always Find Your Way Back Home  Hannah Montana      NaN

      bonus_track promotional_release single_release  ... time_signature  \
18170      NaN      NaT      2009-03-24  ...      4.0
18171      NaN      NaT      2009-03-24  ...      4.0

      duration_ms  explicit  key_name  mode_name  key_mode  line  \
18170      224187.0      0.0      G#      major  G# major      58
18171      224187.0      0.0      G#      major  G# major      59

      lyric  element  \
18170      You'll always, you'll always find your way      Outro
18171  You'll always find your way (Back home) back home      Outro

      element_artist
18170      Taylor Swift
18171      Taylor Swift

[2 rows x 32 columns]
```

4.4. Exploración

```
[68]: songs2.iloc[:100].to_csv('muestra_dataset2.csv')
```

4.4.1. Valores ausentes

```
[69]: is_na_df(songs2)
```

```
[69]:
```

	columna	n_missing	total	prop_missing	prop_original
0	album_name	1779	18172	0.097898	0.902102
1	ep	1779	18172	0.097898	0.902102
2	album_release	1779	18172	0.097898	0.902102
3	track_number	1779	18172	0.097898	0.902102
4	featuring	15898	18172	0.874862	0.125138
5	bonus_track	1779	18172	0.097898	0.902102
6	promotional_release	15836	18172	0.871451	0.128549
7	single_release	14222	18172	0.782633	0.217367
8	danceability	579	18172	0.031862	0.968138
9	energy	579	18172	0.031862	0.968138
10	key	579	18172	0.031862	0.968138
11	loudness	579	18172	0.031862	0.968138
12	mode	579	18172	0.031862	0.968138
13	speechiness	579	18172	0.031862	0.968138
14	acousticness	579	18172	0.031862	0.968138
15	instrumentalness	579	18172	0.031862	0.968138
16	liveness	579	18172	0.031862	0.968138
17	valence	579	18172	0.031862	0.968138
18	tempo	579	18172	0.031862	0.968138
19	time_signature	579	18172	0.031862	0.968138
20	duration_ms	579	18172	0.031862	0.968138
21	explicit	579	18172	0.031862	0.968138
22	key_name	579	18172	0.031862	0.968138
23	mode_name	579	18172	0.031862	0.968138
24	key_mode	579	18172	0.031862	0.968138
25	TOTAL	64694	581504	0.111253	0.888747

```
[70]: is_na_df(songs2).query('prop_missing > 0.15')
```

```
[70]:
```

	columna	n_missing	total	prop_missing	prop_original
4	featuring	15898	18172	0.874862	0.125138
6	promotional_release	15836	18172	0.871451	0.128549
7	single_release	14222	18172	0.782633	0.217367

```
[71]: print('Moda: {0:.2%}'.format(stats.mode(is_na_df(songs2).prop_missing)[0]))  
print('Media: {0:.2%}'.format(np.mean(is_na_df(songs2).prop_missing)))
```

Moda: 3.19%

Media: 14.12%

Tenemos 11% de valores ausentes en total del dataset, la moda de valores ausentes es de 3.19%. La media se eleva a 14.12% por variables como featuring, promotional_release y single_release que tienen más del 75% de valores ausentes, vamos a revisar si es necesario eliminar dichas variables.

```
[72]: dict_single_release = {x: i for i, x in enumerate(songs2['single_release'].
    ↪unique().dropna())}
songs2['single_release_rec'] = songs2['single_release'].map(dict_single_release)
songs2.groupby('track_name')['single_release_rec'].mean().map(lambda x: 1 if x_
    ↪>= 0 else 'nan').value_counts()
```

```
[72]: single_release_rec
nan      287
1         76
Name: count, dtype: int64
```

```
[73]: dict_promotional_release = {x: i for i, x in_
    ↪enumerate(songs2['promotional_release'].unique().dropna())}
songs2['promotional_release_rec'] = songs2['promotional_release'].
    ↪map(dict_promotional_release)
songs2.groupby('track_name')['promotional_release_rec'].mean().map(lambda x: 1_
    ↪if x >= 0 else 'nan').value_counts()
```

```
[73]: promotional_release_rec
nan      318
1         45
Name: count, dtype: int64
```

```
[74]: dict_featuring = {x: i for i, x in enumerate(pd.Series(songs2['featuring'].
    ↪unique().dropna()))}

print('Artistas que cantaron con Taylor en la pista:', len(dict_featuring))

songs2['featuring_rec'] = songs2['featuring'].map(dict_featuring)
songs2.groupby('track_name')['featuring_rec'].mean().map(lambda x: 1 if x >= 0_
    ↪else 'nan').value_counts()
```

Artistas que cantaron con Taylor en la pista: 27

```
[74]: featuring_rec
nan      319
1         44
Name: count, dtype: int64
```

Los conteos anteriores confirman que conviene eliminarlas debido a su alta concentración de valores ausentes (>15 %).

```
[75]: songs2.
    ↪drop(['featuring', 'promotional_release', 'single_release', 'featuring_rec', 'promotional_release'
    ↪axis=1, inplace=True)
```

```
[76]: print('Media: {0:.2%}'.format(np.mean(is_na_df(songs2).prop_missing)))
```

Media: 4.64%

Eliminando esas variables ya vemos que la media de valores ausentes se asimila a la moda.

4.4.2. Tipo de dato por variable

```
[77]: songs2.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 18172 entries, 0 to 18171
Data columns (total 29 columns):
#   Column                Non-Null Count  Dtype
---  -
0   album_name            16393 non-null  object
1   ep                    16393 non-null  float64
2   album_release         16393 non-null  datetime64[ns]
3   track_number          16393 non-null  float64
4   track_name            18172 non-null  object
5   artist                18172 non-null  object
6   bonus_track           16393 non-null  float64
7   track_release         18172 non-null  datetime64[ns]
8   danceability          17593 non-null  float64
9   energy                17593 non-null  float64
10  key                   17593 non-null  float64
11  loudness              17593 non-null  float64
12  mode                  17593 non-null  float64
13  speechiness           17593 non-null  float64
14  acousticness          17593 non-null  float64
15  instrumentalness       17593 non-null  float64
16  liveness              17593 non-null  float64
17  valence               17593 non-null  float64
18  tempo                 17593 non-null  float64
19  time_signature         17593 non-null  float64
20  duration_ms           17593 non-null  float64
21  explicit              17593 non-null  float64
22  key_name              17593 non-null  object
23  mode_name             17593 non-null  object
24  key_mode              17593 non-null  object
25  line                  18172 non-null  int64
26  lyric                 18172 non-null  object
27  element               18172 non-null  object
28  element_artist        18172 non-null  object
dtypes: datetime64[ns](2), float64(17), int64(1), object(9)
memory usage: 4.0+ MB
```

```
[78]: # Clave o llave del track para agrupar líneas de letras (si hay varias filas por
      ↪ canción)
track_key = ["artist", "album_name", "track_number", "track_name"]
```



```
track_key = [c for c in track_key if c in songs2.columns]
```

```
[79]: track_key
```

```
[79]: ['artist', 'album_name', 'track_number', 'track_name']
```

```
[80]: songs2.loc[:, track_key]
```

```
[80]:
```

	artist	album_name	track_number	\
0	Taylor Swift	Taylor Swift	1.0	
1	Taylor Swift	Taylor Swift	1.0	
2	Taylor Swift	Taylor Swift	1.0	
3	Taylor Swift	Taylor Swift	1.0	
4	Taylor Swift	Taylor Swift	1.0	
...	...	...	...	
18167	Hannah Montana	NaN	NaN	
18168	Hannah Montana	NaN	NaN	
18169	Hannah Montana	NaN	NaN	
18170	Hannah Montana	NaN	NaN	
18171	Hannah Montana	NaN	NaN	

	track_name
0	Tim McGraw
1	Tim McGraw
2	Tim McGraw
3	Tim McGraw
4	Tim McGraw
...	...
18167	You'll Always Find Your Way Back Home
18168	You'll Always Find Your Way Back Home
18169	You'll Always Find Your Way Back Home
18170	You'll Always Find Your Way Back Home
18171	You'll Always Find Your Way Back Home

[18172 rows x 4 columns]

```
[81]: songs2[songs2['track_name']=="You'll Always Find Your Way Back Home"].head()
```

```
[81]:
```

	album_name	ep	album_release	track_number	\
18113	NaN	NaN	NaT	NaN	
18114	NaN	NaN	NaT	NaN	
18115	NaN	NaN	NaT	NaN	
18116	NaN	NaN	NaT	NaN	
18117	NaN	NaN	NaT	NaN	

	track_name	artist	bonus_track	\
18113	You'll Always Find Your Way Back Home	Hannah Montana	NaN	

18114	You'll Always Find Your Way Back Home	Hannah Montana	NaN
18115	You'll Always Find Your Way Back Home	Hannah Montana	NaN
18116	You'll Always Find Your Way Back Home	Hannah Montana	NaN
18117	You'll Always Find Your Way Back Home	Hannah Montana	NaN

	track_release	danceability	energy	...	time_signature	duration_ms	\
18113	2009-03-24	0.577	0.906	...	4.0	224187.0	
18114	2009-03-24	0.577	0.906	...	4.0	224187.0	
18115	2009-03-24	0.577	0.906	...	4.0	224187.0	
18116	2009-03-24	0.577	0.906	...	4.0	224187.0	
18117	2009-03-24	0.577	0.906	...	4.0	224187.0	

	explicit	key_name	mode_name	key_mode	line	\
18113	0.0	G#	major	G# major	1	
18114	0.0	G#	major	G# major	2	
18115	0.0	G#	major	G# major	3	
18116	0.0	G#	major	G# major	4	
18117	0.0	G#	major	G# major	5	

	lyric	element	element_artist
18113	Woo!	Intro	Taylor Swift
18114	You wake up	Verse 1	Taylor Swift
18115	It's raining and it's Monday	Verse 1	Taylor Swift
18116	Looks like one of those rough days	Verse 1	Taylor Swift
18117	Time's up	Verse 1	Taylor Swift

[5 rows x 29 columns]

```
[82]: songs2[songs2['track_name']=="You'll Always Find Your Way Back Home"].shape
```

```
[82]: (59, 29)
```

```
[83]: songs2[songs2['track_name']=="You'll Always Find Your Way Back Home"]['line'].
      ↪max()
```

```
[83]: 59
```

El dataset viene a nivel lírica o línea de cada canción por lo que establecimos 'artist', 'album_name', 'track_number', 'track_name' como la llave para agrupar y obtener las características o métricas de cada canción para poder clusterizar.

```
[84]: songs2.query("artist == 'Taylor Swift'", inplace=True) # canciones de taylor_
      ↪swift como artista principal
```

```
[85]: songs2['artist'].unique() # validación que sólo traigamos a Taylor como artista
```

```
[85]: array(['Taylor Swift'], dtype=object)
```

```
[86]: songs2.shape # dimensión
```

```
[86]: (17204, 29)
```

4.5. Tratamiento de Datos

4.5.1. Funciones para obtener métricas de la letra y canciones

```
[87]: # instalar en primera ejecución
# nltk.download("stopwords"); nltk.download("punkt"); nltk.download('punkt_tab')
```

```
[88]: # campos principales (llave) por canción
track_key = [c for c in ["artist", "album_name", "track_number", "track_name"] if c_
↳ in songs2.columns]

# stopwords y vocales
stop_es = set(stopwords.words("spanish"))
stop_en = set(stopwords.words("english"))
stop_all = stop_es | stop_en

_vowels = re.compile(r"[aeiouáéíóúü]+", re.IGNORECASE)

def count_syllables(word: str) -> int:
    return max(1, len(_vowels.findall(word)))

def text_features(s: str) -> dict:
    if not isinstance(s, str) or not s.strip():
        return {"n_words":0, "n_unique":0, "lex_density":0.0, "avg_word_len":0.
↳ 0, "avg_syllables":0.0, "stop_ratio":0.0}
    toks = [t.lower() for t in word_tokenize(s) if re.match(r"\w+", t)]
    if not toks:
        return {"n_words":0, "n_unique":0, "lex_density":0.0, "avg_word_len":0.
↳ 0, "avg_syllables":0.0, "stop_ratio":0.0}

    n_words = len(toks)
    unique = len(set(toks))
    avg_len = float(np.mean([len(w) for w in toks]))
    avg_syl = float(np.mean([count_syllables(w) for w in toks]))
    n_stop = sum(w in stop_all for w in toks)
    lex_density = 1 - (n_stop / n_words)
    stop_ratio = n_stop / n_words

    return {
        "n_words": n_words,
        "n_unique": unique,
        "lex_density": lex_density,
        "avg_word_len": avg_len,
```

```

    "avg_syllables": avg_syl,
    "stop_ratio": stop_ratio,
}

```

```

[89]: # Métricas de letra por canción

agg_text = (
    songs2.groupby(track_key)["lyric"]
        .apply(lambda s: "\n".join(s.dropna().astype(str))) # concatenar letra
        ↪ completa
        .reset_index()
)

agg_text["n_lines"] = agg_text["lyric"].str.count(r"\n").fillna(0).astype(int) #
    ↪ líneas por canción
agg_text_feats = agg_text["lyric"].apply(text_features).apply(pd.Series) #
    ↪ features de la letra
agg_text["lyric"] = agg_text["lyric"].str.replace('\n', ' ') # sustituir saltos
    ↪ de línea por espacios
agg_text = pd.concat([agg_text[track_key + ['lyric', 'n_lines']],
    ↪ agg_text_feats], axis=1) # concatenar features de la letra por canción

print('Dimensión', agg_text.shape, end='\n\n')
display(agg_text.head(3))

```

Dimensión (326, 12)

	artist	album_name	track_number	track_name	\
0	Taylor Swift	1989	1.0	Welcome To New York	
1	Taylor Swift	1989	2.0	Blank Space	
2	Taylor Swift	1989	3.0	Style	

	lyric	n_lines	n_words	\
0	Walking through a crowd, the village is aglow ...	43	348.0	
1	Nice to meet you, where you been? I could show...	83	509.0	
2	Midnight You come and pick me up, no headlight...	47	383.0	

	n_unique	lex_density	avg_word_len	avg_syllables	stop_ratio
0	91.0	0.557471	4.135057	1.399425	0.442529
1	175.0	0.514735	3.555992	1.249509	0.485265
2	120.0	0.522193	3.605744	1.245431	0.477807

```

[90]: # Proporción de elementos (coro, verso, intro...)

elem_props = (
    songs2.groupby(track_key + ["element"]).size().unstack(fill_value=0)
)

```

```

elem_props = elem_props.div(elem_props.sum(axis=1), axis=0) # normalizar a
↳proporciones
elem_props.columns = [f"elem_prop_{c}" for c in elem_props.columns]

print('Dimensión', elem_props.shape, end='\n\n')
display(elem_props.head(3))

```

Dimensión (326, 29)

				elem_prop_Break \
artist	album_name	track_number	track_name	
Taylor Swift 1989		1.0	Welcome To New York	0.0
		2.0	Blank Space	0.0
		3.0	Style	0.0

				elem_prop_Breakdown \
artist	album_name	track_number	track_name	
Taylor Swift 1989		1.0	Welcome To New York	0.0
		2.0	Blank Space	0.0
		3.0	Style	0.0

				elem_prop_Bridge \
artist	album_name	track_number	track_name	
Taylor Swift 1989		1.0	Welcome To New York	0.090909
		2.0	Blank Space	0.047619
		3.0	Style	0.104167

				elem_prop_Bridge 1 \
artist	album_name	track_number	track_name	
Taylor Swift 1989		1.0	Welcome To New York	0.0
		2.0	Blank Space	0.0
		3.0	Style	0.0

				elem_prop_Bridge 2 \
artist	album_name	track_number	track_name	
Taylor Swift 1989		1.0	Welcome To New York	0.0
		2.0	Blank Space	0.0
		3.0	Style	0.0

				elem_prop_Buildup \
artist	album_name	track_number	track_name	
Taylor Swift 1989		1.0	Welcome To New York	0.0
		2.0	Blank Space	0.0
		3.0	Style	0.0

				elem_prop_Chorus \
artist	album_name	track_number	track_name	

Taylor Swift	1989	1.0	Welcome To New York	0.681818
		2.0	Blank Space	0.571429
		3.0	Style	0.479167

elem_prop_Chorus -

↪Variation \

artist	album_name	track_number	track_name	
Taylor Swift	1989	1.0	Welcome To New York	
↪0.0		2.0	Blank Space	
↪0.0		3.0	Style	
↪0.0				

elem_prop_Chorus 1 \

artist	album_name	track_number	track_name	
Taylor Swift	1989	1.0	Welcome To New York	0.0
		2.0	Blank Space	0.0
		3.0	Style	0.0

elem_prop_Chorus 2 \

artist	album_name	track_number	track_name	
Taylor Swift	1989	1.0	Welcome To New York	0.0
		2.0	Blank Space	0.0
		3.0	Style	0.0

... \

artist	album_name	track_number	track_name	
Taylor Swift	1989	1.0	Welcome To New York	
		2.0	Blank Space	
		3.0	Style	

elem_prop_Pre-Chorus 2 \

artist	album_name	track_number	track_name	
Taylor Swift	1989	1.0	Welcome To New York	0.0
		2.0	Blank Space	0.0
		3.0	Style	0.0

elem_prop_Pre-Chorus 3 \

artist	album_name	track_number	track_name	
Taylor Swift	1989	1.0	Welcome To New York	0.0
		2.0	Blank Space	0.0
		3.0	Style	0.0

elem_prop_Refrain \

artist	album_name	track_number	track_name	
Taylor Swift	1989	1.0	Welcome To New York	0.0

		2.0	Blank Space	0.0
		3.0	Style	0.0
				elem_prop_Spoken Outro \
artist	album_name	track_number	track_name	
Taylor Swift	1989	1.0	Welcome To New York	0.0
		2.0	Blank Space	0.0
		3.0	Style	0.0
				elem_prop_Verse \
artist	album_name	track_number	track_name	
Taylor Swift	1989	1.0	Welcome To New York	0.0
		2.0	Blank Space	0.0
		3.0	Style	0.0
				elem_prop_Verse 1 \
artist	album_name	track_number	track_name	
Taylor Swift	1989	1.0	Welcome To New York	0.113636
		2.0	Blank Space	0.095238
		3.0	Style	0.208333
				elem_prop_Verse 2 \
artist	album_name	track_number	track_name	
Taylor Swift	1989	1.0	Welcome To New York	0.113636
		2.0	Blank Space	0.095238
		3.0	Style	0.208333
				elem_prop_Verse 3 \
artist	album_name	track_number	track_name	
Taylor Swift	1989	1.0	Welcome To New York	0.000000
		2.0	Blank Space	0.095238
		3.0	Style	0.000000
				elem_prop_Verse 4 \
artist	album_name	track_number	track_name	
Taylor Swift	1989	1.0	Welcome To New York	0.000000
		2.0	Blank Space	0.095238
		3.0	Style	0.000000
				elem_prop_Verse 5
artist	album_name	track_number	track_name	
Taylor Swift	1989	1.0	Welcome To New York	0.0
		2.0	Blank Space	0.0
		3.0	Style	0.0

[3 rows x 29 columns]

```
[91]: # Variables musicales (se repiten en todas las líneas + tomar promedio o el
      ↳ primero)

music_feats = (
    songs2.groupby(track_key)
    ↳
    ↳ ["danceability", "energy", "valence", "tempo", "duration_ms", "acousticness", "instrumentalness",
    ↳ .mean()
)

print('Dimensión', music_feats.shape, end='\n\n')
display(music_feats.head(3))
```

Dimensión (326, 8)

artist	album_name	track_number	track_name	danceability \
Taylor Swift	1989	1.0	Welcome To New York	0.789
		2.0	Blank Space	0.760
		3.0	Style	0.588

artist	album_name	track_number	track_name	energy	valence \
Taylor Swift	1989	1.0	Welcome To New York	0.634	0.658
		2.0	Blank Space	0.703	0.570
		3.0	Style	0.791	0.487

artist	album_name	track_number	track_name	tempo \
Taylor Swift	1989	1.0	Welcome To New York	116.992
		2.0	Blank Space	95.997
		3.0	Style	94.933

artist	album_name	track_number	track_name	duration_ms \
Taylor Swift	1989	1.0	Welcome To New York	212600.0
		2.0	Blank Space	231827.0
		3.0	Style	231000.0

artist	album_name	track_number	track_name	acousticness \
Taylor Swift	1989	1.0	Welcome To New York	0.03480
		2.0	Blank Space	0.10300
		3.0	Style	0.00245

artist	album_name	track_number	track_name	instrumentalness \
--------	------------	--------------	------------	--------------------

Taylor Swift 1989	1.0	Welcome To New York	0.000002
	2.0	Blank Space	0.000000
	3.0	Style	0.002580

artist	album_name	track_number	track_name	liveness
Taylor Swift 1989		1.0	Welcome To New York	0.3020
		2.0	Blank Space	0.0913
		3.0	Style	0.1180

```
[92]: # Variables categóricas

cat_feats = (
    songs2.groupby(track_key)
        [["explicit", "mode", "time_signature"]]
        .mean()
)

print('Dimensión:', cat_feats.shape, end='\n\n')
display(cat_feats.head(3))
```

Dimensión: (326, 3)

artist	album_name	track_number	track_name	explicit	mode \
Taylor Swift 1989		1.0	Welcome To New York	0.0	1.0
		2.0	Blank Space	0.0	1.0
		3.0	Style	0.0	1.0

artist	album_name	track_number	track_name	time_signature
Taylor Swift 1989		1.0	Welcome To New York	4.0
		2.0	Blank Space	4.0
		3.0	Style	4.0

```
[93]: # Dataset final a nivel canción (se unen las variables musicales; los elementos
      ↪ por coro, verso, etc.; y, variables numéricas de texto)

df_song = music_feats.join(elem_props, how="left").reset_index()
df_song = df_song.merge(agg_text, on=track_key, how="left")
df_song = df_song.merge(cat_feats, on=track_key, how="left")

print("Dimensión final:", df_song.shape, end='\n\n')
df_song.head(3)
```

Dimensión final: (326, 52)

```
[93]:
```

	artist	album_name	track_number	track_name	danceability	\
0	Taylor Swift	1989	1.0	Welcome To New York	0.789	
1	Taylor Swift	1989	2.0	Blank Space	0.760	
2	Taylor Swift	1989	3.0	Style	0.588	

	energy	valence	tempo	duration_ms	acousticness	...	n_lines	n_words	\
0	0.634	0.658	116.992	212600.0	0.03480	...	43	348.0	
1	0.703	0.570	95.997	231827.0	0.10300	...	83	509.0	
2	0.791	0.487	94.933	231000.0	0.00245	...	47	383.0	

	n_unique	lex_density	avg_word_len	avg_syllables	stop_ratio	explicit	\
0	91.0	0.557471	4.135057	1.399425	0.442529	0.0	
1	175.0	0.514735	3.555992	1.249509	0.485265	0.0	
2	120.0	0.522193	3.605744	1.245431	0.477807	0.0	

	mode	time_signature
0	1.0	4.0
1	1.0	4.0
2	1.0	4.0

[3 rows x 52 columns]

```
[94]: metricas = {
    'd_letra': list(set(agg_text)-set(track_key+['lyric'])),
    'musicales': list(set(music_feats)-set(track_key+['lyric'])),
    'd_elementos': list(set(elem_props)-set(track_key+['lyric']))
}
```

```
[95]: print('métricas de letra\n\n' + ' | '.join(metricas['d_letra']))
print('\nmétricas de musicales\n\n' + ' | '.join(metricas['musicales']))
print("\nmétricas de elementos\n\n" + ' | '.join(metricas['d_elementos']))
```

métricas de letra

stop_ratio | n_unique | n_words | avg_word_len | lex_density | avg_syllables |
n_lines

métricas de musicales

tempo | energy | valence | acousticness | duration_ms | liveness |
instrumentalness | danceability

métricas de elementos

elem_prop_Intro | elem_prop_Verse 4 | elem_prop_Break | elem_prop_Spoken Outro |
elem_prop_Bridge 2 | elem_prop_Verse 5 | elem_prop_Verse | elem_prop_Bridge 1 |
elem_prop_Chorus 3 | elem_prop_Interlude | elem_prop_Pre-Chorus 1 |

```
elem_prop_Verse 1 | elem_prop_Refrain | elem_prop_Pre-Chorus | elem_prop_Post-
Chorus | elem_prop_Chorus 2 | elem_prop_Outro | elem_prop_Breakdown |
elem_prop_Pre-Chorus 3 | elem_prop_Verse 2 | elem_prop_Chorus | elem_prop_Bridge
| elem_prop_Buildup | elem_prop_Verse 3 | elem_prop_Pre-Chorus 2 |
elem_prop_Post-Chorus 1 | elem_prop_Chorus 1 | elem_prop_Post-Chorus 2 |
elem_prop_Chorus - Variation
```

4.5.2. Estadísticos

```
[96]: num_cols = [item for sublist in metricas.values() for item in sublist]
cat_cols = ["explicit", "mode", "time_signature"] # excluimos
↳ "album_name", "track_number", "track_name" porque traerá muchos valores
↳ distintos
```

```
[97]: analisis_songs2 = analyze(df_song, num_cols, cat_cols)
```

```
[98]: analisis_songs2.plot_num()
```

```
[99]: analisis_songs2.plot_cat()
```

- **explicit**: Logical. True si la letra contiene letra explícita.
- **time_signature**: Compás general estimado
- **mode**: Modalidad de una pista. 0 es menor, 1 es mayor

La mayoría de las variables categóricas que decidimos dejar se concentra en más del 80 % de la información en un valor por lo que no nos va a dar tanta información al interpretar. Por otra parte, las variables numéricas que corresponden a elementos de las canciones (coros, versos, puentes, etc.) tienen buena distribución algunas y otras se concentra en un intervalo de 0.02 y 0.05, prácticamente 0 variabilidad.

```
[100]: df_song.drop(["explicit", "mode", "time_signature"], axis=1, inplace=True)
```

```
[101]: print("Dimensión final:", df_song.shape, end='\n\n')
df_song.head(3)
```

Dimensión final: (326, 49)

```
[101]:
```

	artist	album_name	track_number	track_name	danceability	\
0	Taylor Swift	1989	1.0	Welcome To New York	0.789	
1	Taylor Swift	1989	2.0	Blank Space	0.760	
2	Taylor Swift	1989	3.0	Style	0.588	

	energy	valence	tempo	duration_ms	acousticness	...	\
0	0.634	0.658	116.992	212600.0	0.03480	...	
1	0.703	0.570	95.997	231827.0	0.10300	...	
2	0.791	0.487	94.933	231000.0	0.00245	...	


```
elem_prop_Verse 4 elem_prop_Verse 5 \
```

0	0.000000	0.0
1	0.095238	0.0
2	0.000000	0.0

	lyric	n_lines	n_words	\
0	Walking through a crowd, the village is aglow ...	43	348.0	
1	Nice to meet you, where you been? I could show...	83	509.0	
2	Midnight You come and pick me up, no headlight...	47	383.0	

	n_unique	lex_density	avg_word_len	avg_syllables	stop_ratio
0	91.0	0.557471	4.135057	1.399425	0.442529
1	175.0	0.514735	3.555992	1.249509	0.485265
2	120.0	0.522193	3.605744	1.245431	0.477807

[3 rows x 49 columns]

```
[102]: is_na_df(df_song)
```

	columna	n_missing	total	prop_missing	prop_original
0	danceability	9	326	0.027607	0.972393
1	energy	9	326	0.027607	0.972393
2	valence	9	326	0.027607	0.972393
3	tempo	9	326	0.027607	0.972393
4	duration_ms	9	326	0.027607	0.972393
5	acousticness	9	326	0.027607	0.972393
6	instrumentalness	9	326	0.027607	0.972393
7	liveness	9	326	0.027607	0.972393
8	TOTAL	72	15974	0.004507	0.995493

4.5.3. Escalado e Imputación de valores ausentes

```
[103]: num_cols = list(set(num_cols)-set(['mode','explicit','time_signature']))
```

```
[104]: print('Variables numéricas continuas:', len(num_cols))
```

Variables numéricas continuas: 44

```
[105]: # Pipelines
num_pipe = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler())
])

# ColumnTransformer
pre = ColumnTransformer(
    transformers=[
        ("num", num_pipe, num_cols)
```

```

],
remainder="drop",
verbose_feature_names_out=False
)

X_scaled = pre.fit_transform(df_song)
feature_names = pre.get_feature_names_out()
print(X.shape, len(feature_names))

```

(162, 39) 44

```
[106]: is_na_df(pd.DataFrame(X_scaled))
```

```
[106]:
```

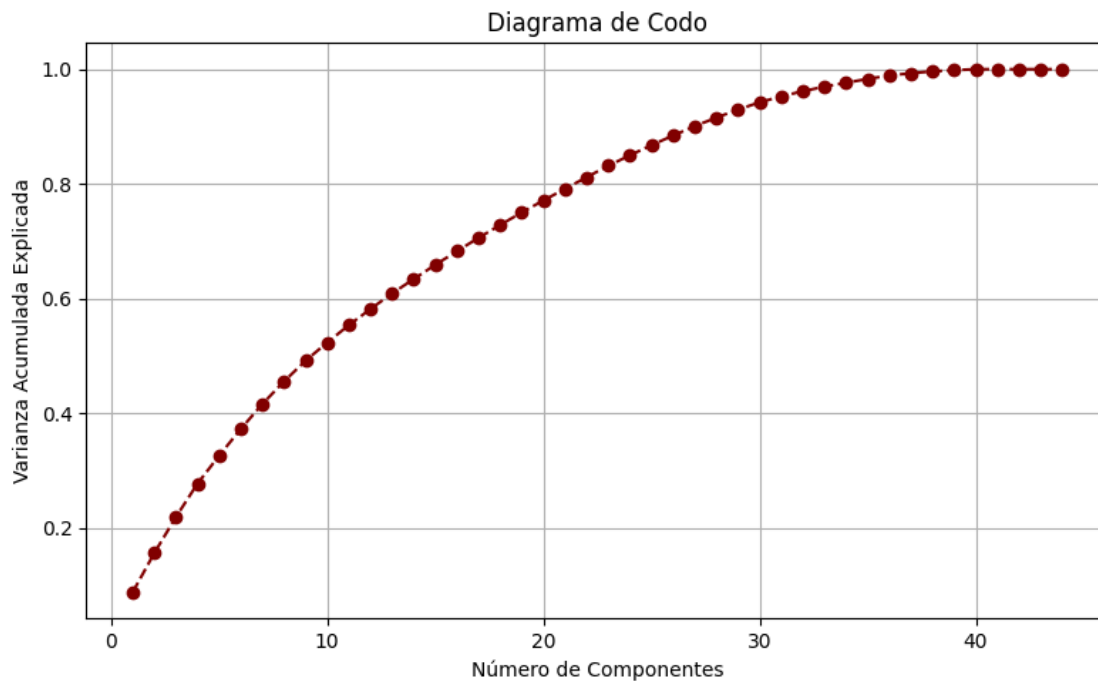
	columna	n_missing	total	prop_missing	prop_original
0	TOTAL	0	14344	0.0	1.0

Se imputa las variables numéricas con valores ausentes con la mediana de la muestra y posteriormente se escala (pipeline).

4.5.4. Componentes Principales

```
[107]: pca_02 = rPCA(pd.DataFrame(X_scaled, columns=num_cols), num_cols, 0.90)
```

```
[108]: pca_02.plot_elbow()
```



```
[109]: print("Componentes:", pca_02.r_pca()[0].shape[1])
```

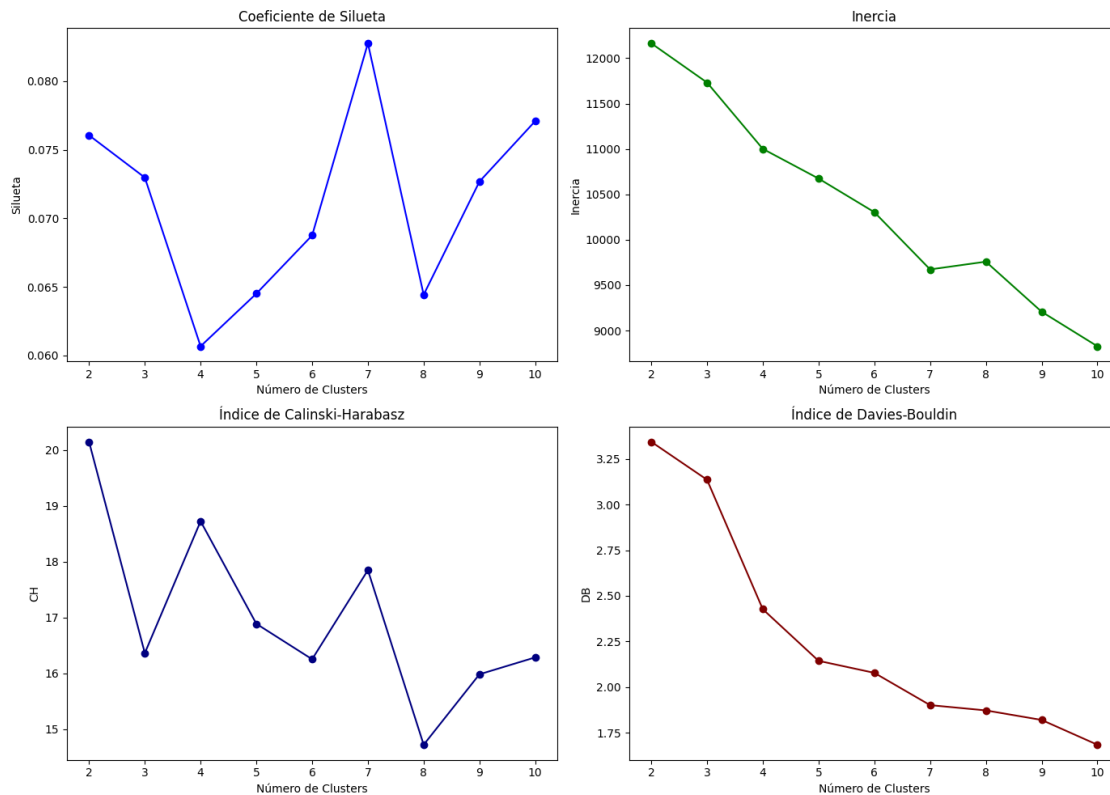
Componentes: 27

4.5.5. Vecinos Cercanos

```
[110]: data2 = pca_02.r_pca()[0]
```

```
[111]: kmeans2 = rKMeans(dataframe=data2, cols=None, max_clusters=10, random_state=42)
```

```
[112]: kmeans2.optimal_clusters()
```



```
[112]:
```

	Clusters	Silueta	Inercia	Calinski-Harabasz	Davies-Bouldin
0	2	0.076056	12163.457536	20.146084	3.344483
1	3	0.072973	11731.376311	16.360057	3.136407
2	4	0.060673	11000.874123	18.722308	2.426509
3	5	0.064513	10673.899417	16.885236	2.143457
4	6	0.068757	10303.595077	16.250188	2.077648
5	7	0.082760	9672.866610	17.846536	1.900678
6	8	0.064431	9758.211924	14.718391	1.871741
7	9	0.072679	9206.747342	15.980521	1.819460
8	10	0.077089	8826.401791	16.283282	1.683486

4.5.6. Interpretación de Gráficos

El análisis de métricas de validación interna sugiere que la elección del número de clusters más apropiado se encuentra entre **2 y 7**. El valor de silueta alcanza su máximo en 7 clusters, lo que indica una buena separación relativa entre grupos. La inercia también muestra un punto de inflexión en torno a 6–7 clusters, mientras que el índice de Davies-Bouldin disminuye de forma consistente hasta estabilizarse a partir de ese rango. Por otro lado, el índice de Calinski-Harabasz señala que con **2 clusters** se logra la mejor compacidad, aunque con **7 clusters** se conserva una definición aceptable. En conjunto, se puede concluir que la solución de **7 clusters** es adecuada si se busca una segmentación más granular, mientras que **2 clusters** representaría una división más general.

```
[113]: pca = PCA(n_components=0.90, random_state=42) # conserva 90% de varianza
X_pca = pca.fit_transform(X_scaled)

# Dataset reducido (ya con PCA o escalado)
X_data = X_pca

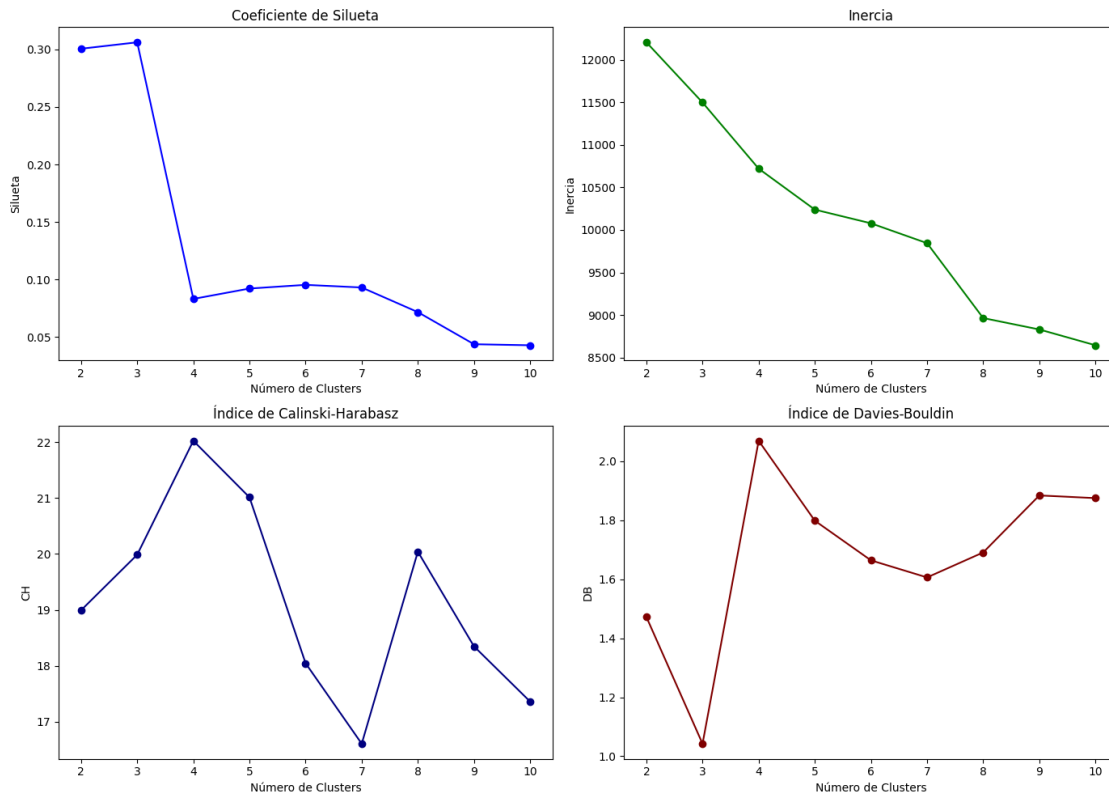
best_score = -1.0
best_state = None
best_model = None
scores = []

for rs in range(100):
    km = KMeans(n_clusters=7, random_state=rs, n_init="auto")
    labels = km.fit_predict(X_data)
    score = silhouette_score(X_data, labels)
    scores.append((rs, score))
    if score > best_score:
        best_score = score
        best_state = rs
        best_model = km

print(f"- Mejor silueta: {best_score:.4f} con random_state = {best_state}")
```

- Mejor silueta: 0.0931 con random_state = 83

```
[114]: kmeans2.optimal_clusters(random_state=83)
```



```
[114]:
```

	Clusters	Silueta	Inercia	Calinski-Harabasz	Davies-Bouldin
0	2	0.300551	12204.325005	18.993675	1.471946
1	3	0.306131	11496.943354	19.986782	1.042381
2	4	0.083179	10720.106612	22.023799	2.069108
3	5	0.092202	10238.846765	21.012551	1.798316
4	6	0.095416	10078.095281	18.045805	1.663953
5	7	0.093113	9844.743704	16.606733	1.606044
6	8	0.071713	8964.869503	20.041072	1.690739
7	9	0.043828	8830.742622	18.348152	1.883892
8	10	0.042913	8645.125376	17.360952	1.874833

Probando con la mejor semilla para 7 clusters vemos mejor silueta para el grupo de 2 clusters.

```
[115]: n_clusters_list = [2, 3, 7]          # columnas
random_states = [11, 42, 83, 95]         # filas

fig, axes = plt.subplots(len(random_states), len(n_clusters_list),
                          figsize=(15, 15), sharex=True, sharey=True)

for i, rs in enumerate(random_states):
    for j, k in enumerate(n_clusters_list):
        ax = axes[i, j]
```



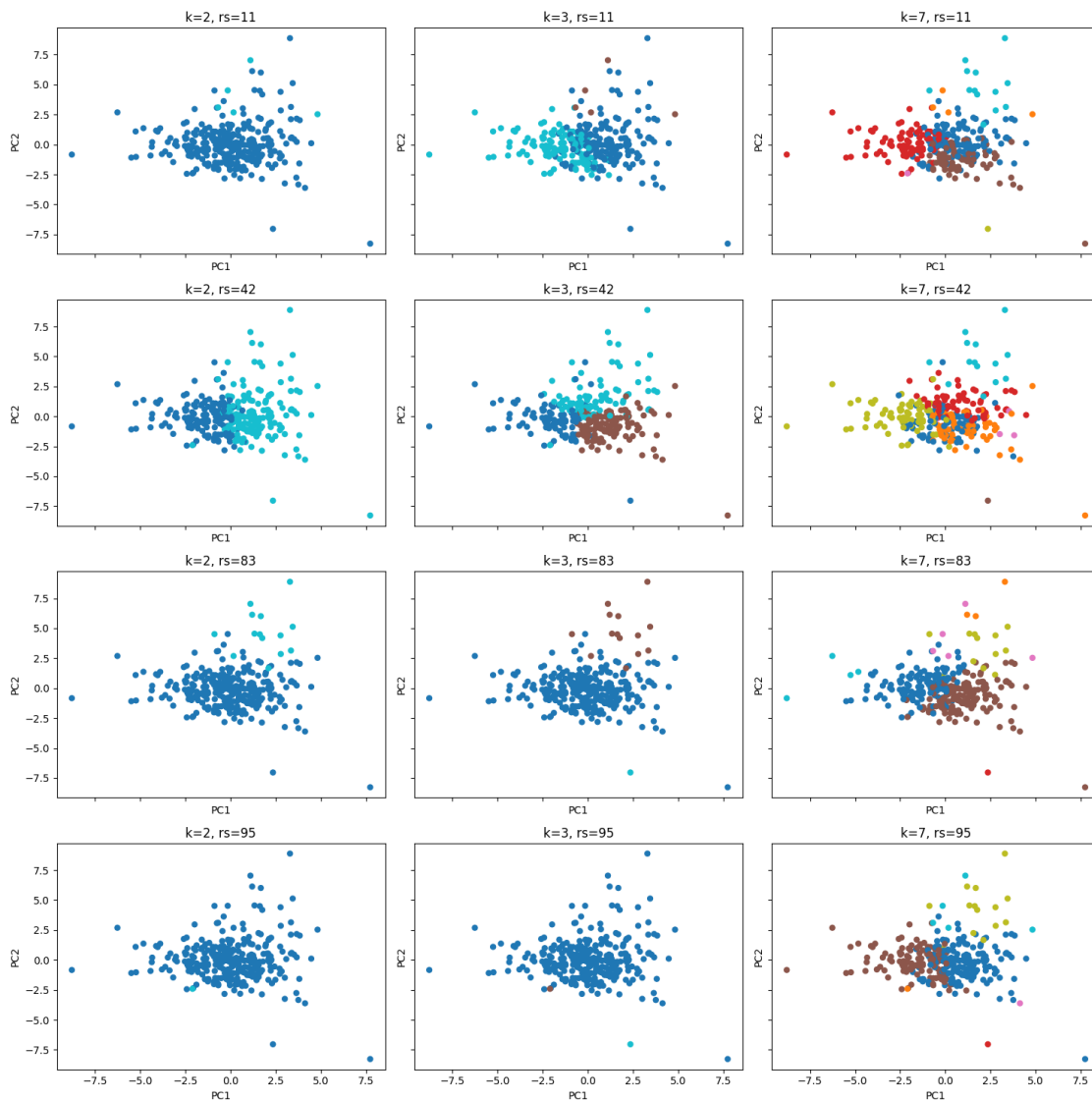
```

kmeans = KMeans(n_clusters=k, random_state=rs, n_init="auto")
labels = kmeans.fit_predict(X_pca)

scatter = ax.scatter(X_pca[:,0], X_pca[:,1],
                    c=labels, cmap="tab10", s=25)
ax.set_title(f"k={k}, rs={rs}")
ax.set_xlabel("PC1")
ax.set_ylabel("PC2")

plt.tight_layout()
plt.show()

```



- Aunque en el corrido de simulaciones para encontrar la mejor semilla para 7 clusters (semilla

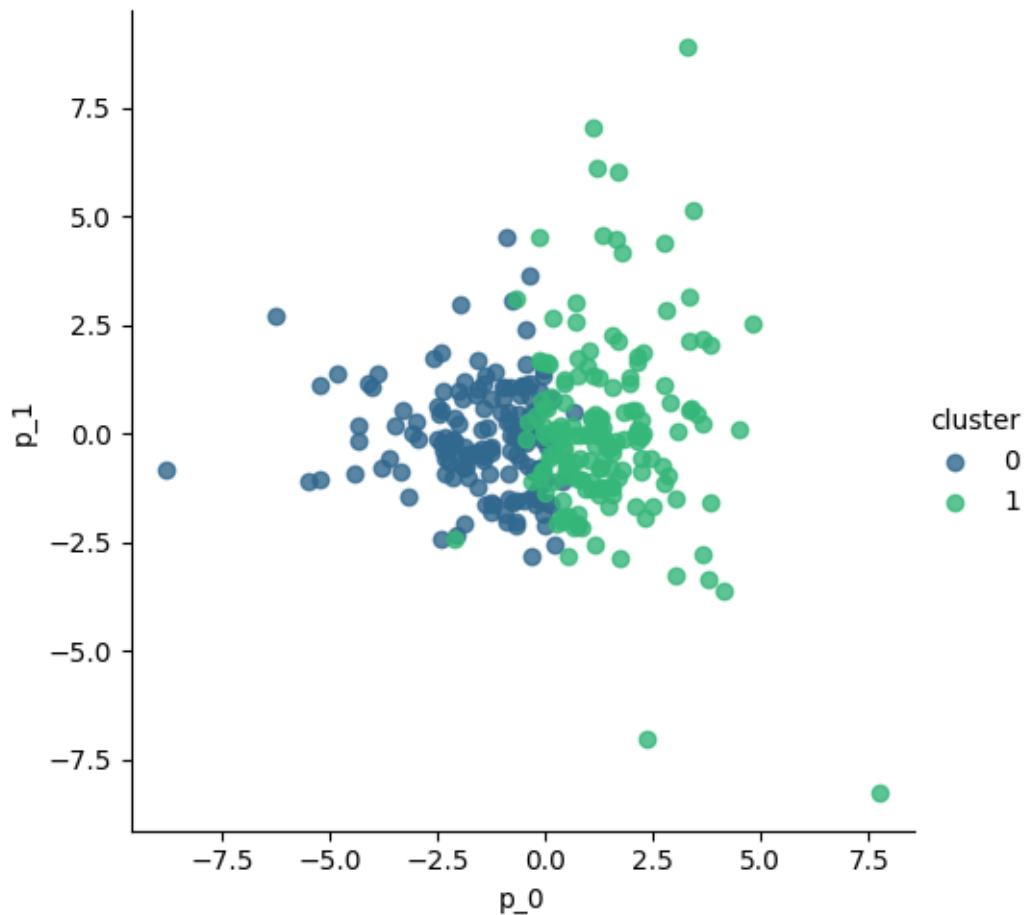
83) vemos en el gráfico 2D que los puntos se sobreponen y es más, observamos mejor silueta para el grupo de 2 clusters.

- A pesar de probar configuraciones con 2, 3 y 7 clusters, los resultados muestran un alto solapamiento entre grupos. La visualización en componentes principales evidencia que los clusters no presentan fronteras claras, lo cual coincide con los bajos valores de silueta.
- No hay una buena segmentación de clusters y tiene sentido porque hablamos de un sólo artista por lo que esperamos un sólo género musical, tonalidades similares en las canciones, bailabilidad e incluso letras similares (misma longitud, misma cantidad de versos, misma cantidad de vocales o sílabas usadas, entre otras métricas).
- Con el fin de ver los resultados con dos clusters elijiremos el de 2 con semilla 42 para ver las características de cada cluster aunque por el punto anterior consideramos también que nos referimos en realidad a un sólo cluster (un sólo artista o género musical en el dataset).

```
[116]: kmeans2.fit(n_clusters=2, random_state=42)
kmeans2.plot(data2)
df_song['cluster'] = kmeans2.get_df()['cluster']

display(pd.concat([df_song['cluster'].value_counts(), df_song['cluster'].
↪value_counts(normalize=True).apply(lambda x: '{:.1%}'.format(x))], axis=1))
```

	count	proportion
cluster		
1	169	51.8%
0	157	48.2%



```
[117]: perfil2 = Perfil(df_song, var_cat, num_cols)
```

```
[118]: aux2 = perfil2.kw_test().sort_values(by='p-value', ascending=False).
        ↪reset_index(drop=True)

var_sig2 = aux2.loc[aux2['p-value'] < 0.05, "Variable"].tolist()
var_nosig2 = aux2.loc[aux2['p-value'] >= 0.05, "Variable"].tolist()

print('\nVariables no significativas\n')
display(aux2.query(f"Variable == {var_nosig2}"))

print('\nVariables significativas\n')
display(aux2.query(f"Variable == {var_sig2}"))
```

Variables no significativas

Variable	p-value
----------	---------

0	elem_prop_Break	0.95493
1	danceability	0.68390
2	duration_ms	0.41058
3	elem_prop_Interlude	0.35651
4	n_unique	0.34043
5	elem_prop_Bridge 2	0.33512
6	elem_prop_Verse 5	0.33512
7	elem_prop_Verse	0.33512
8	elem_prop_Bridge 1	0.33512
9	elem_prop_Chorus 3	0.33512
10	elem_prop_Post-Chorus 2	0.33512
11	elem_prop_Buildup	0.33512
12	elem_prop_Post-Chorus 1	0.33512
13	liveness	0.33324
14	instrumentalness	0.31465
15	elem_prop_Spoken Outro	0.29950
16	elem_prop_Refrain	0.29734
17	elem_prop_Breakdown	0.26459
18	tempo	0.21459
19	elem_prop_Outro	0.16095
20	elem_prop_Chorus - Variation	0.14169
21	elem_prop_Bridge	0.12615
22	elem_prop_Chorus	0.05971
23	elem_prop_Pre-Chorus 3	0.05279

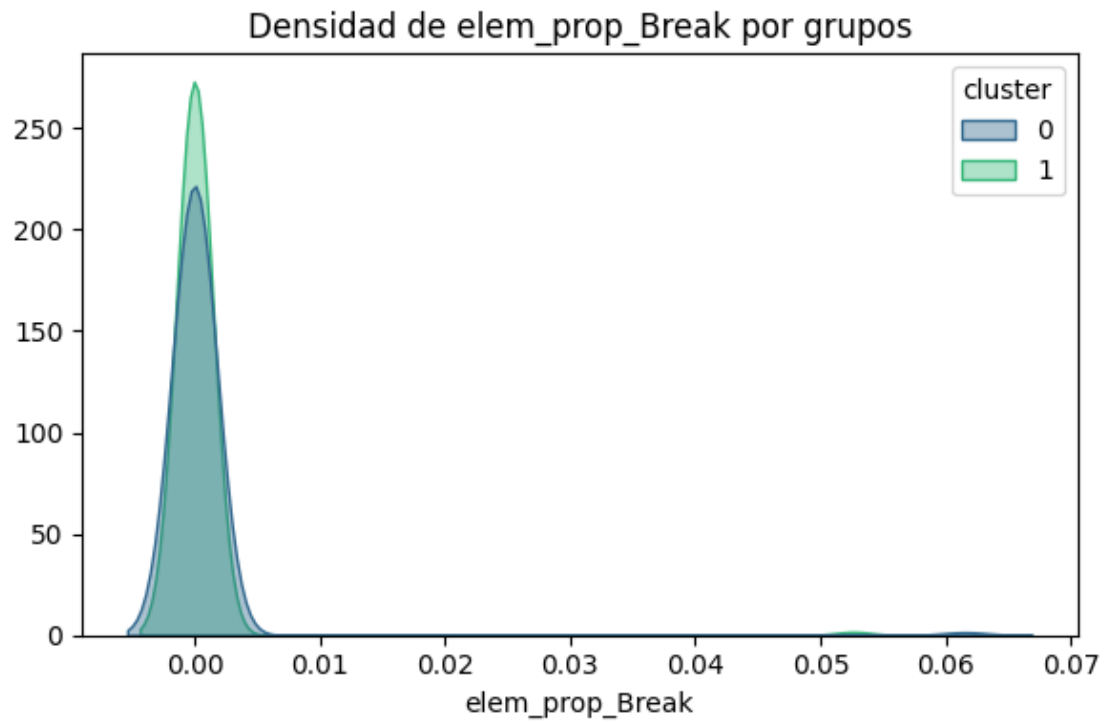
Variables significativas

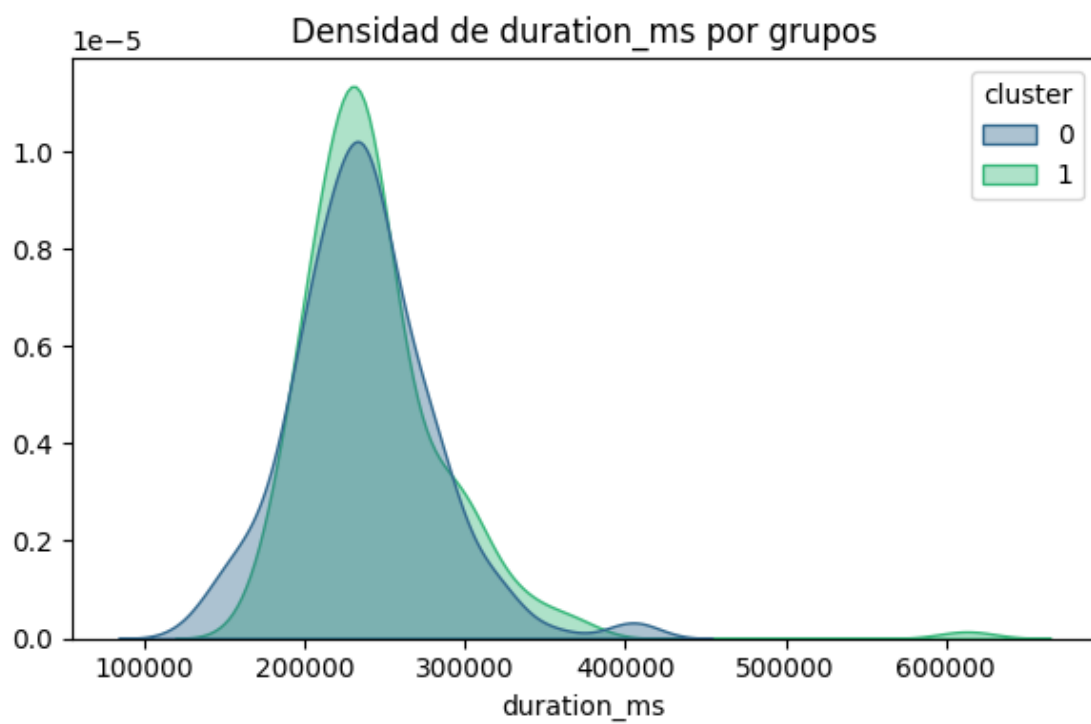
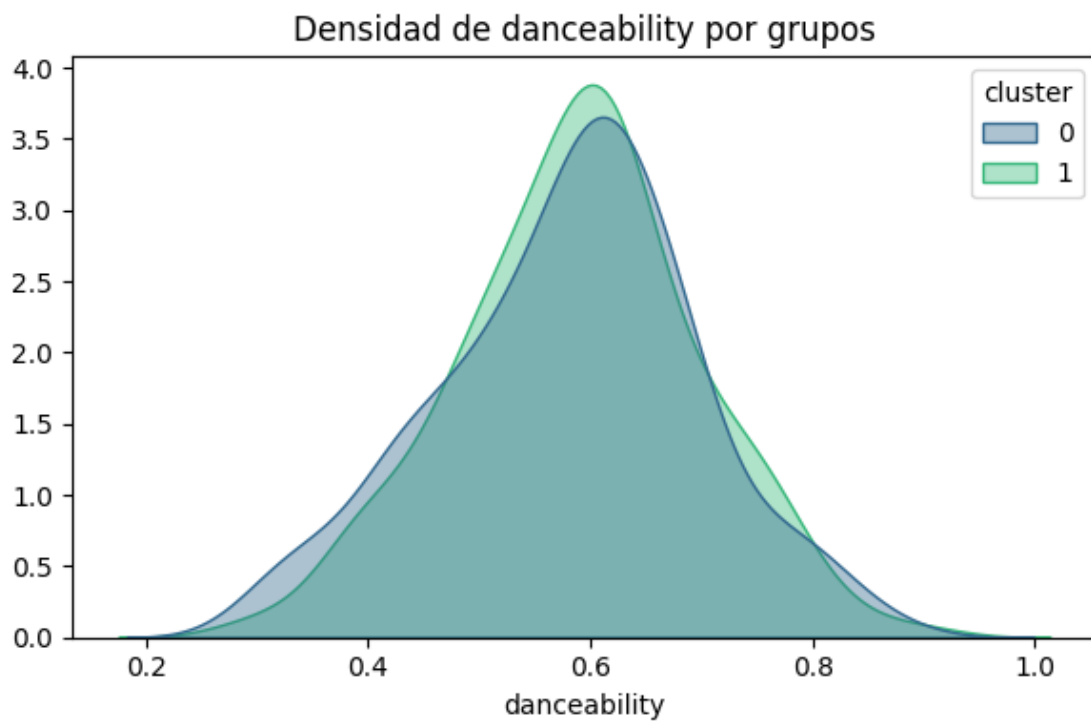
	Variable	p-value
24	elem_prop_Chorus 1	0.03012
25	elem_prop_Verse 4	0.03012
26	elem_prop_Chorus 2	0.00586
27	valence	0.00363
28	elem_prop_Pre-Chorus 1	0.00334
29	elem_prop_Post-Chorus	0.00180
30	elem_prop_Verse 3	0.00120
31	elem_prop_Pre-Chorus 2	0.00069
32	elem_prop_Intro	0.00032
33	stop_ratio	0.00001
34	lex_density	0.00001
35	elem_prop_Verse 2	0.00000
36	avg_word_len	0.00000
37	acousticness	0.00000
38	energy	0.00000
39	avg_syllables	0.00000
40	elem_prop_Pre-Chorus	0.00000
41	elem_prop_Verse 1	0.00000
42	n_lines	0.00000

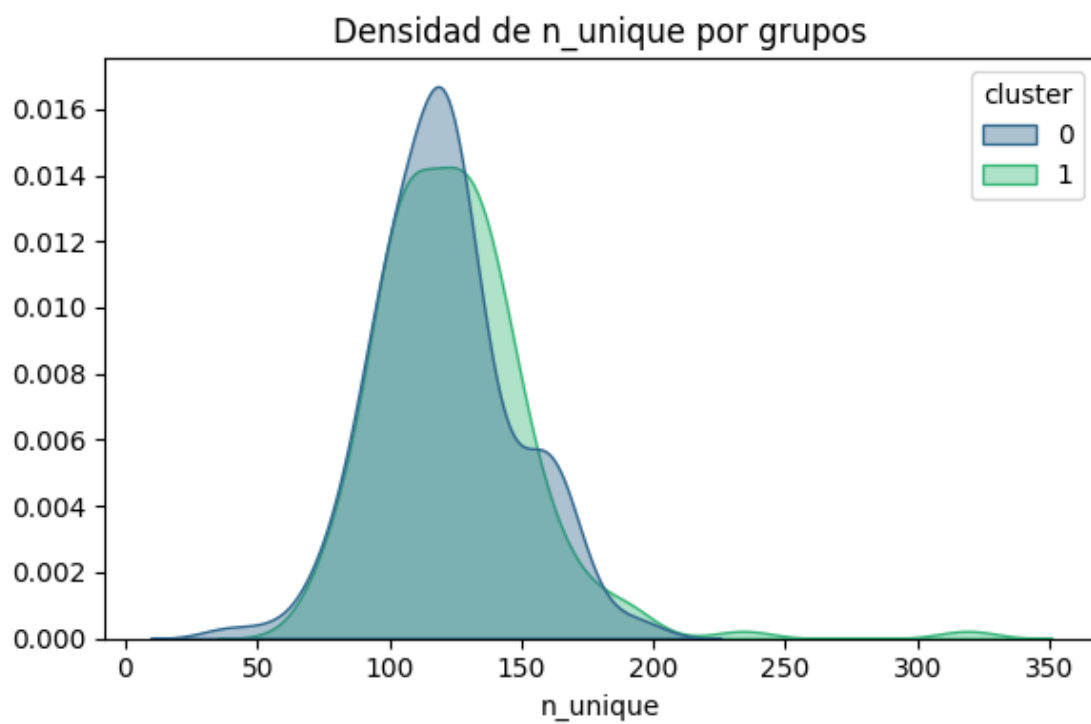
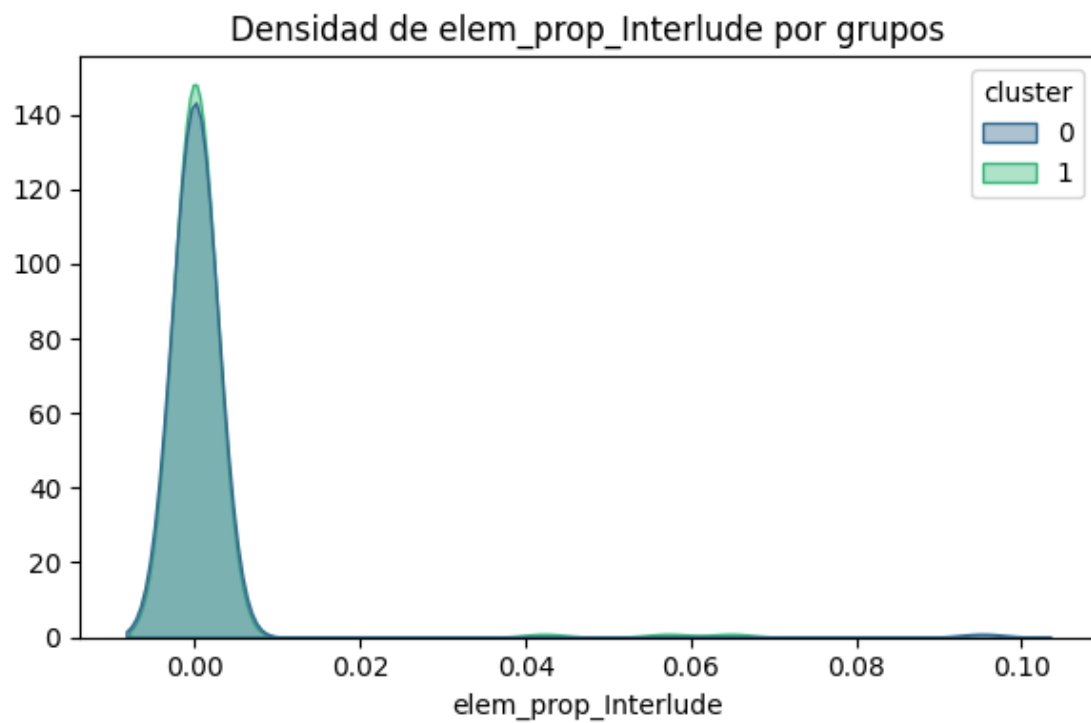
43

n_words 0.00000

```
[119]: # Distribución de variables no significativas  
perfil2.plot_dist(var_nosig2)
```

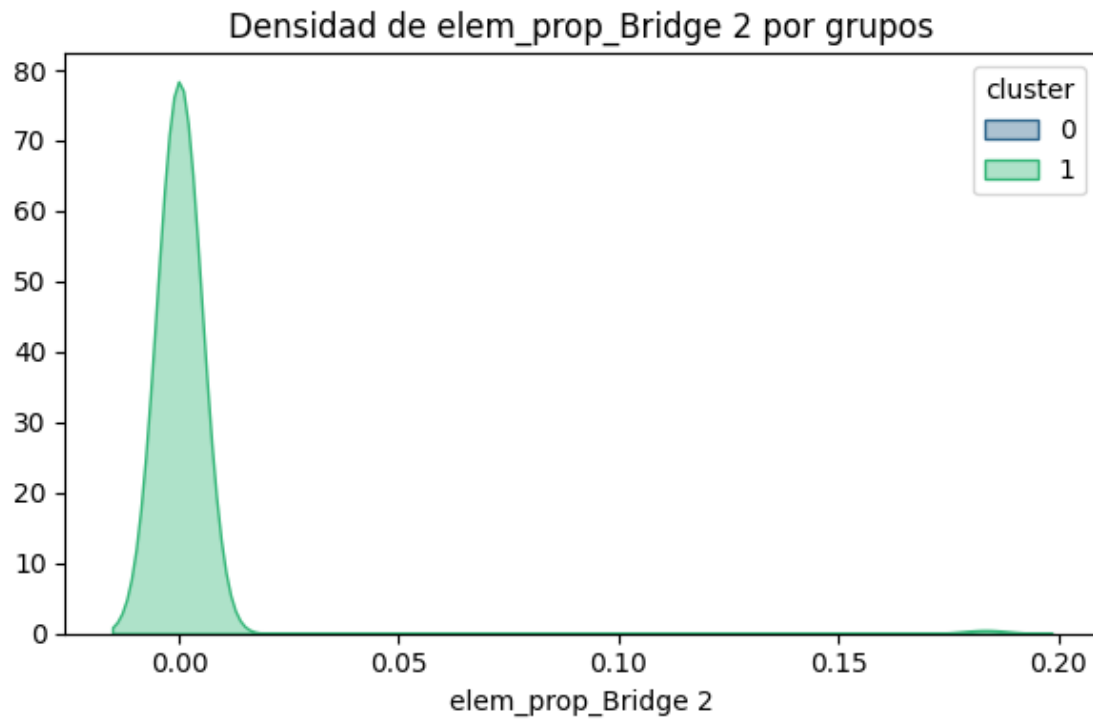






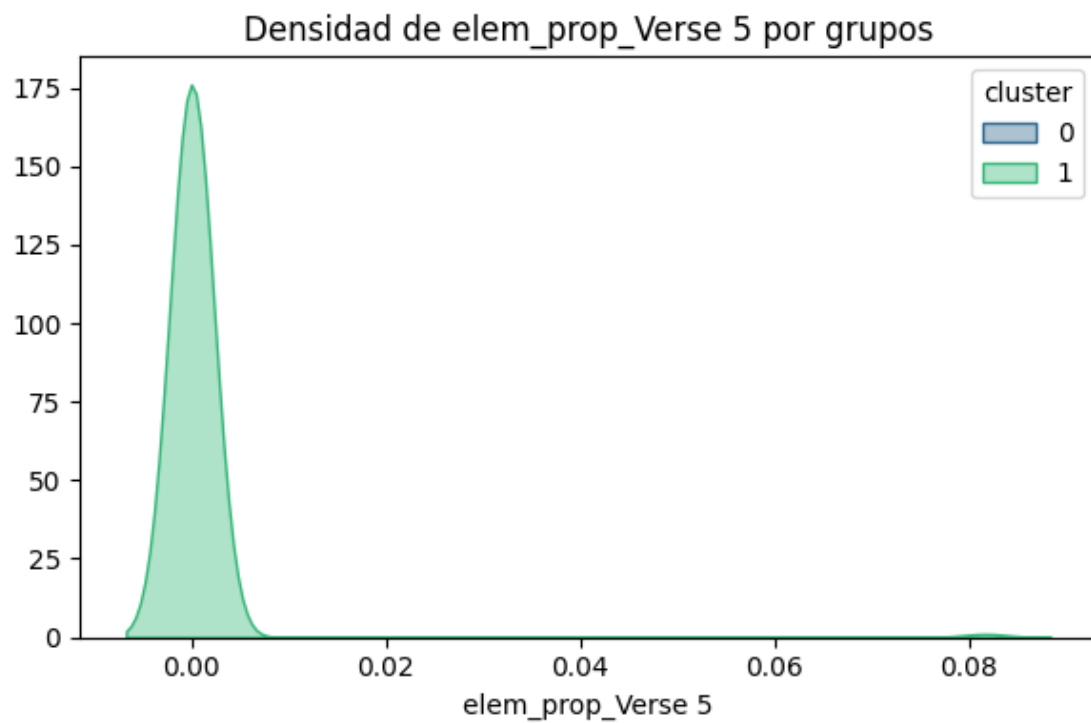
/tmp/ipykernel_3403/2607923038.py:15: UserWarning:

Dataset has 0 variance; skipping density estimate. Pass `warn\_singular=False` to disable this warning.



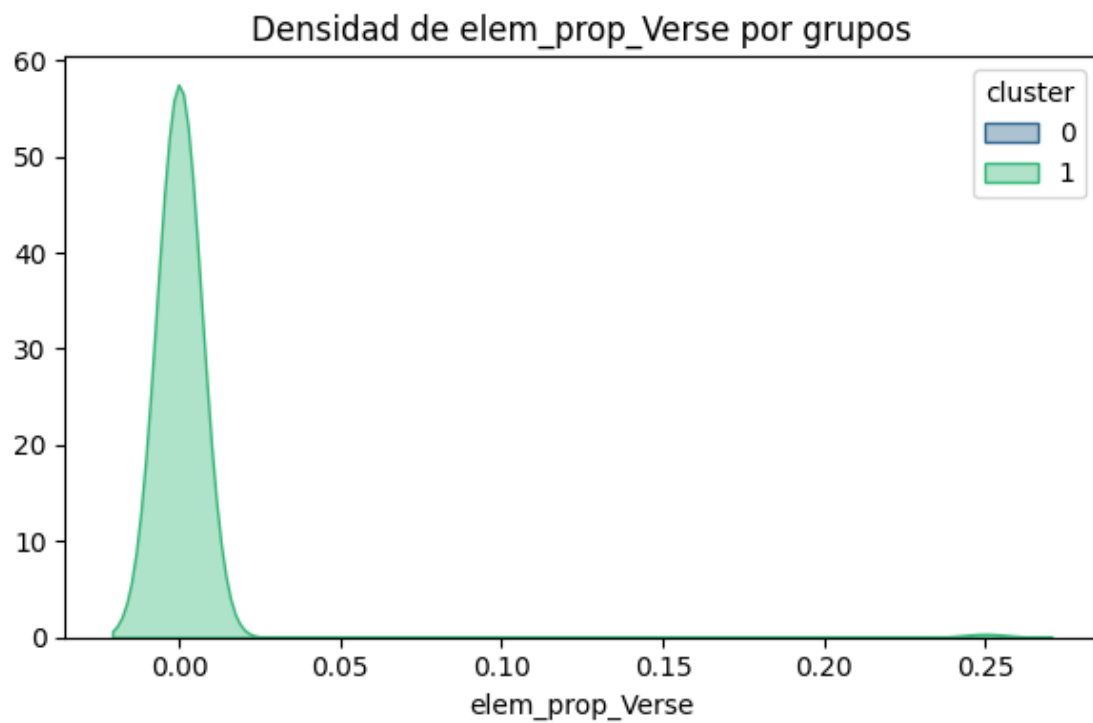
/tmp/ipykernel_3403/2607923038.py:15: UserWarning:

Dataset has 0 variance; skipping density estimate. Pass `warn\_singular=False` to disable this warning.



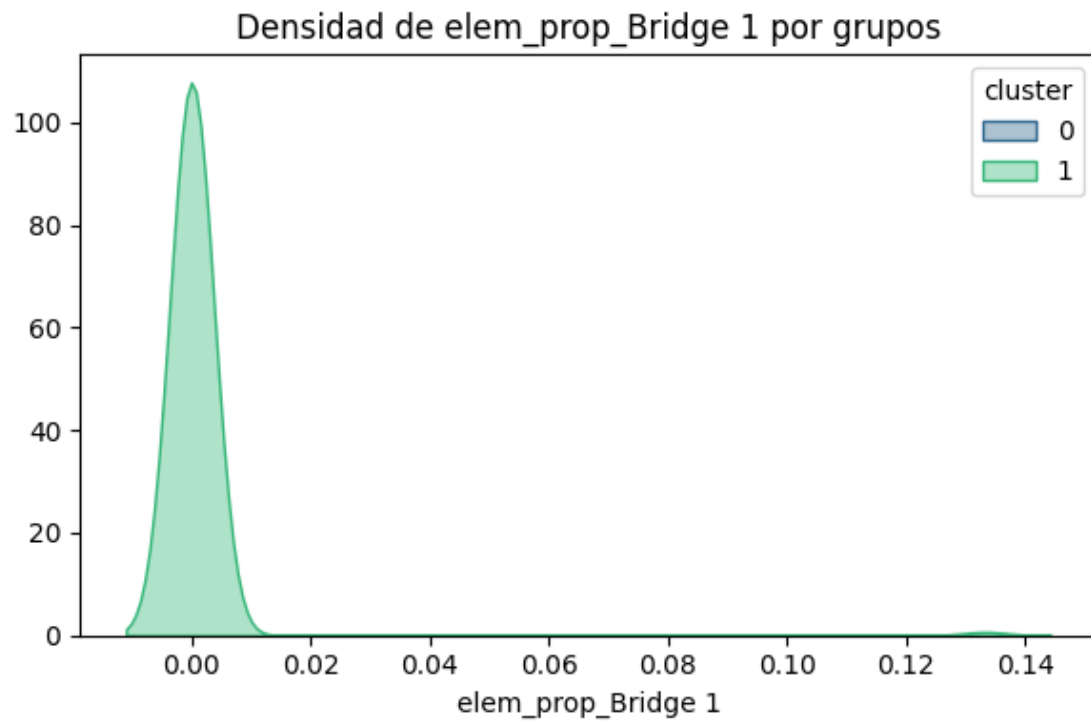
```
/tmp/ipykernel_3403/2607923038.py:15: UserWarning:
```

```
Dataset has 0 variance; skipping density estimate. Pass `warn_singular=False` to  
disable this warning.
```



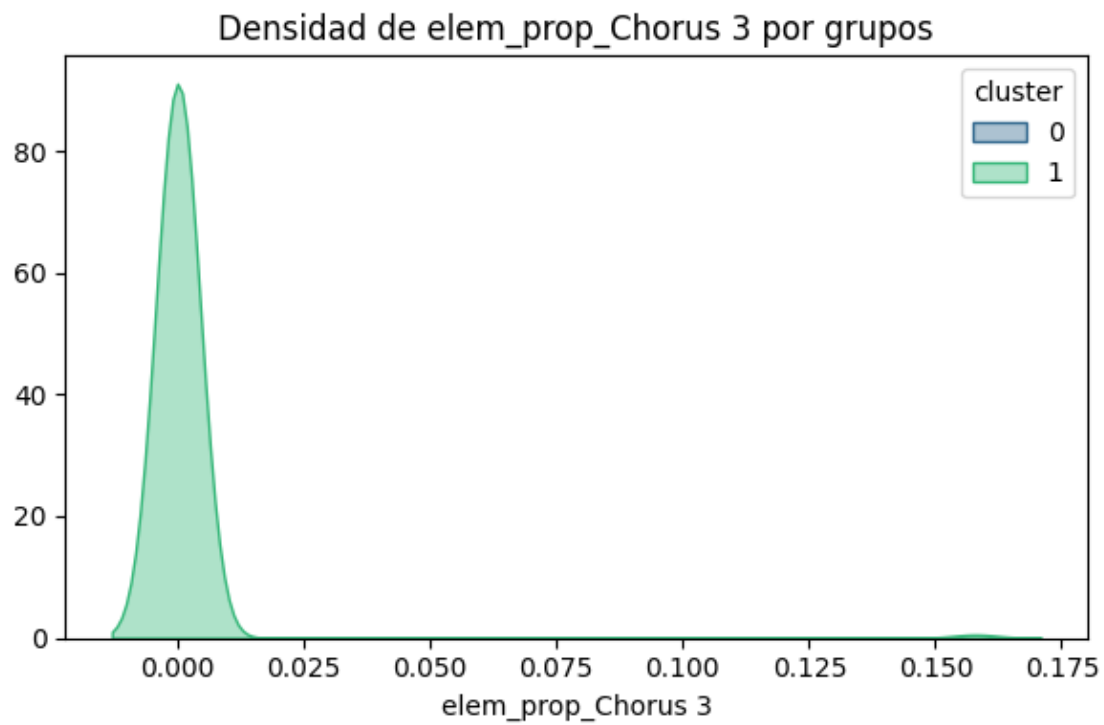
```
/tmp/ipykernel_3403/2607923038.py:15: UserWarning:
```

```
Dataset has 0 variance; skipping density estimate. Pass `warn_singular=False` to  
disable this warning.
```



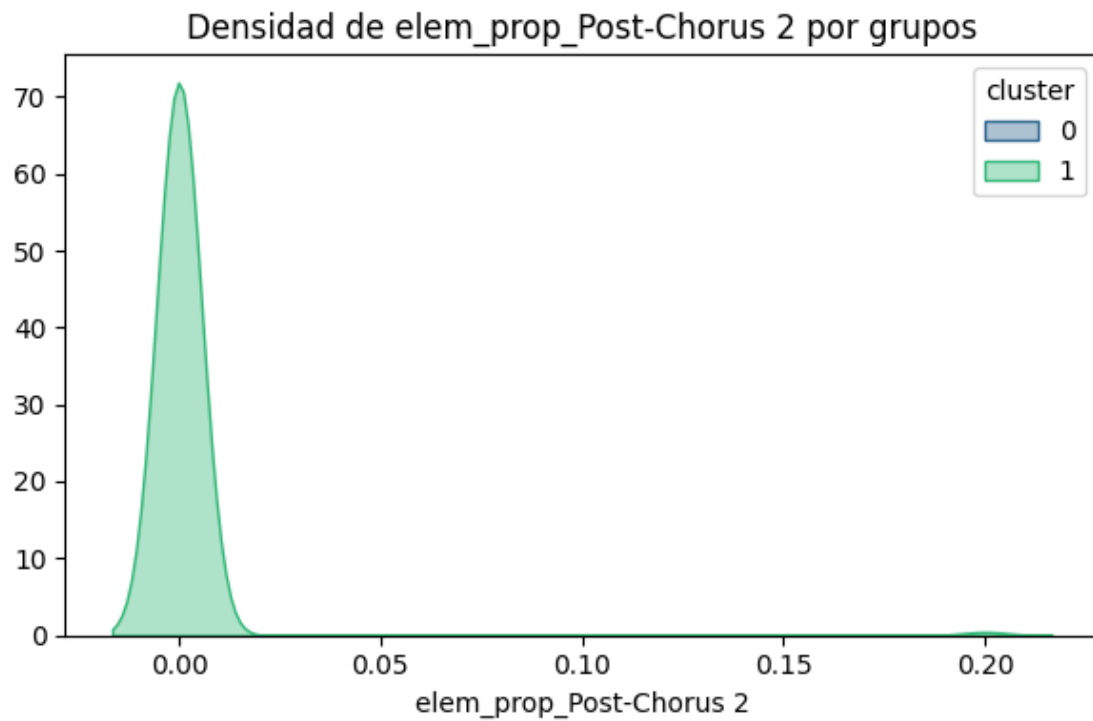
/tmp/ipykernel_3403/2607923038.py:15: UserWarning:

Dataset has 0 variance; skipping density estimate. Pass `warn_singular=False` to disable this warning.



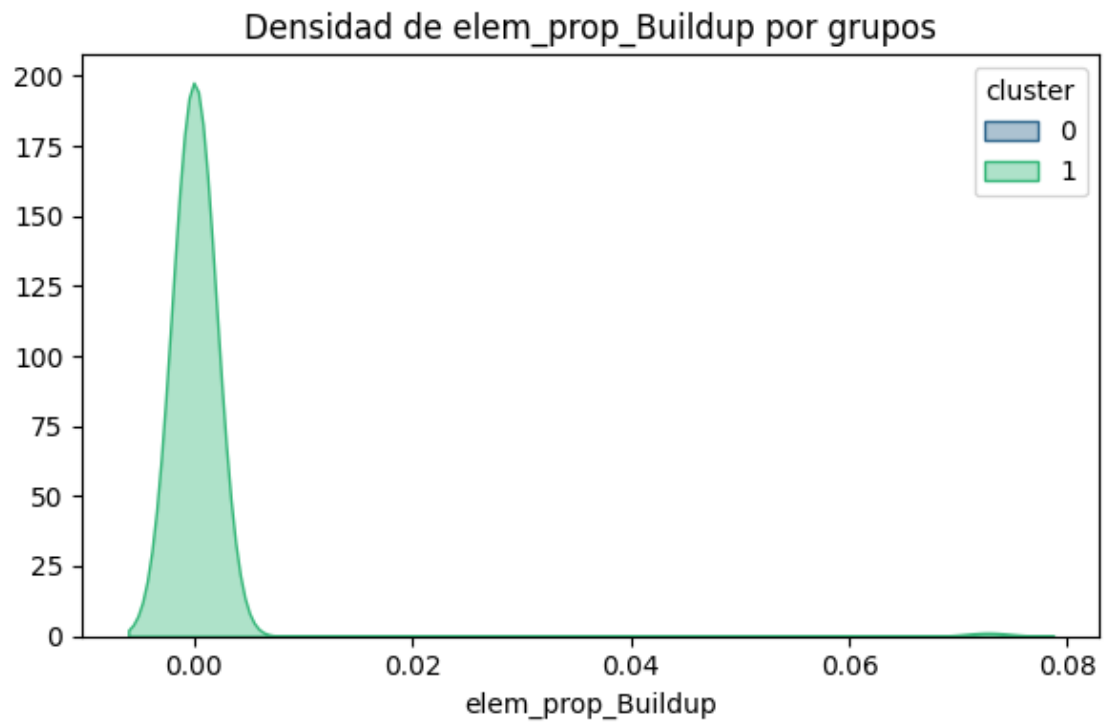
```
/tmp/ipykernel_3403/2607923038.py:15: UserWarning:
```

```
Dataset has 0 variance; skipping density estimate. Pass `warn_singular=False` to  
disable this warning.
```



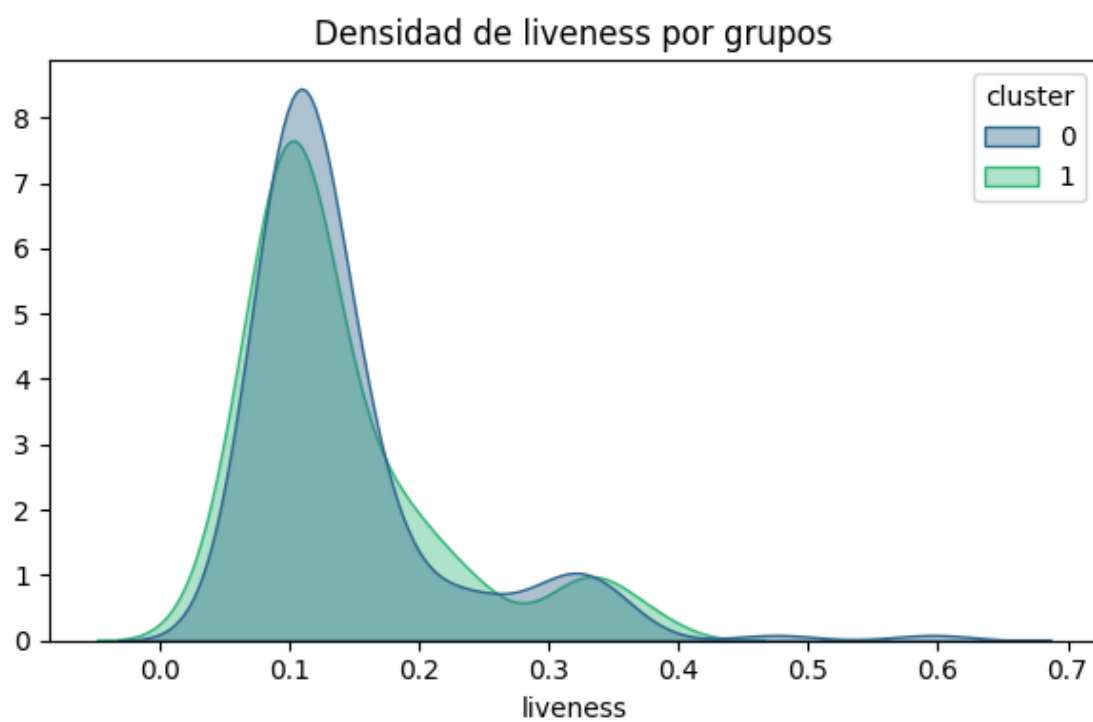
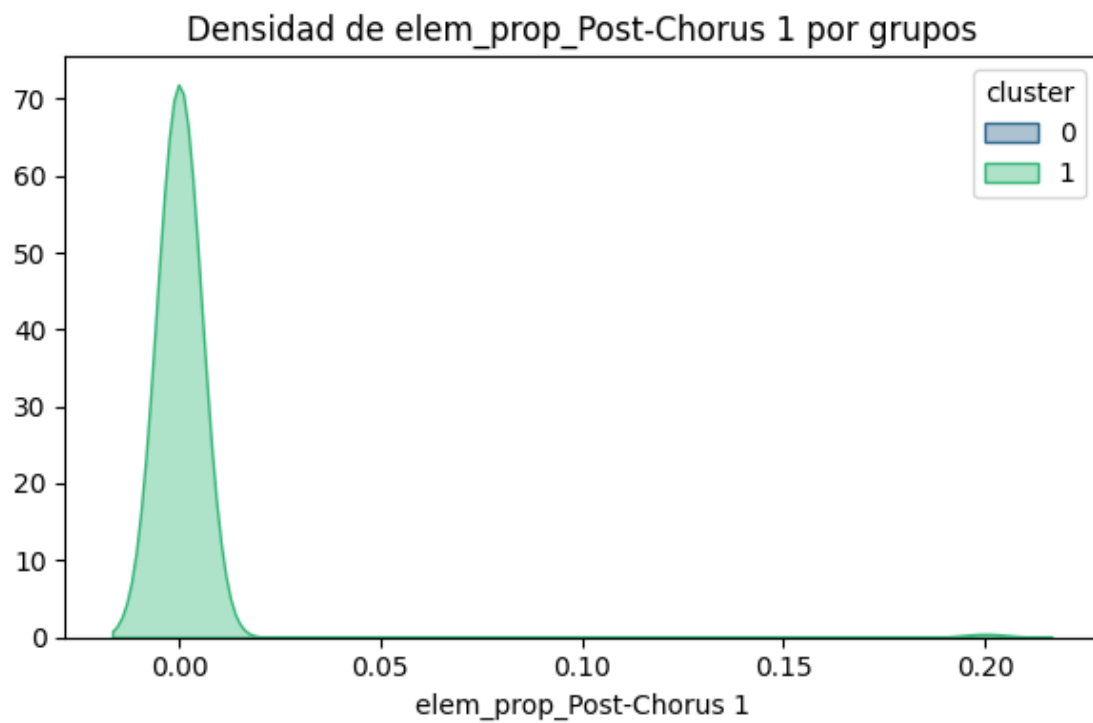
```
/tmp/ipykernel_3403/2607923038.py:15: UserWarning:
```

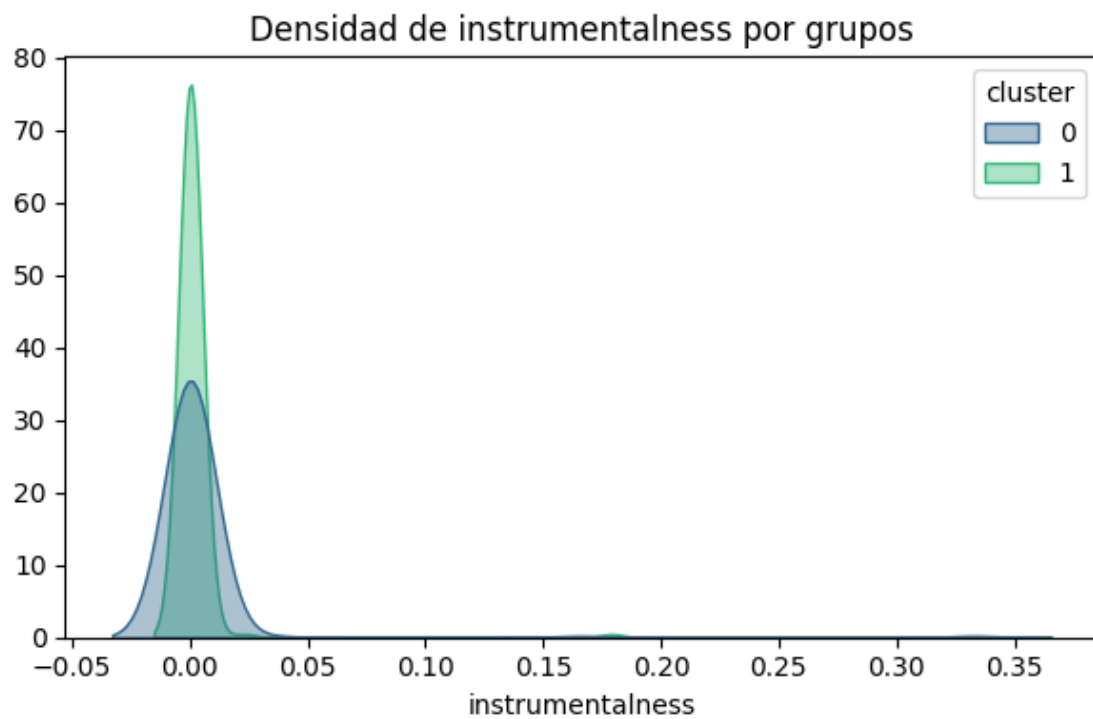
```
Dataset has 0 variance; skipping density estimate. Pass `warn_singular=False` to  
disable this warning.
```



/tmp/ipykernel_3403/2607923038.py:15: UserWarning:

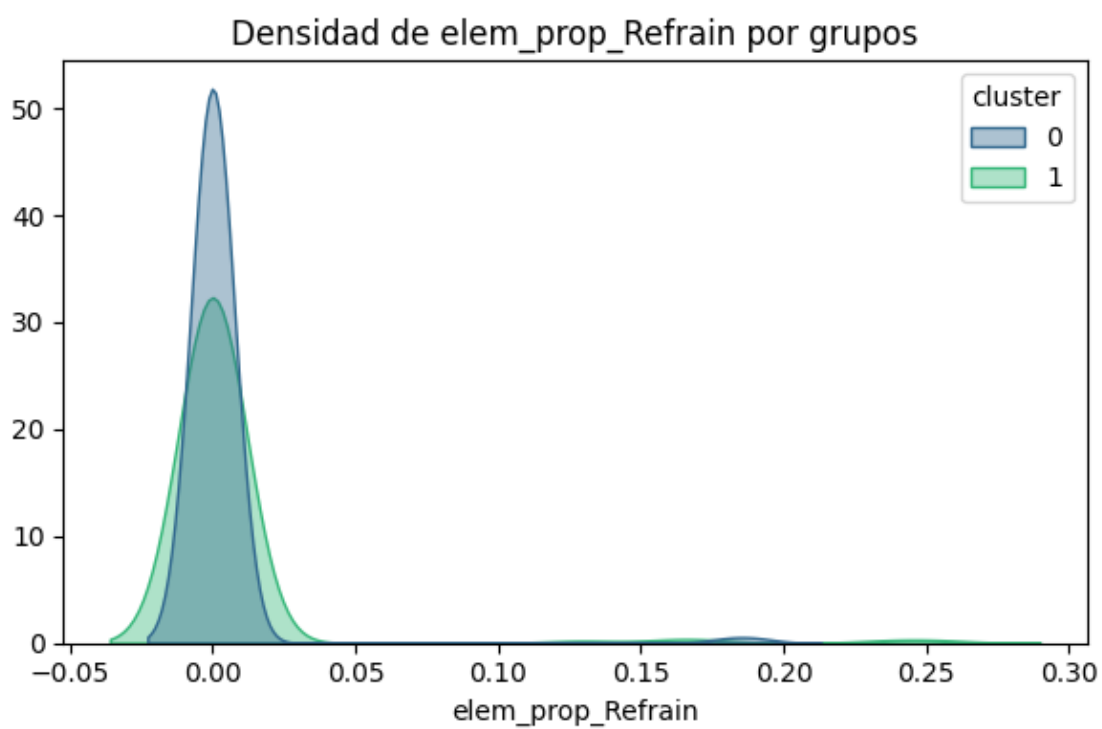
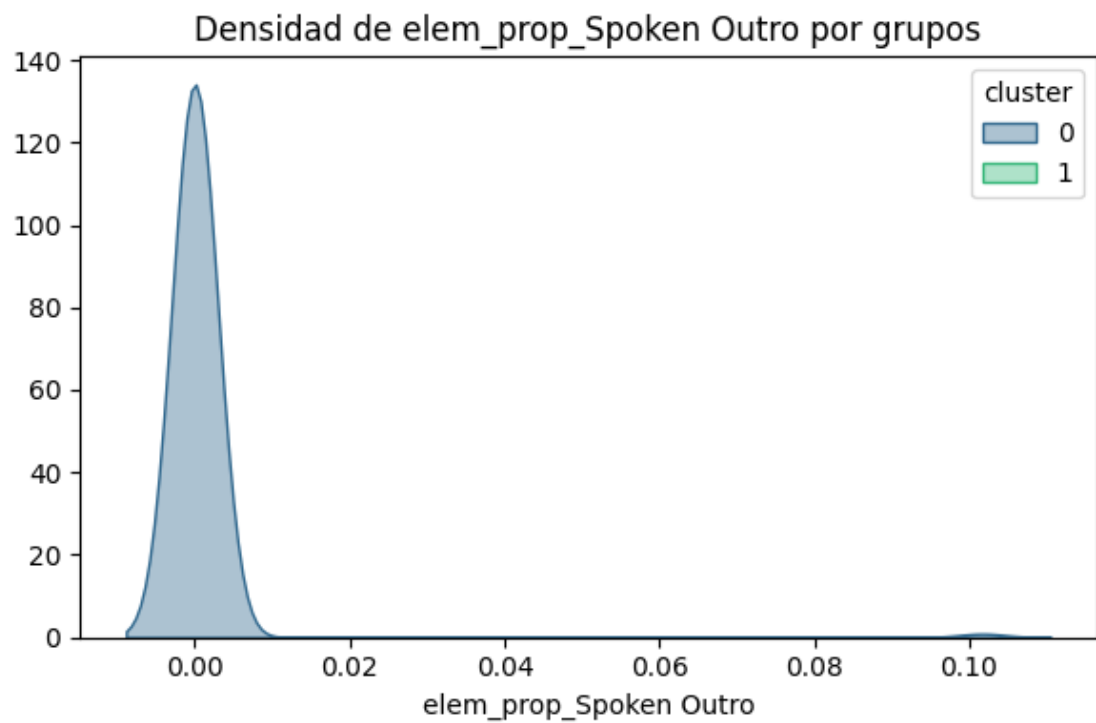
Dataset has 0 variance; skipping density estimate. Pass `warn\_singular=False` to disable this warning.

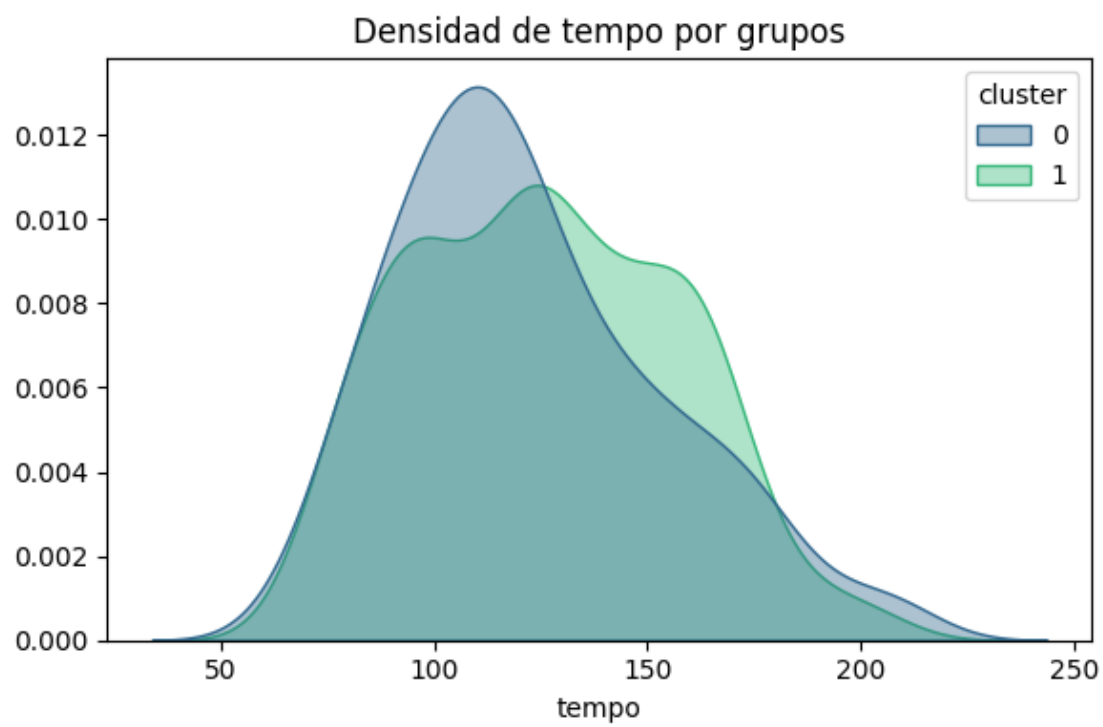
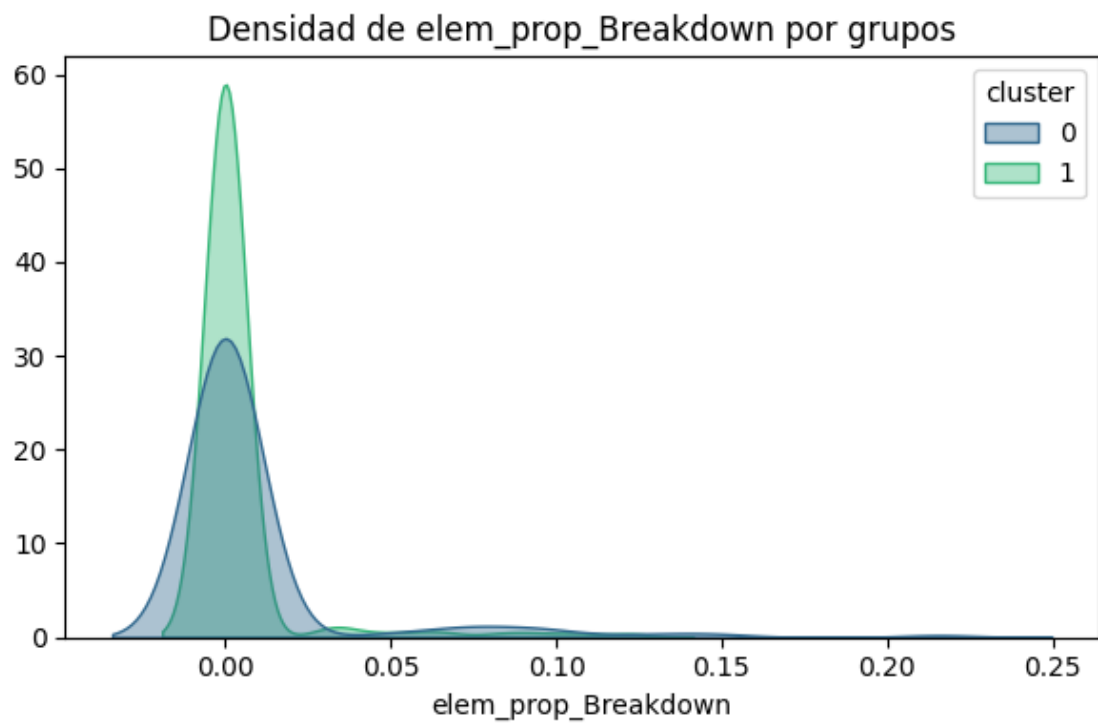


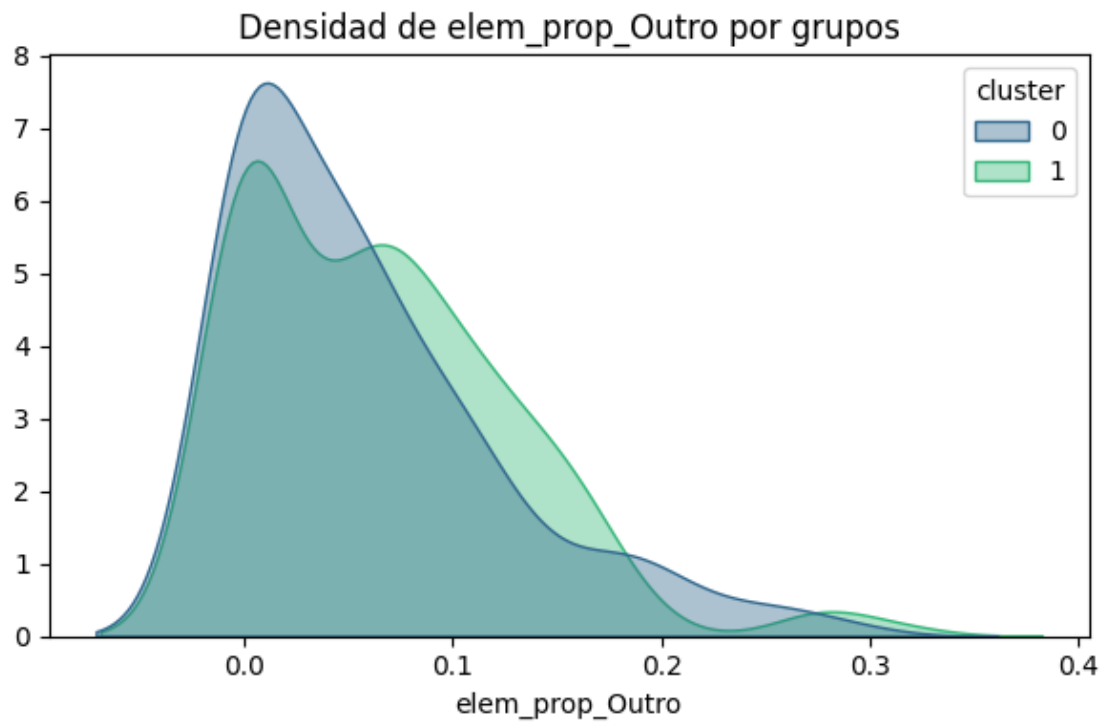


```
/tmp/ipykernel_3403/2607923038.py:15: UserWarning:
```

```
Dataset has 0 variance; skipping density estimate. Pass `warn_singular=False` to  
disable this warning.
```

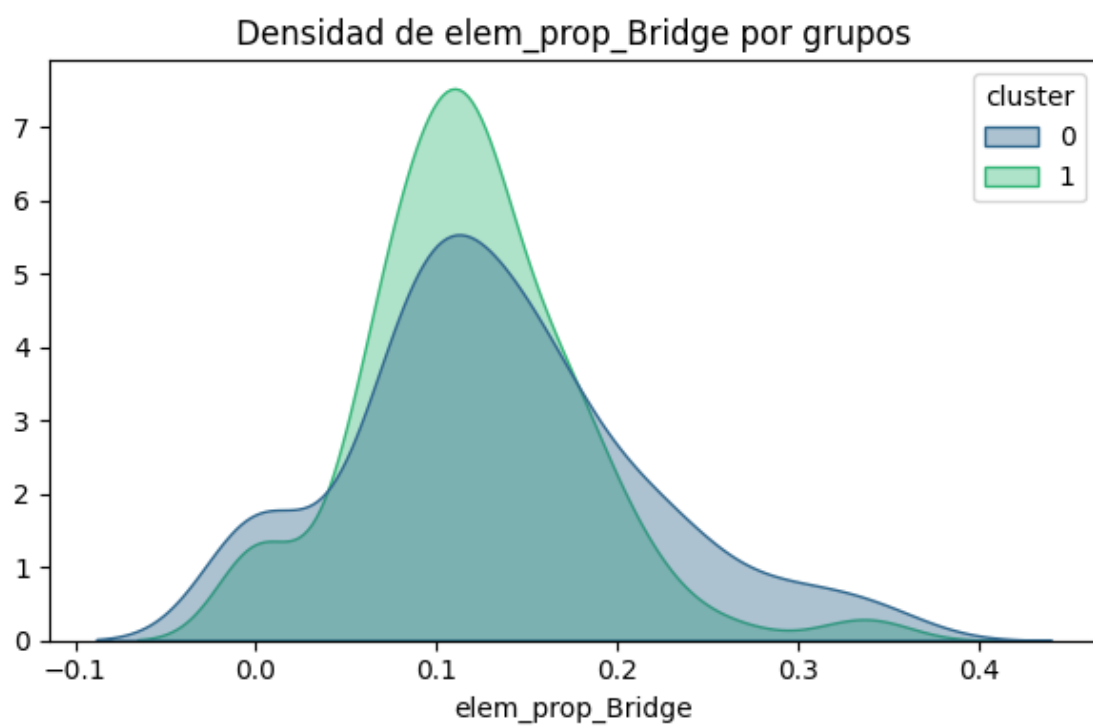
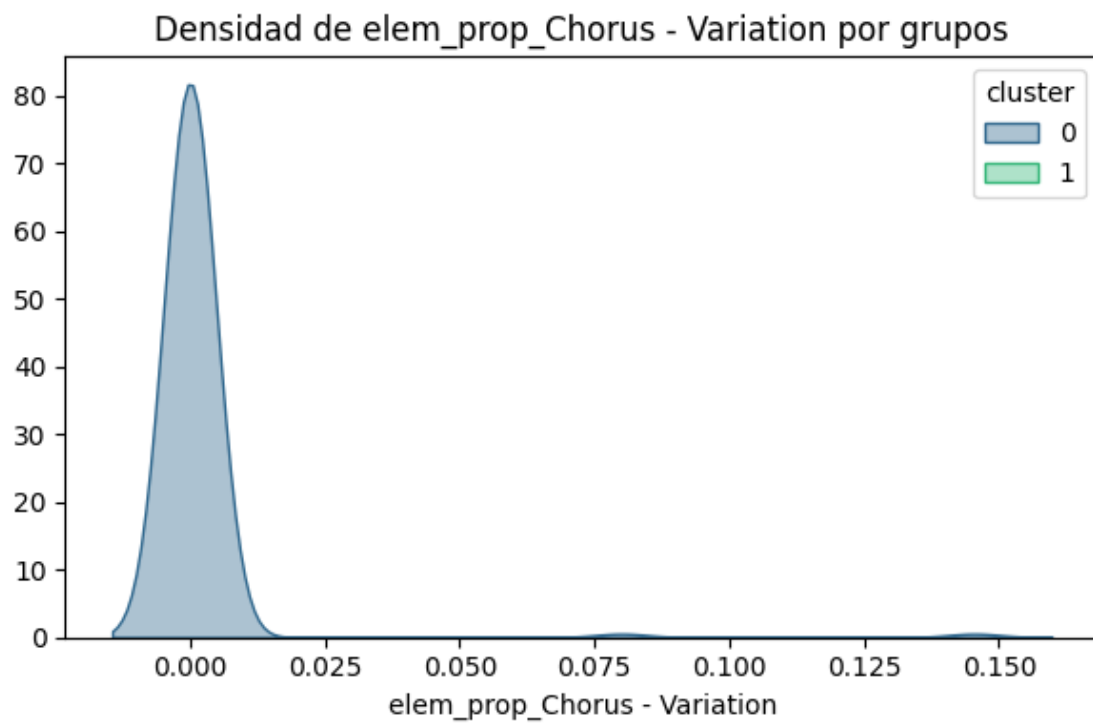



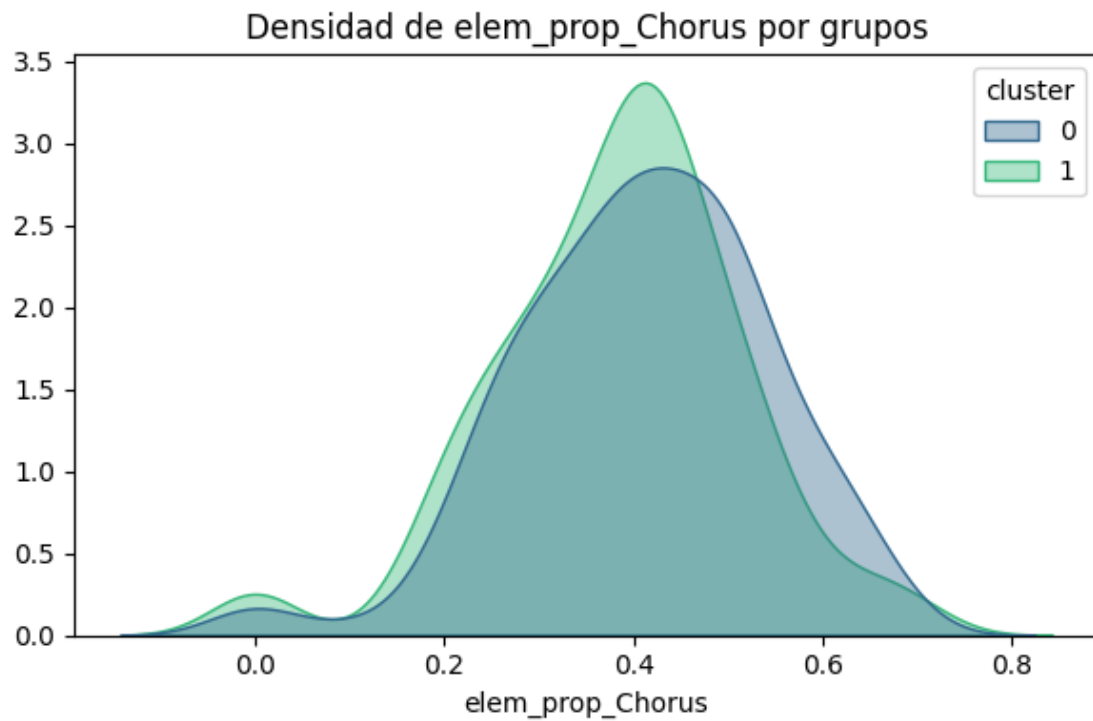




/tmp/ipykernel_3403/2607923038.py:15: UserWarning:

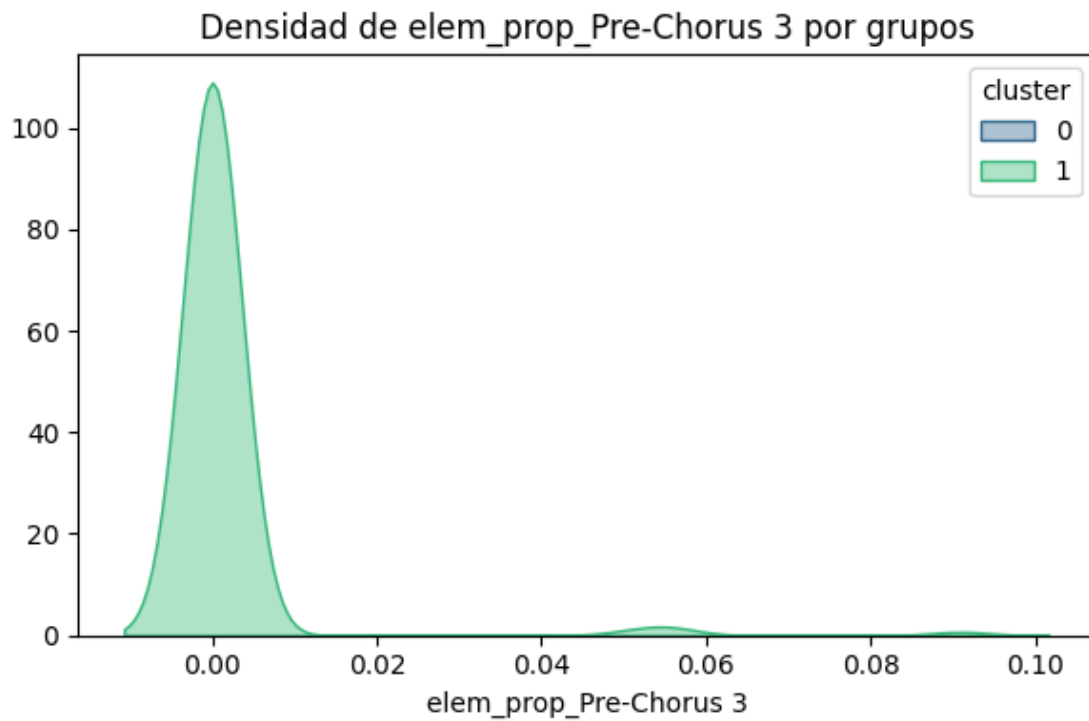
Dataset has 0 variance; skipping density estimate. Pass `warn\_singular=False` to disable this warning.





```
/tmp/ipykernel_3403/2607923038.py:15: UserWarning:
```

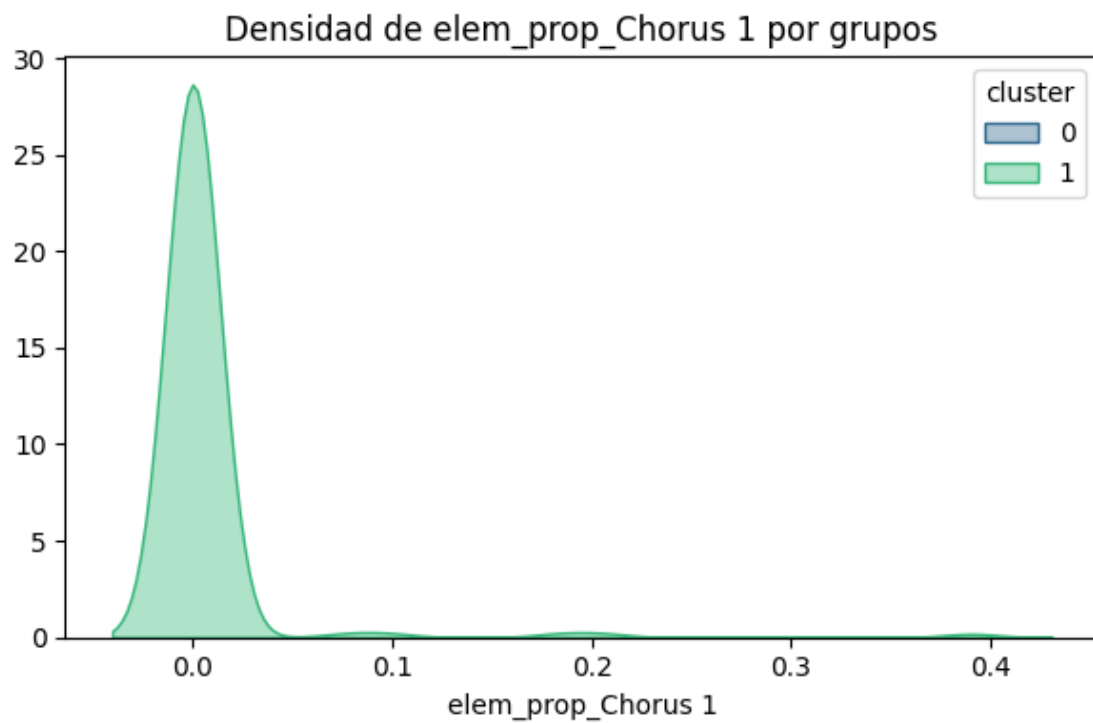
```
Dataset has 0 variance; skipping density estimate. Pass `warn_singular=False` to  
disable this warning.
```



```
[120]: # Distribución de variables significativas
perfil2.plot_dist(var_sig2)
```

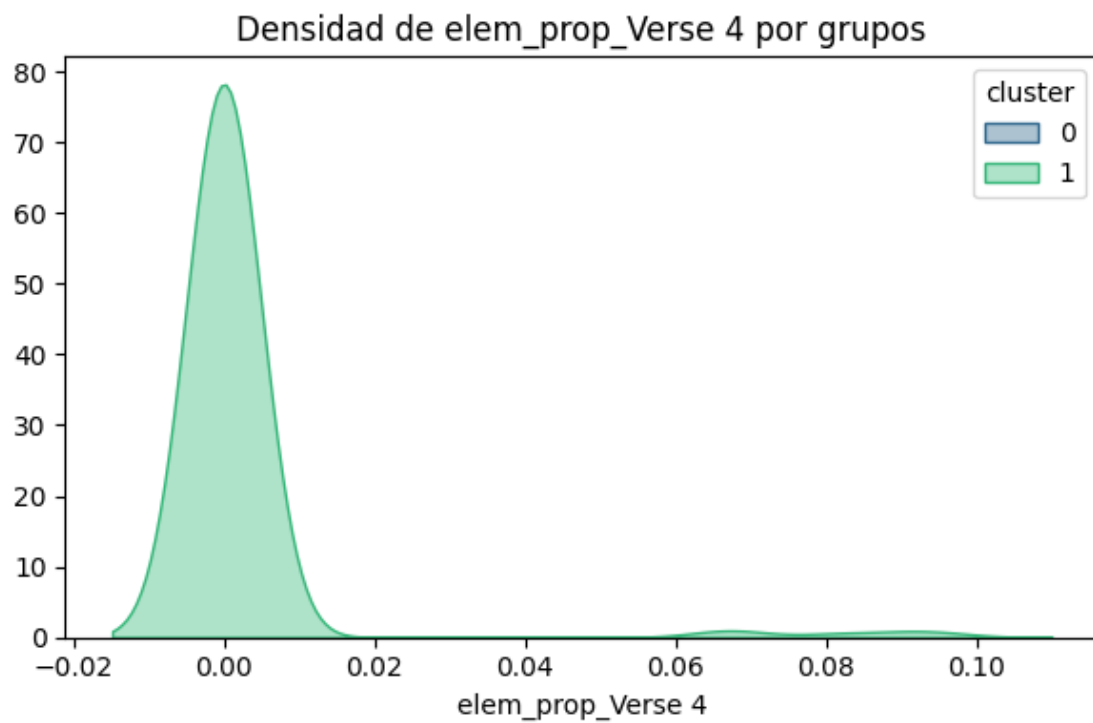
/tmp/ipykernel_3403/2607923038.py:15: UserWarning:

Dataset has 0 variance; skipping density estimate. Pass `warn\_singular=False` to disable this warning.



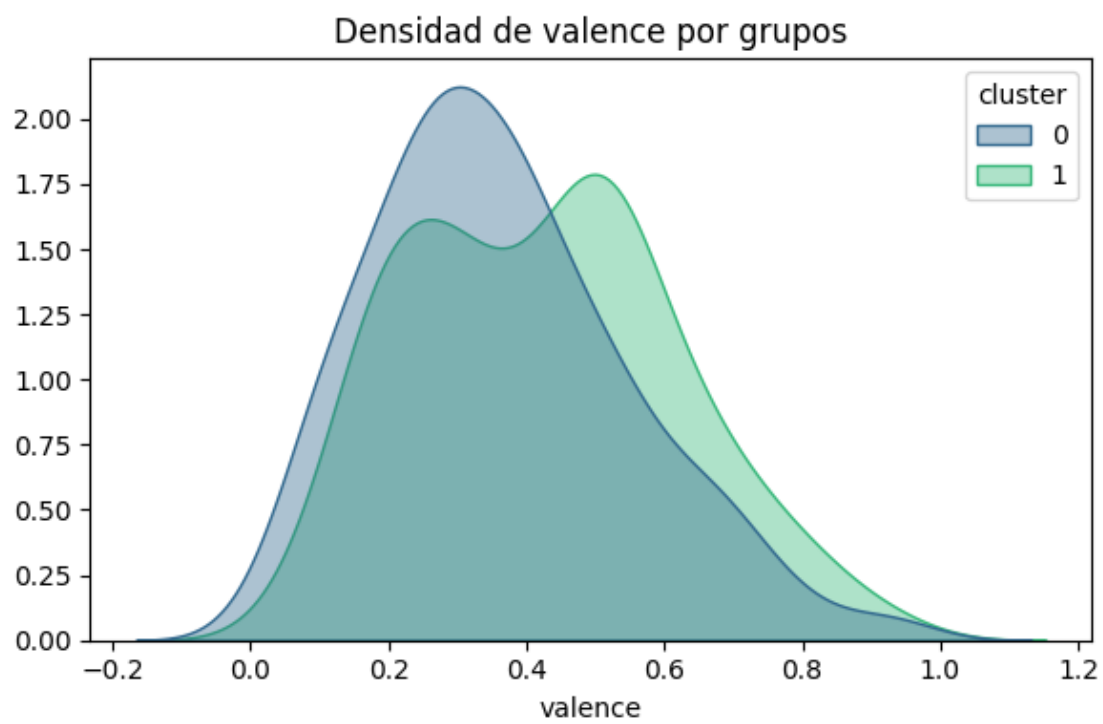
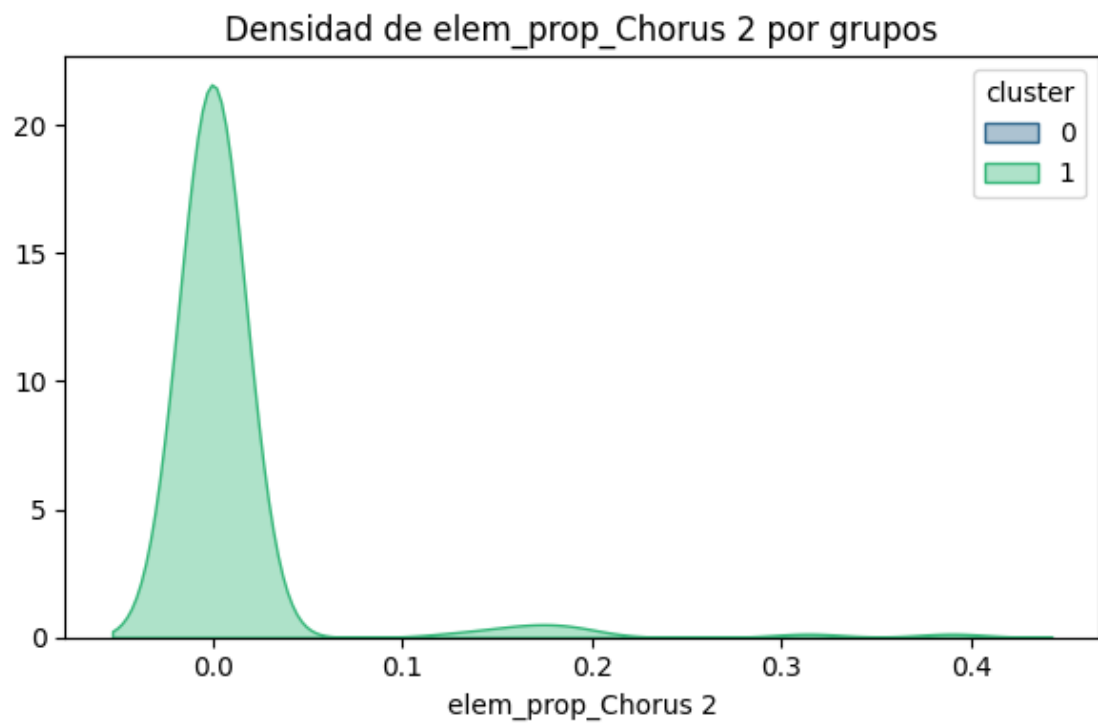
```
/tmp/ipykernel_3403/2607923038.py:15: UserWarning:
```

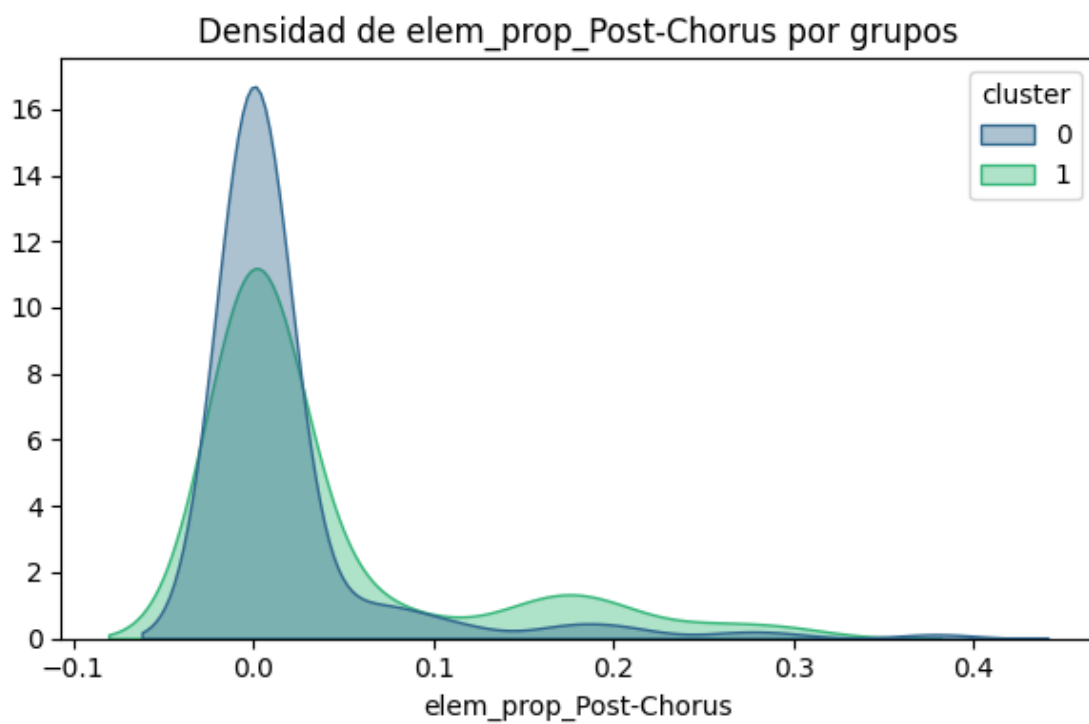
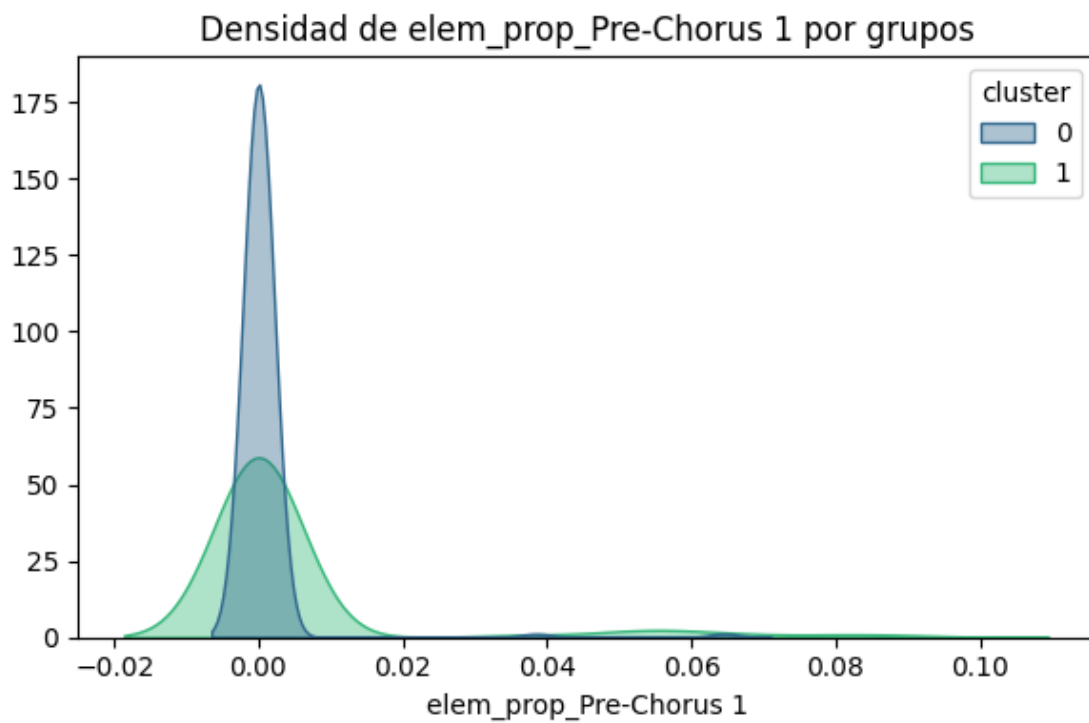
```
Dataset has 0 variance; skipping density estimate. Pass `warn_singular=False` to  
disable this warning.
```

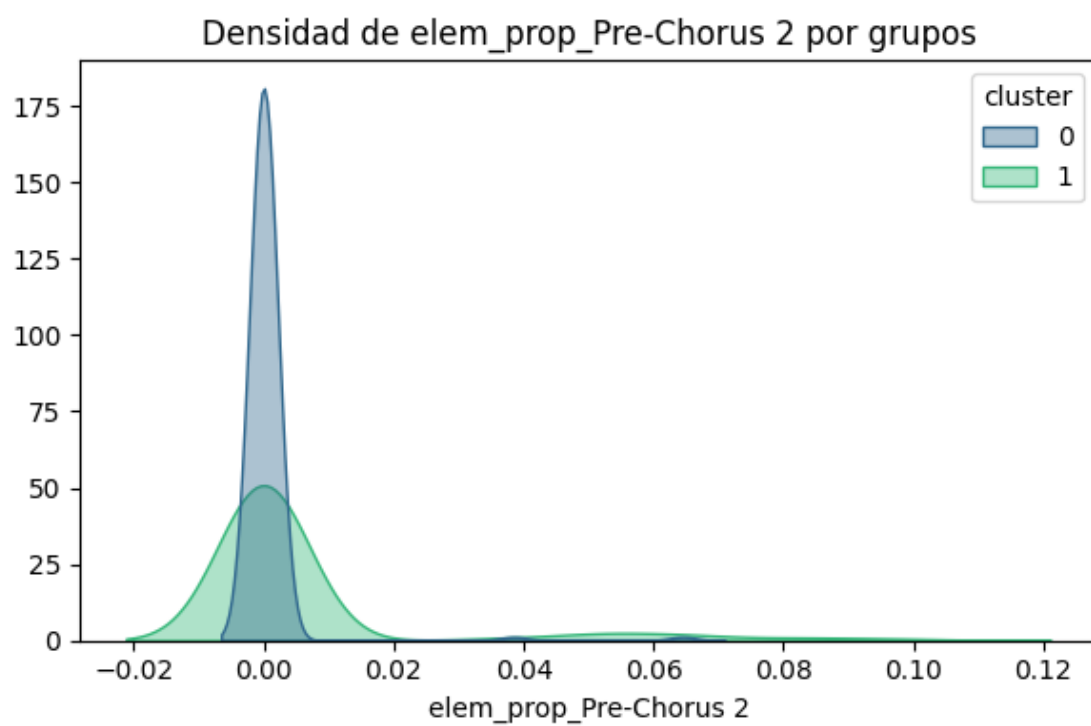
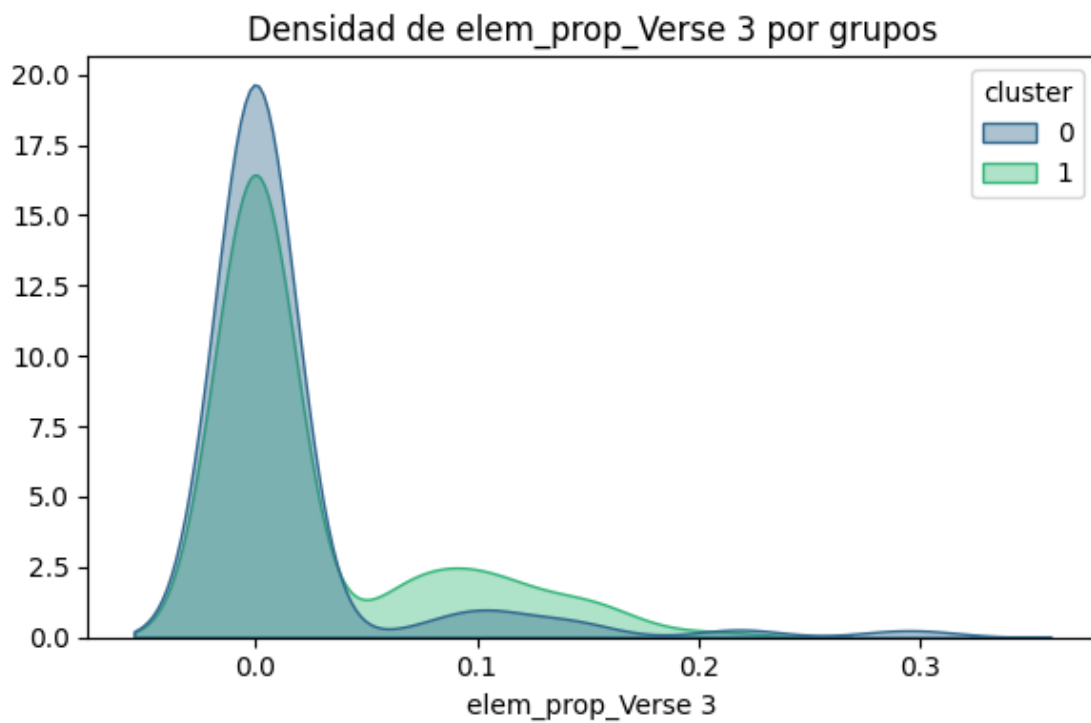


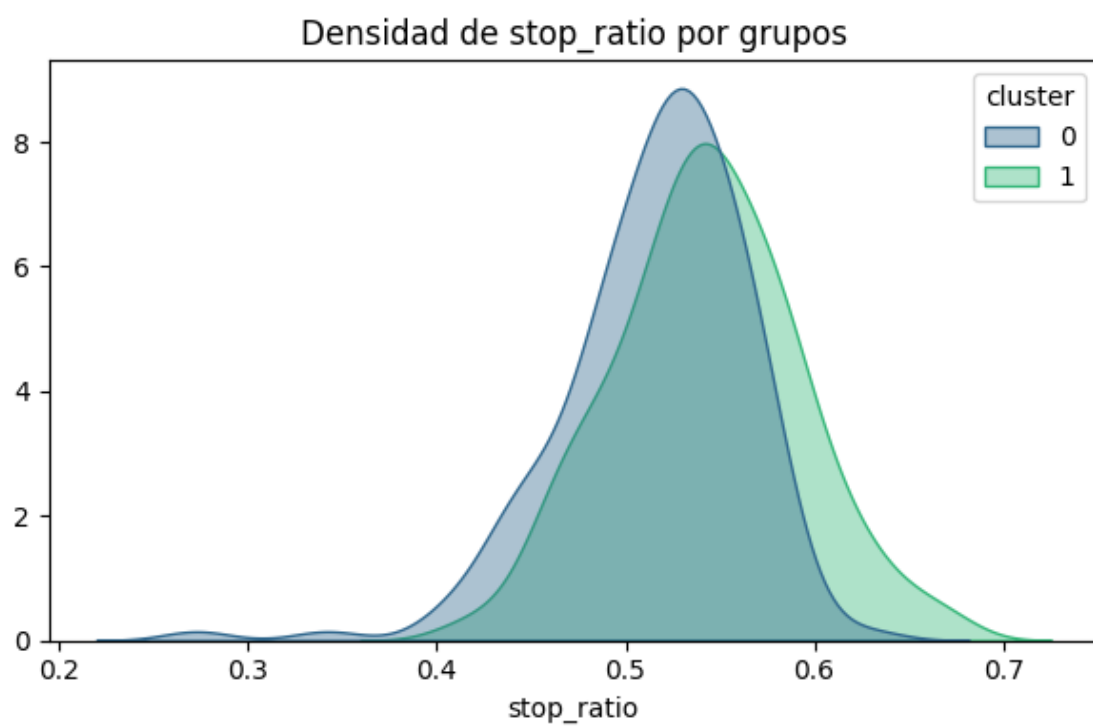
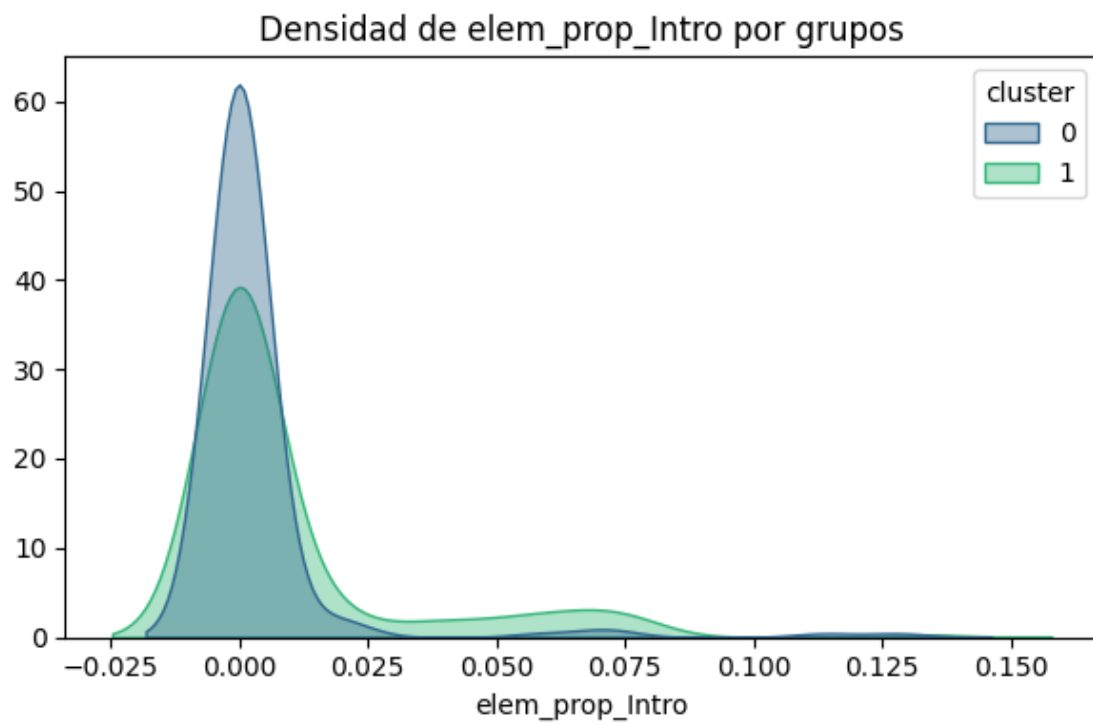
/tmp/ipykernel_3403/2607923038.py:15: UserWarning:

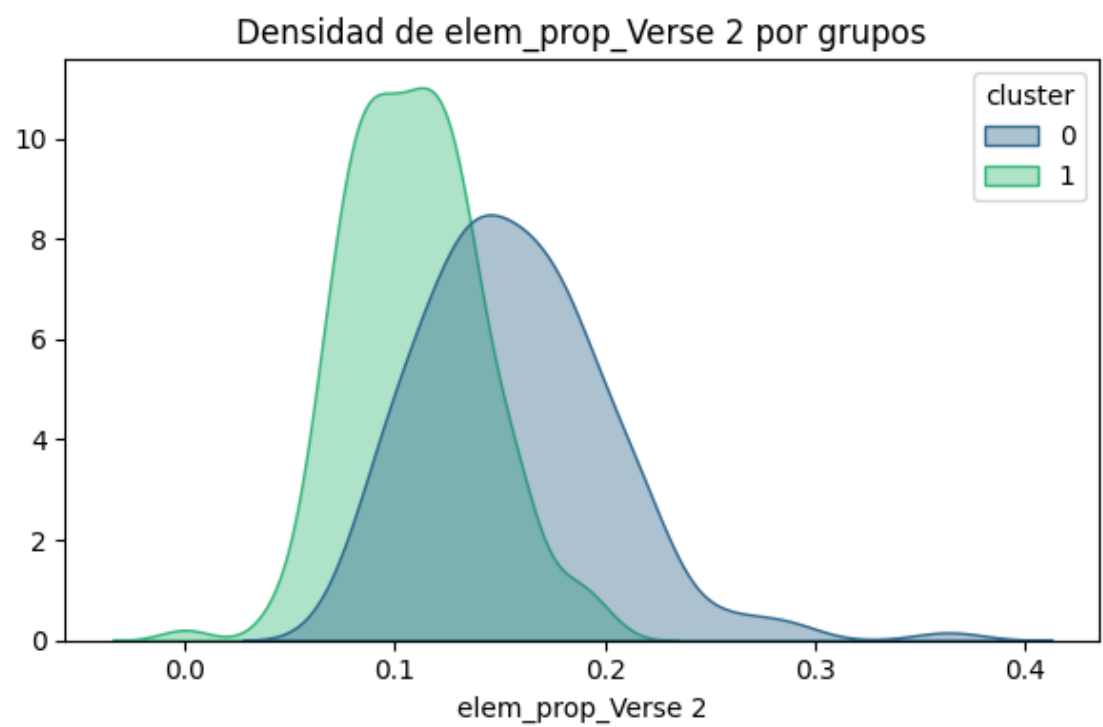
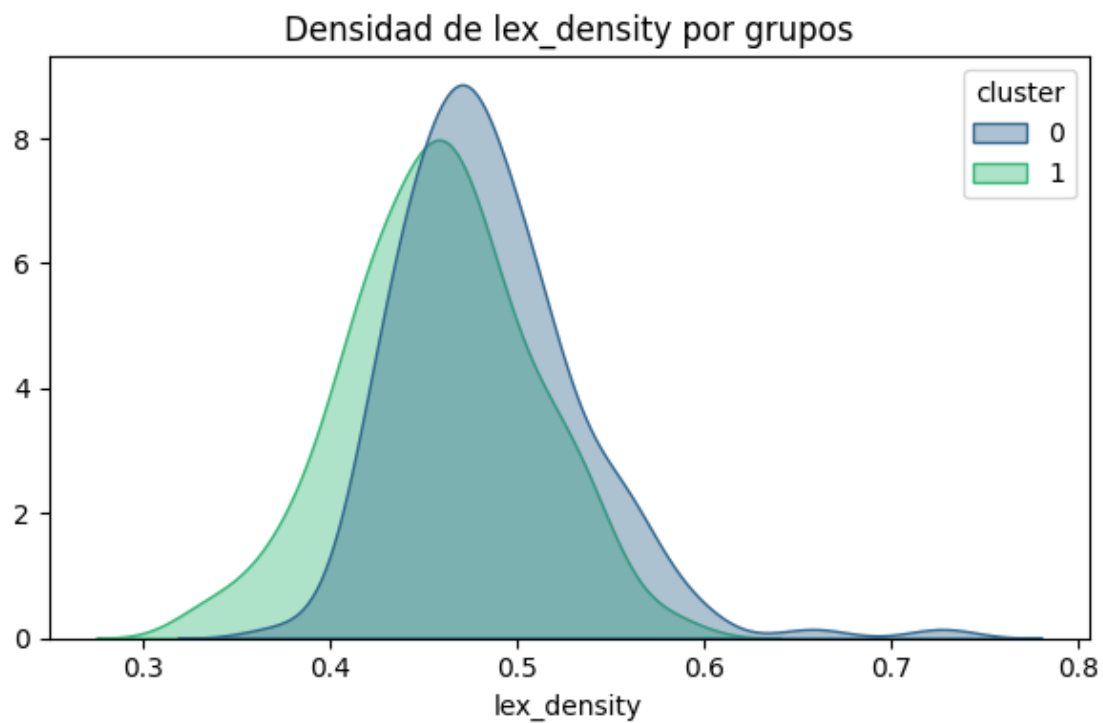
Dataset has 0 variance; skipping density estimate. Pass `warn_singular=False` to disable this warning.

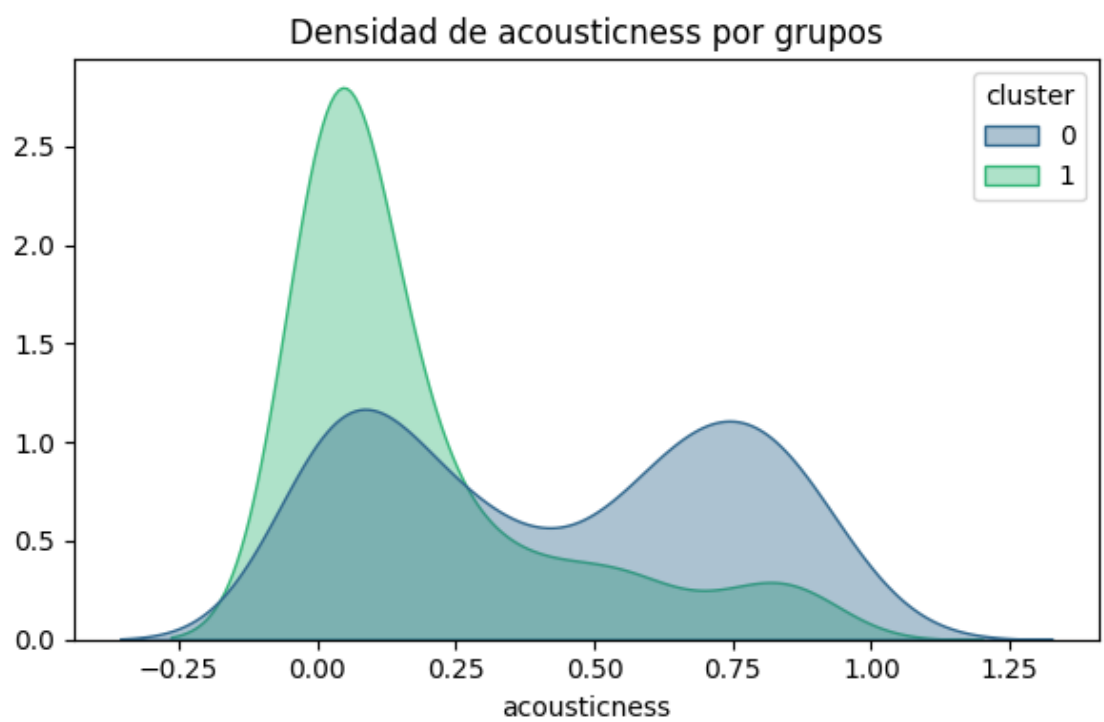
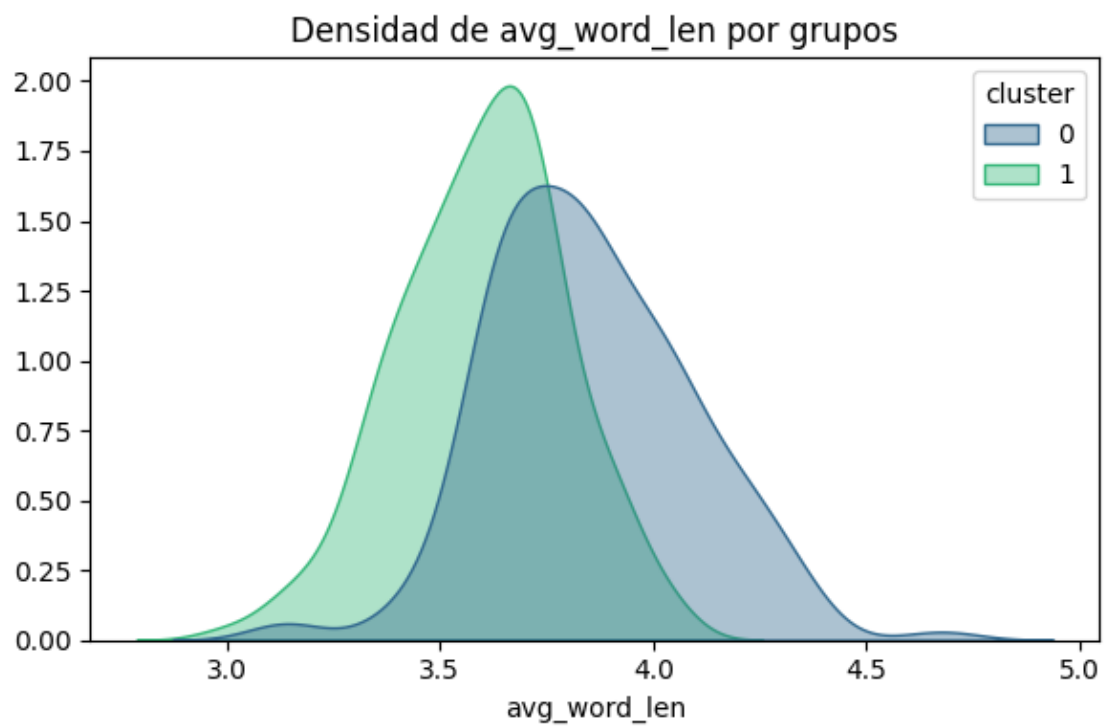


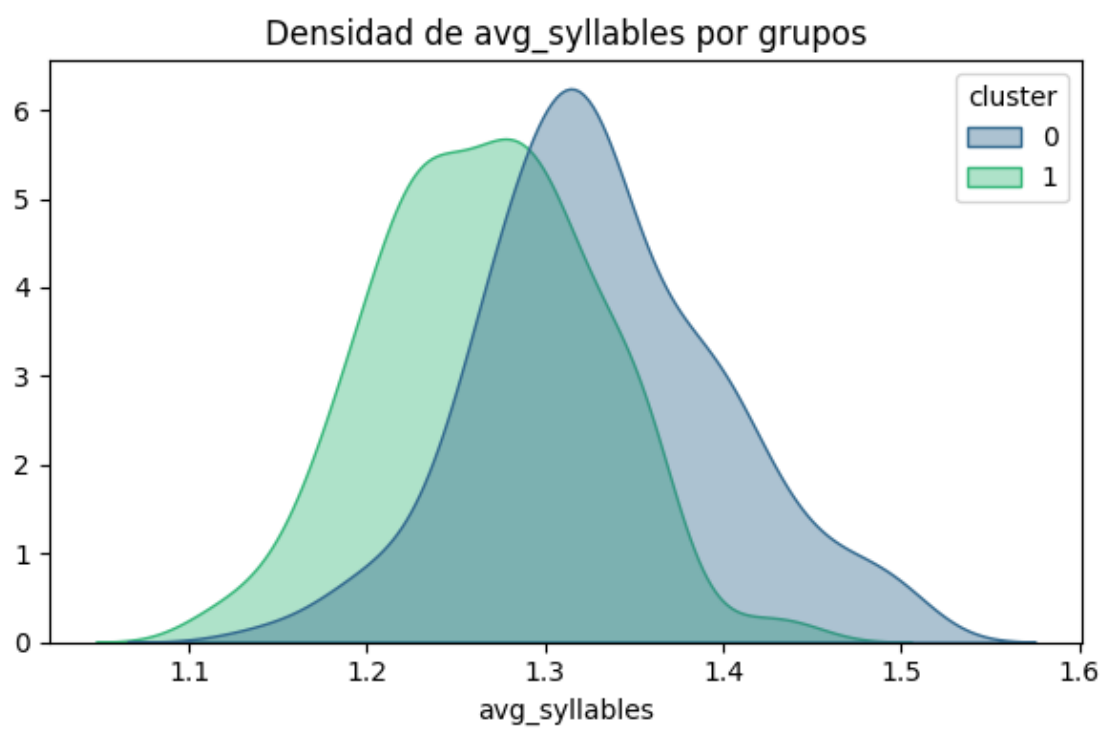
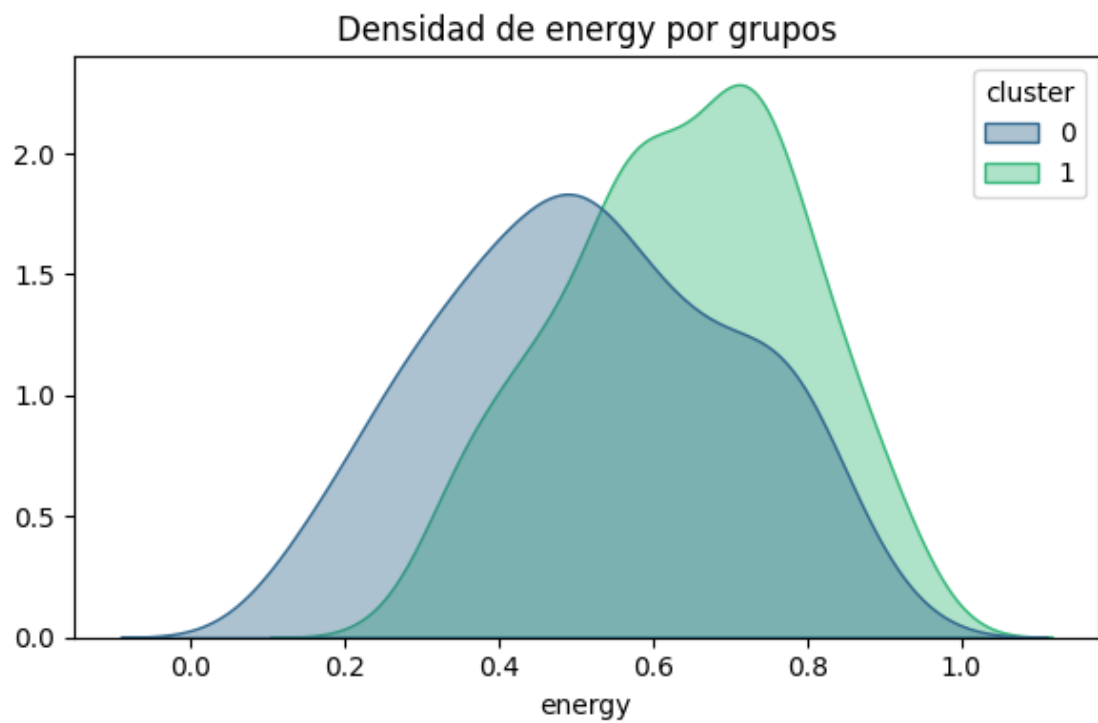


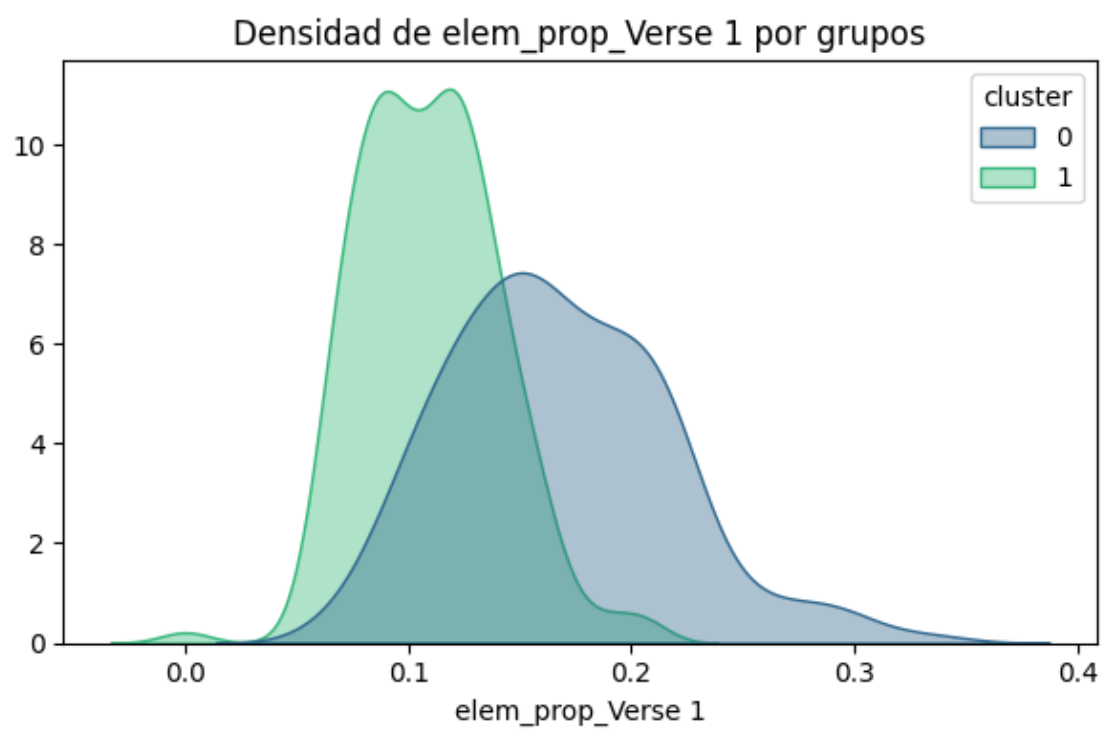
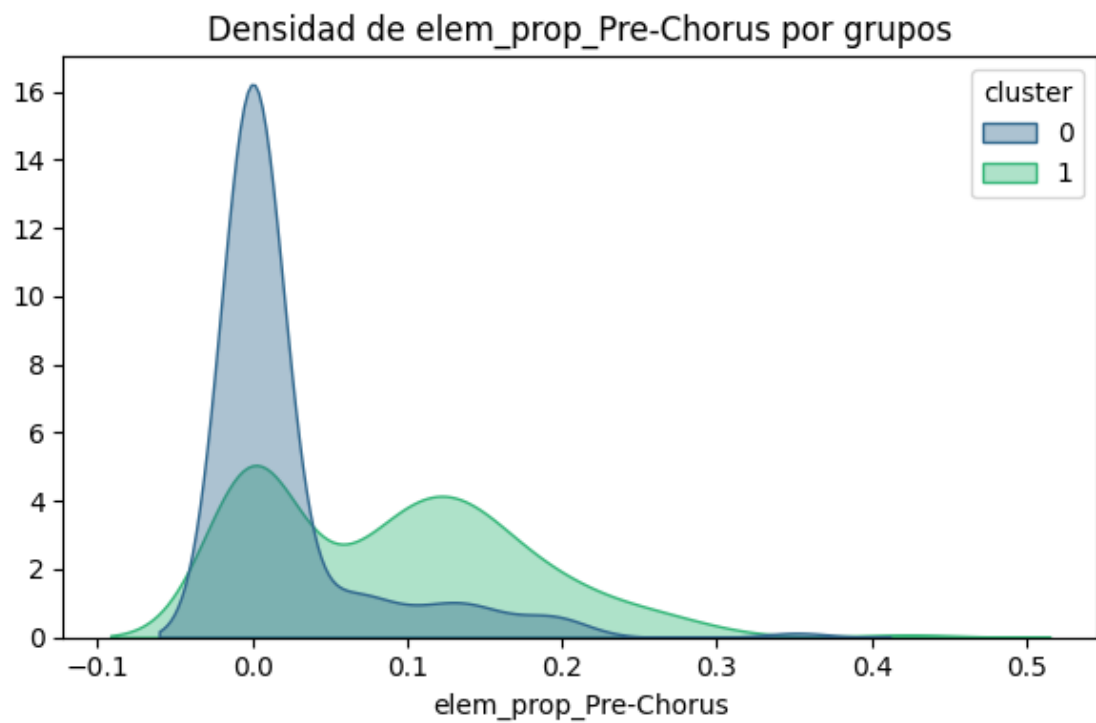


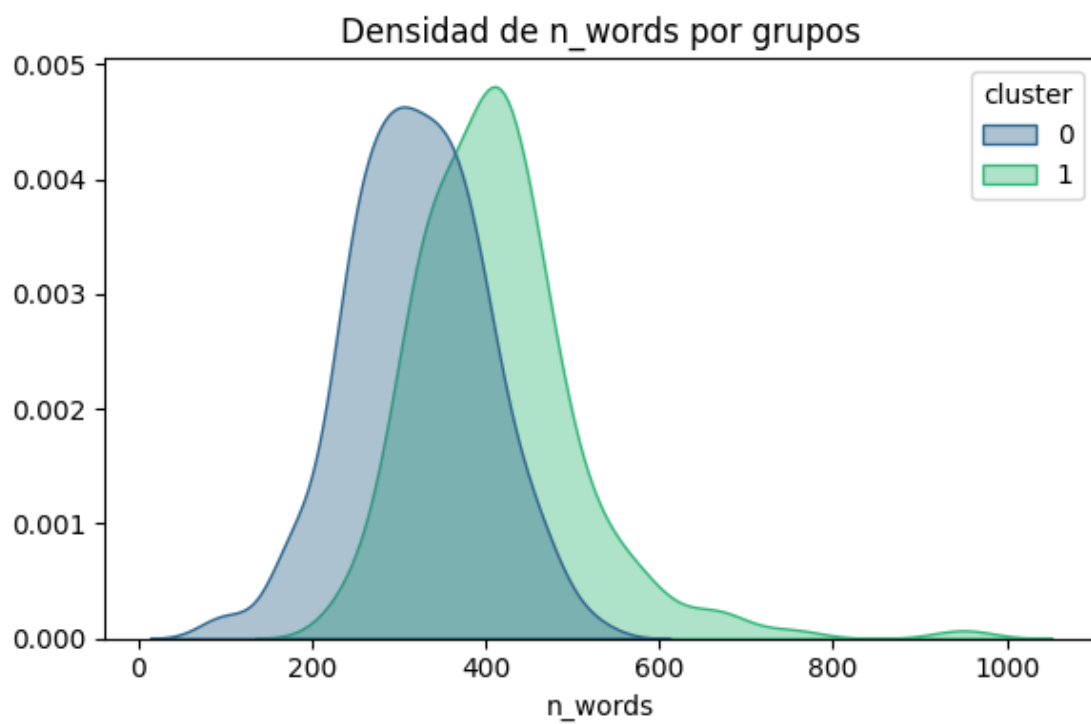
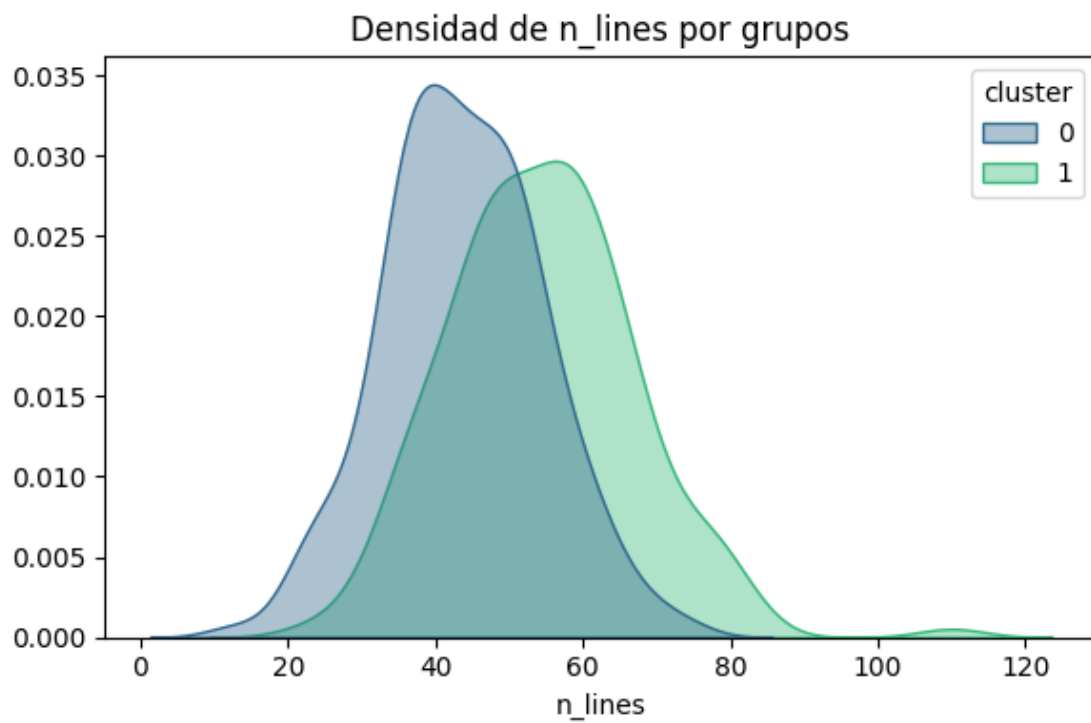












```
[121]: # Pipelines
num_pipe = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="median"))
])

# ColumnTransformer
pre = ColumnTransformer(
    transformers=[
        ("num", num_pipe, num_cols)
    ],
    remainder="drop",
    verbose_feature_names_out=False
)
```

```
[122]: values = pre.fit_transform(df_song)
imputed_df = pd.DataFrame(data=values, columns=pre.get_feature_names_out())
```

```
[123]: imputed_df['cluster'] = df_song['cluster']
```

```
[124]: clase2 = playlist(
    df=imputed_df,
    ↵
    ↪selection=['danceability', 'duration_min', 'liveness', 'tempo', 'n_words', 'n_lines', 'elem_prop_Pr
    ↪1', 'acousticness', 'energy', 'valence', 'elem_prop_Verse 1'],
    sig_var=var_sig2,
    other_var=['danceability', 'duration_ms', 'liveness', 'tempo']
    #group_var=['cluster']
)
```

```
[125]: clase2.resumen(agg='mean')
```

```
[125]:
```

	cluster	count	count	valence	elem_prop_Bridge 2	elem_prop_Verse \
0	0	157	120.579618	0.362562	0.000000	0.000000
1	1	169	124.284024	0.424459	0.001085	0.001479

	elem_prop_Bridge 1	elem_prop_Chorus 3	elem_prop_Interlude	lex_density \
0	0.000000	0.000000	0.000607	0.484839
1	0.000789	0.000934	0.000972	0.457677

	...	elem_prop_Outro	acousticness	elem_prop_Chorus	duration_ms \
0	...	0.058111	0.42572	0.410044	236252.089172
1	...	0.065406	0.18429	0.383500	243218.349112

	elem_prop_Buildup	elem_prop_Pre-Chorus 2	danceability	n_lines \
0	0.00000	0.000656	0.580478	43.713376
1	0.00043	0.006299	0.588734	54.461538

	elem_prop_Post-Chorus 2	elem_prop_Chorus - Variation
0	0.000000	0.001436
1	0.001183	0.000000

[2 rows x 46 columns]

```
[126]: clase2.select(0)
```

```
[126]:
```

	danceability	duration_min	liveness	tempo	n_words	n_lines	\
0	0.789	3 min and 32 s	0.3020	116.992	348.0	43.0	
2	0.588	3 min and 51 s	0.1180	94.933	383.0	47.0	
12	0.815	4 min and 31 s	0.1000	103.970	340.0	51.0	
14	0.474	4 min and 27 s	0.0903	170.109	409.0	52.0	
16	0.760	3 min and 32 s	0.3250	116.989	348.0	39.0	
..	...	...	...	...	...	...	
306	0.317	4 min and 49 s	0.0867	93.933	185.0	23.0	
308	0.644	3 min and 54 s	0.0909	150.072	287.0	32.0	
309	0.676	3 min and 40 s	0.1350	118.840	245.0	37.0	
310	0.313	3 min and 31 s	0.0948	179.947	235.0	27.0	
325	0.661	3 min and 55 s	0.1300	94.922	377.0	43.0	

	elem_prop_Pre-Chorus 1	acousticness	energy	valence	elem_prop_Verse 1
0	0.000000	0.03480	0.634	0.658	0.113636
2	0.000000	0.00245	0.791	0.487	0.208333
12	0.038462	0.23200	0.377	0.211	0.096154
14	0.000000	0.70700	0.480	0.319	0.207547
16	0.000000	0.00661	0.602	0.685	0.125000
..	...	...	...	...	...
306	0.000000	0.72700	0.254	0.103	0.208333
308	0.000000	0.91600	0.284	0.328	0.212121
309	0.000000	0.96400	0.178	0.404	0.210526
310	0.000000	0.84200	0.258	0.268	0.107143
325	0.000000	0.92100	0.151	0.230	0.090909

[157 rows x 11 columns]

```
[127]: clase2.select(0).describe()
```

```
[127]:
```

	danceability	liveness	tempo	n_words	n_lines	\
count	157.000000	157.000000	157.000000	157.000000	157.000000	
mean	0.580478	0.144713	122.411070	319.949045	43.713376	
std	0.114795	0.082415	31.295243	77.654874	10.714267	
min	0.310000	0.059300	68.534000	98.000000	13.000000	
25%	0.511000	0.101000	99.981000	264.000000	36.000000	
50%	0.596000	0.114000	118.043000	315.000000	44.000000	
75%	0.655000	0.148000	142.377000	375.000000	51.000000	
max	0.872000	0.596000	208.918000	527.000000	74.000000	

	elem_prop_Pre-Chorus 1	acousticness	energy	valence \
count	157.000000	157.000000	157.000000	157.000000
mean	0.000656	0.425720	0.514529	0.362562
std	0.005978	0.323325	0.188625	0.183718
min	0.000000	0.000191	0.118000	0.038400
25 %	0.000000	0.103000	0.379000	0.232000
50 %	0.000000	0.420000	0.509000	0.347000
75 %	0.000000	0.727000	0.646000	0.472000
max	0.064516	0.971000	0.902000	0.928000

	elem_prop_Verse 1
count	157.000000
mean	0.167394
std	0.049691
min	0.067797
25 %	0.129032
50 %	0.160000
75 %	0.200000
max	0.333333

```
[128]: clase2.select(1)
```

```
[128]:
```

	danceability	duration_min	liveness	tempo	n_words	n_lines \
1	0.760	3 min and 51 s	0.0913	95.997	509.0	83.0
3	0.553	3 min and 55 s	0.3410	92.008	662.0	69.0
4	0.605	3 min and 13 s	0.1010	96.970	467.0	66.0
5	0.647	3 min and 39 s	0.3340	160.078	617.0	76.0
6	0.653	3 min and 27 s	0.1020	118.035	590.0	76.0
..	...	...	...	...	...	...
320	0.675	3 min and 34 s	0.0391	110.010	478.0	58.0
321	0.624	3 min and 31 s	0.1380	160.024	433.0	65.0
322	0.719	3 min and 50 s	0.1690	120.085	496.0	66.0
323	0.567	3 min and 27 s	0.3820	163.960	418.0	51.0
324	0.598	3 min and 23 s	0.3400	163.954	509.0	72.0

	elem_prop_Pre-Chorus 1	acousticness	energy	valence	elem_prop_Verse 1
1	0.0	0.103000	0.703	0.5700	0.095238
3	0.0	0.000743	0.841	0.3380	0.128571
4	0.0	0.002010	0.725	0.5390	0.119403
5	0.0	0.064700	0.800	0.9420	0.103896
6	0.0	0.015800	0.893	0.5130	0.064935
..	...	...	...	...	...
320	0.0	0.008820	0.703	0.3140	0.067797
321	0.0	0.060400	0.691	0.2840	0.121212
322	0.0	0.032900	0.469	0.0851	0.059701
323	0.0	0.015600	0.789	0.4380	0.115385

```
324          0.0      0.186000  0.504  0.2520          0.095890
```

```
[169 rows x 11 columns]
```

```
[129]: clase2.select(1).describe()
```

```
[129]:
```

	danceability	liveness	tempo	n_words	n_lines \
count	169.000000	169.000000	169.000000	169.000000	169.000000
mean	0.588734	0.139646	125.698497	412.769231	54.461538
std	0.106570	0.077114	30.290086	95.041031	12.635457
min	0.292000	0.035700	73.942000	237.000000	23.000000
25%	0.521000	0.092000	99.979000	345.000000	46.000000
50%	0.596000	0.114000	123.054000	404.000000	55.000000
75%	0.647000	0.167000	148.035000	454.000000	62.000000
max	0.897000	0.382000	200.391000	950.000000	110.000000

	elem_prop_Pre-Chorus 1	acousticness	energy	valence \
count	169.000000	169.000000	169.000000	169.000000
mean	0.005017	0.184290	0.634408	0.424459
std	0.017328	0.241203	0.152922	0.192463
min	0.000000	0.000197	0.270000	0.049900
25%	0.000000	0.015800	0.535000	0.275000
50%	0.000000	0.073900	0.634000	0.417000
75%	0.000000	0.225000	0.747000	0.545000
max	0.090909	0.937000	0.950000	0.942000

	elem_prop_Verse 1
count	169.000000
mean	0.109360
std	0.031239
min	0.000000
25%	0.085106
50%	0.107143
75%	0.129032
max	0.205128

4.6. Conclusión

La principal diferencia entre estos clusters es la acústica de las canciones, el grupo del cluster 1 se compone de canciones con mayor probabilidad o confianza de que sean acústicas a diferencia del cluster 2.

Otras variables que podríamos considerar para un análisis más complejo son: * Cantidad de vistas
 * Intervalos de tiempo más escuchados en plataformas como YouTube y Spotify * Edad de público oyente

Con esta información sería posible identificar cuáles canciones generan mayor interés en el público y, en particular, cuáles fragmentos de las canciones capturan más su atención. Al disponer también

de las letras, podríamos analizar qué estilos rítmicos y temáticas líricas resultan más atractivos, lo que serviría como base para orientar la composición de futuras canciones hacia esos patrones de preferencia.

En seguimiento con la respuesta anterior, podríamos definir como variable objetivo las canciones más gustadas por el público, la cantidad de vistas de cada canción debería guardar una relación con sus características o métricas musicales, como el ritmo, la letra y los gustos del público, lo cual se refleja directamente en el número de reproducciones.